
Noise-Robust Density Estimation for Tabular Data Anomaly Detection

Dazhi Fu¹ Zhao Zhang² Jicong Fan¹

Abstract

Density-based anomaly detection methods often provide accurate and interpretable predictions but their performance can be severely degraded by inherent noise in the data, such as changes arising from environmental conditions during data collection or background noise. To deal with such noise, we present noise-robust density estimation (NRDE) for tabular data anomaly detection. We aim to estimate the density of pure data with the influence of noise isolated, which is a non-trivial task since the data-generating process is completely unknown. Specifically, NRDE learns a Jacobian-regularized normalizing flow to estimate the sources of data and categorizes sources into two groups, where one group generates pure data and the other generates noise. After generating pure data, we can use the density of such pure data to detect anomalies caused solely by the sources of pure data. Therefore, NRDE is robust to inherent noise. We provide theoretical results to support the effectiveness of NRDE and compare NRDE with 17 baselines on 47 benchmark datasets under different settings, including vanilla anomaly detection, anomaly detection with anomaly contamination, anomaly detection on noisy data, and transductive outlier detection. Our code is available at <https://github.com/fudazhiaka/Noise-Robust-Density-Estimation>.

1. Introduction

In an increasingly data-driven world, the problem of identifying unusual patterns or deviations from expected behavior—known as anomaly detection—has become paramount across diverse domains. Anomaly detection (Chandola et al.,

2009; Pang et al., 2021; Ruff et al., 2021), sometimes also referred to as novelty or outlier detection (Breunig et al., 2000; Pimentel et al., 2014; Fu & Fan, 2026), involves the identification of data points, events, or observations that significantly differ from the majority of the data. These anomalies can signal critical incidents such as fraud (Ahmed et al., 2016), security breaches (Breier & Branišová, 2017), system failures (Du et al., 2017), or novel insights, making their accurate detection essential for timely intervention and decision-making.

In the past few years, many deep learning-based anomaly detection methods have been proposed (Cai & Fan, 2022; Shenkar & Wolf, 2022; Xu et al., 2023; Fu et al., 2024; Yin et al., 2024; Xiao et al., 2025a; Dai et al., 2025; Xiao et al., 2025b). For instance, DeepSVDD (Ruff et al., 2018b) assumes that representations of normal data can be enclosed within a small hypersphere and representations of anomalous data lie outside the hypersphere, where the representations are given by a neural network. ADERH (Durani et al., 2026) constructs ensembles of hyperspheres from randomly paired samples to identify rare points. Although these methods often demonstrate impressive performance in various scenarios, several of them require making assumptions on the structure or distribution of normal and anomalous data, which may not hold or are difficult to guarantee by the training process. For instance, Zhang et al. (2024) analyzed the limitations of the hypersphere assumption in high-dimensional spaces and proposed to project normal data into the region bounded by two hyperspheres. Moreover, some of these methods are proposed to solve the one-class classification problem (Schölkopf et al., 2001), which relies on the assumption that training data originates from a single class or exhibits a single manifold structure. Consequently, these methods can be ineffective when training data encompasses multiple clusters or lies on multiple disconnected manifolds (Khayatkhoie et al., 2018).

It should be noted that density-based methods make no assumptions about the shape or distribution of the data and are capable of modeling complex data structures. This flexibility allows them to be effective even when training data encompasses multiple classes. Traditional density-based methods include Kernel Density Estimation (KDE) (Parzen, 1962), Gaussian Mixture Models (GMM), etc. These methods often suffer from the curse of dimensionality and are not

¹School of Data Science, The Chinese University of Hong Kong, Shenzhen, China ²School of Computer Science and Information Engineering, Hefei University of Technology, Hefei, China. Correspondence to: Jicong Fan <fanjicong@cuhk.edu.cn>.

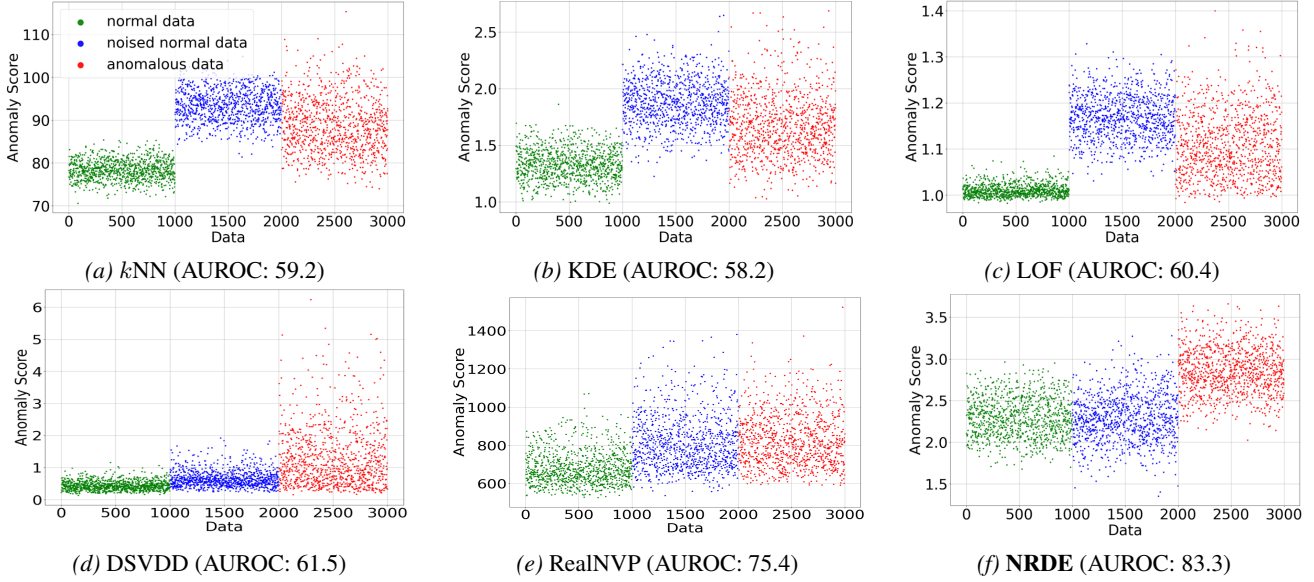


Figure 1. Detection performance on a synthetic dataset. A larger anomaly score indicates that the sample is more likely to be anomalous. The data were generated from a few data sources and many noise sources. Points marked in green, blue, and red represent **normal data**, **noisy normal data** (caused by noise change), and **anomalous data**, respectively. See (8) for definitions. The five compared methods detect most of the noisy normal samples as anomalies, while our NRDE is robust to changes in noise. More details about this experiment are in Appendix F.

effective on complex data. To address the problem, several deep learning based density estimation methods have been proposed. For instance, DAGMM (Zong et al., 2018) combines deep auto-encoders with GMM, utilizing the output density given by GMM in a low-dimensional space to detect anomalies. Normalizing flow (Kobyzev et al., 2020), an effective generative model, is also effective in estimating the density of complex data, and hence is useful for anomaly detection. Some flow-based image anomaly detection methods (Gudovskiy et al., 2022) first employ feature extractors to derive semantic representations and then implement normalizing flow to detect anomalies. In this work, we focus on tabular data since data of other types can be converted into tabular formats using feature extractors.

For standard anomaly detection, density-based methods, including normalizing flow and other shallow and deep models, are sensitive to changes in inherent noise in the data, yielding high false-positive rates. It is noteworthy that such inherent noise can be quite different from artificial noise like Gaussian noise, since it could represent minor changes in equipment or the environment during data collection. More specifically, real data contains inherent noise and can be described as being generated by $\mathbf{x} = G([\mathbf{s}_D; \mathbf{s}_N])$, where $\mathbf{s}_D$ and $\mathbf{s}_N$ denote the pure data source and noise source, respectively, and G is the generating function. The changes of $\mathbf{x}$ caused by $\mathbf{s}_N$ should not be treated as anomalies, or at least should be distinguished from the anomalies of interest, and we call such data noisy normal data for convenience. For instance, in a vehicle monitoring system,

changes in background noise may alter the observed data, but we are only concerned with the status of the vehicle itself. Similarly, in medical diagnosis, we hope that changes in instruments and equipment or the occasional noise do not affect the diagnostic results for diseases. In Figure 1, we use a synthetic dataset to show the influence of inherent noise on the performance of five anomaly detection methods and our proposed method **Noise-Robust Density Estimation** (NRDE). We observe that the five methods fail to distinguish between noisy normal data and real anomalies, exhibiting high false-positive rates and low AUROC values, whereas NRDE is robust to changes in the inherent noise in the data and performs the best.

NRDE trains a neural network to estimate the density of pure data with the influence of noise isolated. Specifically, we propose a Jacobian-regularized normalizing flow that explicitly separates latent sources into data and noise components, allowing us to isolate noise and estimate the density of pure data. As a result, we can detect anomalies that are caused by data sources without being affected by the noise. The framework of NRDE is shown in Figure 2. Our contributions are summarized as follows:

- We propose a novel density-based AD method, NRDE, for tabular data based on a Jacobian-regularized normalizing flow.
- NRDE separates the sources into pure-data sources and noise sources and performs density estimation for the pure data only, making it robust to changes in noise.

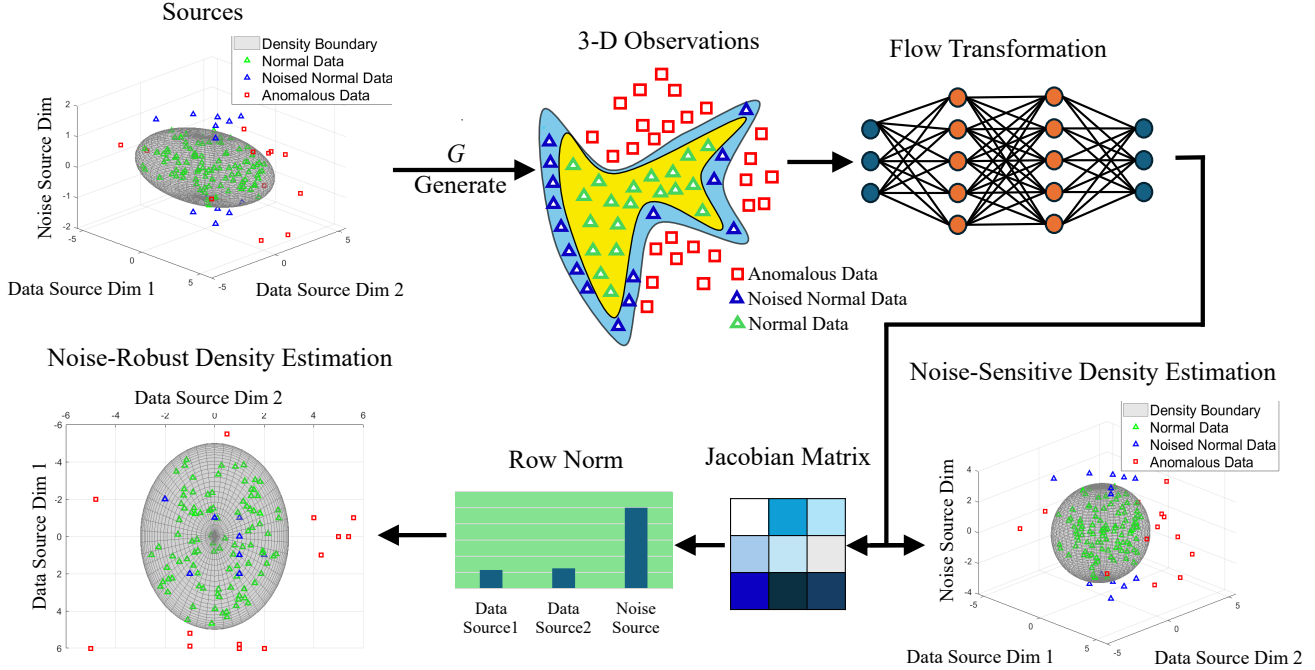


Figure 2. Architecture of the proposed NRDE. NRDE estimates the density of pure data by utilizing a normalizing flow with Jacobian regularization, where the influence of noise sources is isolated. Therefore, NRDE is robust to changes in inherent noise in the data.

- We provide some theoretical analysis for NRDE to support its effectiveness.
- We conduct experiments on 47 datasets to compare NRDE against 17 baseline methods. While primary evaluation is performed under the standard anomaly detection setting, our experimental setup also includes anomaly detection on noisy data, anomaly detection with contaminated data, and outlier detection.

2. Related Work and Preliminary Knowledge

2.1. Generative Models for Anomaly Detection

Deep generative models (Schlegl et al., 2019; Xia et al., 2022; Liu et al., 2025) are useful in anomaly detection due to their ability to model complex data structures. For instance, RobustRealNVP (Liu et al., 2022) ignores low-density points that are likely to be anomalies, by discarding the gradient produced by these points in the training stage, and therefore obtains a robust density function. In (Rozner et al., 2023), the authors found that densities around normal samples are relatively stable and proposed to use an autoregressive probabilistic model to maximize the density of training samples while minimizing their density variance. DTE (Livernoche et al., 2023) estimates the distribution over diffusion time for a given input and uses the mode or mean of this distribution as the anomaly score. TCCM (Li et al., 2025) employs a one-step flow-matching model to learn velocity fields between the normal data distribution and a simple reference distribution, and uses the discrep-

ancy between the predicted and true velocity vectors as the anomaly score. Unfortunately, these works do not address the problem of sensitivity to changes in inherent noise in standard anomaly detection shown by Figure 1.

2.2. ICA and Normalizing Flow

Independent Component Analysis (ICA) (Hyvärinen & Oja, 2000) assumes that observed data is generated by an unknown mixing process of several independent components (sources) drawn from simple distributions, and tries to obtain these components. By categorizing the mixing process, we can divide ICA methods into linear ICA and nonlinear ICA. Linear ICA assumes that the mixing process is linear and the sources are non-Gaussian, and often solves the problem by maximizing the non-Gaussianity. As for nonlinear ICA, the mixing process is assumed to be nonlinear, and the main problem faced by the field is that the model is unidentifiable or the sources are inseparable. In other words, there are infinitely many ways to transform the data into independent components, which may still be a mixture of underlying sources. By utilizing additional structure in the data or introducing auxiliary variables, many methods (Hyvärinen & Pajunen, 1999; Hyvärinen & Morioka, 2016; Zheng et al., 2022) have been developed. It is noteworthy that prior studies on ICA (Jung et al., 2000; Salimi-Khorshidi et al., 2014; Pruim et al., 2015) have shown that source components are not equally informative: some components correspond to meaningful signal sources, while others mainly capture

noise, nuisance, or background factors.

Here, we briefly review the foundational concept of normalizing flows, which is closely related to ICA. Given a set of observations, each of which, denoted as $\mathbf{x}$, is drawn from a complex distribution $\mathcal{X}$ in $\mathbb{R}^d$, normalizing flow aims to learn a function $F_{\mathcal{W}} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ composed of a sequence of invertible mappings $\{f_{\mathcal{W}_i}\}_{i=1}^T$, i.e., $F_{\mathcal{W}} = f_{\mathcal{W}_T} \circ \dots \circ f_{\mathcal{W}_2} \circ f_{\mathcal{W}_1}$, that transforms the complex distribution $\mathcal{X}$ into a simpler one, denoted as $\mathcal{Z}$, such as a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Here, T is the number of mappings and $\mathcal{W} = \{\mathcal{W}_1, \dots, \mathcal{W}_T\}$ denotes the set of all neural network parameters. Because $F_{\mathcal{W}}$ is invertible, the density of $\mathbf{x}$, namely $p_{\mathcal{X}}(\mathbf{x})$, can be computed using the change-of-variables formula:

$$p_{\mathcal{X}}(\mathbf{x}) = p_{\mathcal{Z}}(F_{\mathcal{W}}(\mathbf{x})) |\det(\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x}))|, \quad (1)$$

where $\det(\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x}))$ is the determinant of the Jacobian matrix of $F_{\mathcal{W}}$ evaluated at $\mathbf{x}$. One example of a coupling normalizing flow is RealNVP, proposed by (Dinh et al., 2016), where $f_{\mathcal{W}_i}$ is called the coupling transformation. Denoting by $\mathbf{x}^{(i)} \in \mathbb{R}^d$ the input of $f_{\mathcal{W}_i}$, $\mathbf{x}^{(i)}$ is usually split into two parts, i.e., $\mathbf{x}_{\alpha}^{(i)} = [x_{\alpha_1}^{(i)}, x_{\alpha_2}^{(i)}, \dots, x_{\alpha_{q_i}}^{(i)}]^{\top}$ and $\mathbf{x}_{\beta}^{(i)} = [x_{\beta_1}^{(i)}, x_{\beta_2}^{(i)}, \dots, x_{\beta_{d-q_i}}^{(i)}]^{\top}$, where $1 < q_i < d$. Then the output $\mathbf{y}^{(i)}$ of $f_{\mathcal{W}_i}$ is given as

$$\mathbf{y}_{\alpha}^{(i)} = \mathbf{x}_{\alpha}^{(i)}, \quad \mathbf{y}_{\beta}^{(i)} = \mathbf{x}_{\beta}^{(i)} \odot \exp(h_{i1}(\mathbf{x}_{\alpha}^{(i)})) + h_{i2}(\mathbf{x}_{\alpha}^{(i)}), \quad (2)$$

where $h_{i1} : \mathbb{R}^{q_i} \rightarrow \mathbb{R}^{d-q_i}$ and $h_{i2} : \mathbb{R}^{q_i} \rightarrow \mathbb{R}^{d-q_i}$ are two multilayer neural networks.

3. Proposed Method

3.1. Formulation of Noise-Robust Anomaly Detection

Let $\mathcal{D} = \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}\}$ be a set of d -dimensional training data drawn from an unknown distribution $\mathcal{X}$. The primary goal of anomaly detection (AD) is to learn a model $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}$ from the training set $\mathcal{D}$, which can quantify the degree of anomaly or the dissimilarity of a new sample $\mathbf{x}_{\text{new}}$ relative to the distribution $\mathcal{X}$.

From a generative perspective, as described in ICA (Hyvärinen & Oja, 2000), an observation $\mathbf{x}$ can be regarded as being generated by applying an unknown invertible linear or non-linear transformation, denoted as $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$, to some unknown source $\mathbf{s} \in \mathbb{R}^d$, i.e.,

$$\mathbf{x} = G(\mathbf{s}), \quad (3)$$

where $\mathbf{s} \sim \mathcal{S}$. As mentioned in other ICA and normalizing flow techniques, the source distribution $\mathcal{S}$ is usually simple and each dimension of $\mathcal{S}$ is independent. For instance¹,

¹Although the standard linear ICA requires an assumption that

consider $\mathcal{S} = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_d]^{\top}$, $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_d^2)$, and $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$. For convenience, we consider that the primary distinction among these sources resides in their variances, leading to the specification $\mathcal{S} = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. Based on G , the ideal normalizing flow $F_{\mathcal{W}}^*$ can be formulated as:

$$F_{\mathcal{W}}^*(\mathbf{x}) := \boldsymbol{\Sigma}^{-\frac{1}{2}} G^{-1}(\mathbf{x}), \quad (4)$$

where $\mathbf{z} = F_{\mathcal{W}}^*(\mathbf{x}) = \boldsymbol{\Sigma}^{-\frac{1}{2}} \mathbf{s} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Inspired by prior studies (Salimi-Khorshidi et al., 2014; Pruim et al., 2015), we split the source $\mathbf{s}$ into two distinct parts:

$$\mathbf{s} = [\mathbf{s}_D; \mathbf{s}_N], \quad (5)$$

where $\mathbf{s}_D \in \mathbb{R}^m$ denotes the pure data (or signal) source and $\mathbf{s}_N \in \mathbb{R}^{d-m}$ denotes the noise source. It is natural to assume that the variances of $\mathbf{s}_D$ are much greater than those of $\mathbf{s}_N$, namely,

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m > c\sigma_{m+1} \geq c\sigma_{m+2} \geq \dots \geq c\sigma_d, \quad (6)$$

where c is some constant much greater than 1. The data with noise removed, i.e., the pure data, is

$$\mathbf{x}_{\text{pure}} = G([\mathbf{s}_D; \mathbf{0}]). \quad (7)$$

Thus, the inherent noise in the data is $\epsilon := \mathbf{x} - \mathbf{x}_{\text{pure}}$. Letting $\mathcal{T}$ denote the signal source distribution deemed as normal, we have the following categorization for the data:

$$\begin{aligned} \text{pure normal data : } \mathbf{x}_{\text{pure}} &= G([\mathbf{s}_D; \mathbf{0}]), \mathbf{s}_D \sim \mathcal{T}, \\ \text{noisy normal data : } \mathbf{x}_{\text{norm}} &= G([\mathbf{s}_D; \mathbf{s}_N]), \mathbf{s}_D \sim \mathcal{T}, \\ \text{anomalous data : } \mathbf{x}_{\text{anom}} &= G([\hat{\mathbf{s}}_D; \tilde{\mathbf{s}}_N]), \hat{\mathbf{s}}_D \not\sim \mathcal{T}, \end{aligned} \quad (8)$$

where $\mathbf{s}_N \neq \mathbf{0}$ and $\tilde{\mathbf{s}}_N$ is arbitrary. In this work, given the observation $\mathbf{x}$, the task of noise-robust anomaly detection is to recover $\mathbf{x}_{\text{pure}}$ and evaluate whether $\mathbf{x}_{\text{pure}}$ is normal or anomalous, which is determined by $\mathbf{s}_D$ only and is irrelevant to $\mathbf{s}_N$.

Rationale for the assumption in (6): Prior studies typically rely on external structural information, metadata, and labeled data to train classifiers for identifying data sources. In contrast, we focus solely on the signal itself. This assumption is reasonable because a meaningful signal, by definition, should contain structured information and variation that differentiates it from the background. Noise, often arising from random and uncorrelated processes, tends to have its energy dispersed without a dominant structure. Therefore, the variance of the signal, which captures its total power and variability, is expected to be higher than that of the noise. This

the sources are non-Gaussian, the Gaussian assumption in this work makes sense because G may first convert each source to non-Gaussian and then perform mixing.

is a common and often necessary condition for the signal to be detectable and analyzable amid random fluctuations. For instance, in machine learning and statistics, PCA (Jolliffe & Cadima, 2016) assumes the most important data patterns are the directions with the highest variance, effectively treating them as the "signal" and discarding low-variance "noise." In signal processing, denoising filters work by removing low-power (low-variance) frequencies assumed to be noise, while preserving high-power (high-variance) frequencies considered to be the signal. More experimental results to verify our assumption and motivation are provided in Appendix H. For instance, the performance of baselines on pure data is provided in Table 13.

3.2. Signal and Noise Isolation

To realize the aforementioned noise-robust anomaly detection, we need to calculate $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ or $p_{\bar{\mathcal{X}}}(\mathbf{x}_{\text{pure}})$, where $\mathcal{X}$ denotes the distribution of $\mathbf{x}$ and $\bar{\mathcal{X}}$ denotes the distribution of $\mathbf{x}_{\text{pure}}$ defined on the m -dimensional manifold embedded in $\mathbb{R}^d$. When $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ or $p_{\bar{\mathcal{X}}}(\mathbf{x}_{\text{pure}})$ is smaller, $\mathbf{x}_{\text{pure}}$, as well as the corresponding noisy counterpart $\mathbf{x}$, is more likely to be anomalous.

Let $F_{\mathcal{W}}$ be the flow model learned from $\mathcal{D}$ and suppose $\mathbf{x}_{\text{pure}}$ can be identified from $\mathbf{x}$, we can obtain

$$p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) = p_{\mathcal{Z}}(F_{\mathcal{W}}(\mathbf{x}_{\text{pure}})) |\det(\nabla_{\mathbf{x}_{\text{pure}}} F_{\mathcal{W}}(\mathbf{x}_{\text{pure}}))|. \quad (9)$$

Using (4), we have the ideal case for $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$, i.e.,

$$\begin{aligned} & \log p_{\mathcal{X}}^*(\mathbf{x}_{\text{pure}}) \\ &= \log p_{\mathcal{Z}}(F_{\mathcal{W}}^*(\mathbf{x}_{\text{pure}})) + \log |\det(\nabla_{\mathbf{x}_{\text{pure}}} F_{\mathcal{W}}^*(\mathbf{x}_{\text{pure}}))| \\ &= \log |\det(\nabla_{\mathbf{x}_{\text{pure}}} F_{\mathcal{W}}^*(\mathbf{x}_{\text{pure}}))| - \sum_{i=1}^m \frac{s_i^2}{2\sigma_i^2} - \frac{d}{2} \log(2\pi), \end{aligned} \quad (10)$$

where we have used the fact that $F_{\mathcal{W}}^*(\mathbf{x}_{\text{pure}}) = [\mathbf{z}_D; \mathbf{0}]$ and $\mathbf{z} = \Sigma^{-\frac{1}{2}} \mathbf{s}$.

Similarly, for $p_{\bar{\mathcal{X}}}(\mathbf{x}_{\text{pure}})$, we have

$$p_{\bar{\mathcal{X}}}^*(\mathbf{x}_{\text{pure}}) = \frac{\mathcal{N}(\mathbf{z}_D; \mathbf{0}, \mathbf{I}_m)}{\sqrt{\det[J_{F_{\mathcal{W}}^{*-1}}(\mathbf{z}_D)^\top J_{F_{\mathcal{W}}^{*-1}}(\mathbf{z}_D)]}}, \quad (11)$$

where $J_{F_{\mathcal{W}}^{*-1}}(\mathbf{z}_D)$ denotes $\nabla_{\mathbf{z}_D} F_{\mathcal{W}}^{*-1}([\mathbf{z}_D; \mathbf{0}])$. The detailed derivation is provided in Appendix A.3.

The challenge is that we may never obtain $F_{\mathcal{W}}^*$. The learned $F_{\mathcal{W}}$ from $\mathcal{D}$ can only ensure that $\mathbf{z} = F_{\mathcal{W}}(\mathbf{x}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. It is difficult to determine which of $z_1, \dots, z_d$ correspond to $\mathbf{s}_D$ and which of $z_1, \dots, z_d$ correspond to $\mathbf{s}_N$. Moreover, the number of data sources m is unknown and is not easy to estimate. In the following, we show how to address these problems.

Note that (4) indicates that

$$\frac{\partial z_j}{\partial \mathbf{x}} = \sigma_j^{-1} \times \frac{\partial G_j^{-1}(\mathbf{x})}{\partial \mathbf{x}}. \quad (12)$$

We assume that

$$\gamma - \delta \leq \left\| \frac{\partial G_j^{-1}(\mathbf{x})}{\partial \mathbf{x}} \right\| \leq \gamma + \delta, \quad \forall j \in [d], \quad (13)$$

where γ and δ are some positive constants and $\delta \ll \gamma$. This assumption is reasonable because G usually mixes the sources randomly and uniformly. Moreover, this formulation generalizes the widely adopted assumption in linear ICA (Hyvärinen et al., 2001), which assumes $\mathbf{W}^\top \mathbf{W} = \mathbf{I}$ in $G(\mathbf{s}) = \mathbf{W}\mathbf{s}$, corresponding to the special case $\gamma = 1$ and $\delta = 0$. Combining (12) and (13), we have

$$(\gamma - \delta) \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|^{-1} \leq \sigma_j \leq (\gamma + \delta) \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|^{-1}. \quad (14)$$

If $(\gamma - \delta) \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|^{-1} > c(\gamma + \delta) \left\| \frac{\partial z_{j'}}{\partial \mathbf{x}} \right\|^{-1}$ or $\left\| \frac{\partial z_{j'}}{\partial \mathbf{x}} \right\| > c \frac{\gamma + \delta}{\gamma - \delta} \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|$ equivalently, then $\sigma_j > c\sigma_{j'}$. This indicates that we may compare $\left\| \frac{\partial z_1}{\partial \mathbf{x}} \right\|, \dots, \left\| \frac{\partial z_d}{\partial \mathbf{x}} \right\|$ to distinguish between $\mathbf{s}_D$ and $\mathbf{s}_N$. However, a clear gap may not exist between $\left\| \frac{\partial z_1}{\partial \mathbf{x}} \right\|, \dots, \left\| \frac{\partial z_d}{\partial \mathbf{x}} \right\|$. An intuitive example is shown in Figure 3. The reason is that the source $\mathbf{s}$ in (3) is not identifiable and there are many equivalent problems (Hyvärinen & Pajunen, 1999; Hyvärinen et al., 2019; Zheng et al., 2022). For instance, let $\mathbf{R}$ be an orthonormal matrix and suppose $F_{\mathcal{W}}(\mathbf{x}) = \mathbf{R}F_{\mathcal{W}}^*(\mathbf{x})$ is a normalizing flow learned from $\mathcal{D}$. In this case, $F_{\mathcal{W}}(\mathbf{x}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the estimated density remains unchanged. However, $F_{\mathcal{W}}(\mathbf{x})$ becomes a combination of $\mathbf{z}$, and the row norms of the Jacobian matrix do not explicitly reflect the variances of sources.

However, we can exploit the prior knowledge in (6) to train $F_{\mathcal{W}}$ close to $F_{\mathcal{W}}^*$ and consider the following optimization

$$\begin{aligned} & \underset{\mathcal{W}, A, B}{\text{maximize}} \sum_{\mathbf{x} \in \mathcal{D}} \log(p_{\mathcal{Z}}(F_{\mathcal{W}}(\mathbf{x})) |\det(\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x}))|) \\ & \text{subject to } \min_{j \in A} \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|^{-1} > c' \max_{j \in B} \left\| \frac{\partial z_j}{\partial \mathbf{x}} \right\|^{-1}, \quad \forall \mathbf{x} \in \mathcal{D} \\ & A \cup B = [d], \quad A \cap B = \emptyset, \quad |A| = m \end{aligned} \quad (15)$$

where $c' = c \frac{\gamma + \delta}{\gamma - \delta}$ and A corresponds to $\mathbf{s}_D$ and B corresponds to $\mathbf{s}_N$. It is very difficult to solve (15) because c, γ, δ are unknown and the constraints are related to every $\mathbf{x}$ and the min and max operations. We also need to know m .

3.3. Jacobian-Regularized Normalizing Flow

The constraints in (15) indicate that some rows of the Jacobian matrix $\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x})$ have much smaller norms than other

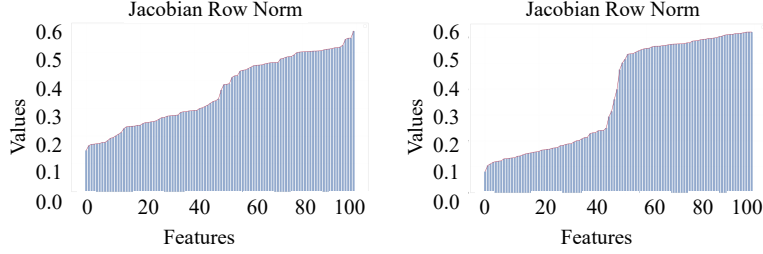


Figure 3. Visualization of row norms of the Jacobian matrix on a synthetic dataset with 50 pure data sources and 50 noise sources. The left panel shows the unregularized case, while the right panel shows the regularized case. More visualization results on real datasets are shown in Appendix I.1.

rows, which is a kind of sparsity. Therefore, we propose to regularize $\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x})$ during the optimization of $F_{\mathcal{W}}$ and hence solve

$$\begin{aligned} \text{minimize}_{\mathcal{W}} \quad & \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{D}} -\log \left(p_{\mathcal{Z}}(F_{\mathcal{W}}(\mathbf{x})) |\det(\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x}))| \right) \\ & + \lambda \mathcal{R} \left(\frac{1}{n} \sum_{\mathbf{x} \in \mathcal{D}} |\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x})| \right), \end{aligned} \quad (16)$$

where $\mathcal{R}$ denotes a sparse regularizer on matrices and $\lambda > 0$ is a hyperparameter. Instead of imposing regularization on each individual $\mathbf{x} \in \mathcal{D}$, we regularize the average absolute Jacobian, since the categories of sources should remain consistent across $\mathcal{D}$. We use the following $\mathcal{R}$:

$$\mathcal{R}(\mathbf{Q}) = \sum_{i=1}^d \sqrt{\|\mathbf{q}_{i:}\|_1}, \quad (17)$$

where $\mathbf{q}_{i:}$ denotes the i th row of $\mathbf{Q} \in \mathbb{R}^{d \times d}$. Note that $\mathcal{R}^2(\mathbf{Q})$ is the $\ell_{1,1/2}$ quasi-norm, which is sharper than the $\ell_{2,1}$ norm widely used in sparse optimization. Figure 3 illustrates the effect of $\mathcal{R}$. More details about $\mathcal{R}$ are provided in Appendix C.

An alternative to (17) is using $\mathbf{R}(\mathbf{Q}) = \sum_{j \in [B]} \|\mathbf{q}_{j:}\| - \sum_{j \in [A]} \|\mathbf{q}_{j:}\|$, where A is the index set of the m rows of $\mathbf{Q}$ with smaller norms and B is the index set of the $d - m$ rows of $\mathbf{Q}$ with larger norms determined in each iteration. This method requires a good estimate of m and is sensitive to the initialization. The performance is not as good as (17).

Although solving (16) makes sense, to obtain the pure data $\mathbf{x}_{\text{pure}}$, we will have to estimate m . Here we provide a simple way to determine $\hat{m}$, an approximation of m as follows: we first compute the average absolute Jacobian

$$J = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{D}} |\nabla F_{\mathcal{W}}(\mathbf{x})|. \quad (18)$$

We then take the row norms $\{\|J_i\|\}_{i=1}^d$ and sort them in ascending order, denoted by $\{\|J_i\|\}_{i=1}^d$. Then we measure

the variance gap by computing $\Delta_i = \frac{J_{(i+1)} - J_i}{J_{(i+1)}}$, and find m by

$$\hat{m} = \arg \max_i \Delta_i, \quad (19)$$

and

$$\|J_i\| \approx \frac{\gamma + \delta}{\sigma_i}. \quad (20)$$

Note that $\|J_i\|$ is smaller for sources with larger variance σ_i , which is more likely to be a data source. After determining $\hat{m}$, we can select $\hat{m}$ sources with the smallest row norms as $\mathbf{s}_{\hat{D}}$ or $\mathbf{z}_{\hat{D}}$ and generate the pure data $\hat{\mathbf{x}}_{\text{pure}}$ by:

$$\hat{\mathbf{x}}_{\text{pure}} = F_{\mathcal{W}}^{-1}([\mathbf{z}_{\hat{D}}; \mathbf{0}]). \quad (21)$$

Now, one can use the density of pure data $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}})$ defined in (9) or $p_{\bar{\mathcal{X}}}(\hat{\mathbf{x}}_{\text{pure}})$ defined in (11) as its anomaly score. For clarity, we refer to the two methods as NRDE and NRDE-M (manifold), respectively, throughout the experiments. More details about $p_{\bar{\mathcal{X}}}(\mathbf{x}_{\text{pure}})$ is provided in Appendix A.3.

In summary, we train a Jacobian-regularized normalizing flow via (16). After the model is well-trained, we can determine the signal sources using (19). Then, for any test data $\mathbf{x}^{\text{new}}$, we can generate its pure data $\hat{\mathbf{x}}_{\text{pure}}^{\text{new}}$ using (21) and then use the density of pure data $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$ or $p_{\bar{\mathcal{X}}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$ as the anomaly score to determine whether $\mathbf{x}^{\text{new}}$ is anomalous or not. More details about the algorithm of NRDE are shown in Appendix F.

3.4. Theoretical Guarantees

We provide the following theoretical guarantee for $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}})$ to approximate $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ and detect anomalies successfully.

Theorem 3.1. *Let $F_{\mathcal{W}} = f_{\mathcal{W}_T} \circ \dots \circ f_{\mathcal{W}_2} \circ f_{\mathcal{W}_1}$ denote an ideal sequence of coupling normalizing flows, where the Lipschitz constant of each $f_{\mathcal{W}_i}$ is bounded above by τ_i^+ and below by τ_i^- and the Lipschitz constant of h_{i1} is bounded above by $\tau_{i\alpha}$. Suppose $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ and $p_{\bar{\mathcal{X}}}(\hat{\mathbf{x}}_{\text{pure}})$ are the estimated densities of pure data $\mathbf{x}_{\text{pure}}$ and $\hat{\mathbf{x}}_{\text{pure}}$ respectively.*

Then the following inequality holds:

$$|\log p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}) - \log p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})| \leq \eta \|[\mathbf{z}_D; \mathbf{0}] - [\mathbf{z}_{\hat{D}}; \mathbf{0}]\| + \frac{\|\mathbf{z}_{\hat{D}}\|^2 - \|\mathbf{z}_D\|^2}{2},$$

where $\eta = \sum_{i=1}^T (\tau_{i\alpha} \sqrt{d} \prod_{j=1}^{i-1} \tau_j^+) \prod_{k=1}^T \frac{1}{\tau_k}$. This theorem indicates that our method can approximate the density of the pure data and the estimation error mainly originates from the difference between estimated data sources $\mathbf{z}_{\hat{D}}$ and true data sources $\mathbf{z}_D$.

Assumption 3.2. For any $\mathbf{x}^a, \mathbf{x}^b \in \mathbb{R}^d$, there exists a constant φ such that $|p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^a) - p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^b)| \geq \varphi \|\mathbf{x}^a - \mathbf{x}^b\|$. Moreover, if $\mathbf{x}^c$ is an anomaly, $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) \leq \max_{\mathbf{x} \in \mathcal{D}} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}})$

This assumption is reasonable since φ can be calculated as $\inf_{\mathbf{x} \in \text{dom}(p_{\mathcal{X}})} \|\nabla_{\mathbf{x}} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}})\|$ and if the pure data of an anomaly has a density exceeding the maximum density observed in the pure data of all training data, its detection becomes considerably more challenging or even impossible.

Theorem 3.3. Let $\mathbf{x}^c$ be an anomaly. Suppose that $\mathbf{x}^a, \mathbf{x}^b \in \mathcal{D}$ such that $\arg \max_{\mathbf{x}} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}) = \mathbf{x}^a$ and $\arg \min_{\mathbf{x}} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}) = \mathbf{x}^b$ and $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^a) = s_1, p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^b) = s_2$. Then, under Assumption 3.2, if $\|\mathbf{x}^c - \mathbf{x}^a\| > \frac{s_1 - s_2}{\varphi}$, $\mathbf{x}^c$ can be detected as an anomaly.

Theorem 3.3 shows that our proposed method can detect anomalies that are significantly distant from normal data. The proofs for the theorems are provided in Appendix A. We also compare the time complexity of density-based methods in Appendix B, while the actual training-time comparison is included in Appendix I.7.

4. Numerical Results

4.1. Experimental Settings

Datasets In our experiments, we evaluate the performance of 17 baseline methods on 47 widely used real-world datasets spanning multiple domains in a popular benchmark for anomaly detection proposed by (Han et al., 2022). Statistical information about these datasets is provided in Appendix G. In anomaly detection tasks, we follow the protocol of (Zong et al., 2018; Bergman & Hoshen, 2020; Shenkar & Wolf, 2022; Xu et al., 2023) by randomly partitioning normal samples in the datasets: 50% for training, while the remaining 50% are combined with all anomalous samples to form the test set. For outlier detection, the model is trained on the entire dataset to identify outliers within it, which is a transductive learning setting. Since our primary focus is on standard anomaly detection, we randomly select 10 datasets for evaluation under other anomaly detection

settings including outlier detection, AD on noisy data, and AD with anomaly contamination.

Baselines The 17 baselines include TCCM (Li et al., 2025), ADERH (Durani et al., 2026), DTE (Livernoche et al., 2023), MCM (Yin et al., 2024), DPAD (Fu et al., 2024), SLAD (Xu et al., 2023), ECOD (Li et al., 2022), ICL (Shenkar & Wolf, 2022), NAD (Qiu et al., 2021), DSVDD (Ruff et al., 2018a), RealNVP (Dinh et al., 2016), IF (Liu et al., 2008), AE (Hinton & Salakhutdinov, 2006), OC-SVM (Schölkopf et al., 2001), LOF (Breunig et al., 2000), k NN (Ramaswamy et al., 2000), KDE (Parzen, 1962). For TCCM, ADERH, DTE, MCM, DPAD, SLAD, ICL, and NAD, we use the code provided by the authors of the papers. For other methods, we use the code from the Python library PyOD (Chen et al., 2024). For all baseline methods and across all experiments, we perform grid search to identify the best-performing hyperparameter configuration on each dataset, unless stated otherwise. The detailed hyperparameter configuration for grid search is in Appendix E.

It should be mentioned that for standard anomaly detection, in addition to grid search, we also compare all methods under two other hyperparameter selection strategies: one is based on AutoUAD (Dai & Fan, 2025), and the other is based on the recommended hyperparameter settings in the original papers or code. The results are shown in Appendix I.2, including Table 15, Table 16, Table 17, and Table 18.

Evaluation Metrics We use the Area Under the Receiver Operating Characteristic Curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC) as evaluation metrics, following (Han et al., 2022; Xu et al., 2023; Yin et al., 2024; Li et al., 2025). These two metrics do not rely on specific decision thresholds and are capable of comprehensively assessing the performance of different methods. More details are provided in Appendix D.

4.2. Results of Standard Anomaly Detection

Table 1 reports the detailed performance of different methods on 14 datasets, and the average AUROC and AUPRC results across 47 datasets, with detailed results for each dataset available in Appendix I.2. NRDE achieves the best performance, outperforming the second-best method by more than 2%. Compared to RealNVP, NRDE demonstrates a significant improvement. Additionally, NRDE and NRDE-M outperform other baseline methods on a larger number of datasets. For example, on yeast, vertebral, and WPBC, baseline methods attain AUROC scores around 50%, indicating anomaly detection is particularly challenging for these datasets. In contrast, our method significantly outperforms baselines, highlighting its effectiveness in complex datasets. Notably, KDE and k NN—two traditional methods—outperform many other baseline methods. We attribute this phenomenon to two main factors. First, as mentioned

Noise-Robust Density Estimation for Tabular Data Anomaly Detection

Table 1. AUROC (%) and AUPRC (%) with standard deviation of each method on 14 tabular datasets from ADBench. The detailed results on 47 datasets is shown in Appendix I.2. Due to the page limitation, We only report the average performance, average rank, and paired p -values over all 47 datasets. The best and second-best results are highlighted in red and orange, respectively. Hyperparameter of each method on each dataset is tuned using grid search.

AUROC	KDE (1962)	kNN (2000)	LOF (2000)	OC-SVM (2001)	IF (2008)	AE (2006)	DSVDD (2018)	RealNVP (2016)	NAD (2021)	ECOD (2022)	ICL (2022)	SLAD (2023)	DPAD (2024)	MCM (2024)	DTE-C (2024)	TCCM (2025)	ADERH (2025)	NRDE (2025)	NRDE-M (2025)
anthyroid	91.4 ± 0.0	94.1 ± 0.0	95.4 ± 0.0	91.7 ± 0.0	93.1 ± 0.0	93.5 ± 2.3	92.3 ± 1.6	96.1 ± 0.5	87.7 ± 3.2	78.7 ± 0.0	83.5 ± 0.5	87.3 ± 0.4	82.8 ± 4.8	98.2 ± 0.0	97.9 ± 0.1	94.1 ± 0.8	91.7 ± 0.4	97.3 ± 0.5	98.4 ± 0.3
cover	95.5 ± 0.0	98.5 ± 0.0	99.3 ± 0.0	98.8 ± 0.0	92.1 ± 0.0	98.4 ± 0.4	49.8 ± 0.5	83.8 ± 0.1	82.2 ± 7.6	92.0 ± 1.1	67.4 ± 3.4	73.1 ± 5.8	50.1 ± 0.0	95.1 ± 0.0	98.1 ± 0.1	98.7 ± 1.0	93.1 ± 0.1	99.9 ± 0.0	99.6 ± 0.1
mammography	89.8 ± 0.0	87.8 ± 0.0	85.0 ± 0.0	88.9 ± 0.0	89.0 ± 0.0	91.5 ± 0.2	89.6 ± 0.2	89.5 ± 0.4	76.1 ± 0.3	90.7 ± 0.0	76.7 ± 1.8	76.0 ± 0.4	78.2 ± 8.9	91.6 ± 0.0	89.3 ± 0.8	86.3 ± 0.5	87.5 ± 0.0	91.7 ± 0.1	92.4 ± 0.7
PageBlocks	95.0 ± 0.0	96.2 ± 0.0	97.5 ± 0.0	96.1 ± 0.0	94.1 ± 0.0	97.0 ± 0.5	96.4 ± 0.1	92.0 ± 0.9	92.9 ± 0.1	91.4 ± 0.0	91.7 ± 0.9	90.0 ± 1.6	90.8 ± 4.1	79.8 ± 0.0	96.7 ± 0.3	95.7 ± 1.4	94.4 ± 0.0	93.5 ± 0.5	97.2 ± 0.5
Pima	78.5 ± 0.0	79.0 ± 0.0	77.3 ± 0.0	76.5 ± 0.0	76.9 ± 0.0	76.2 ± 0.8	75.7 ± 0.4	79.3 ± 0.6	63.5 ± 1.2	61.5 ± 0.0	70.2 ± 0.3	64.7 ± 7.2	75.0 ± 0.7	70.3 ± 0.0	73.1 ± 1.5	74.9 ± 3.5	77.2 ± 0.2	80.0 ± 0.3	81.1 ± 0.8
smtp	88.2 ± 0.0	93.6 ± 0.0	94.9 ± 0.0	87.2 ± 0.0	93.5 ± 0.0	94.6 ± 0.0	91.9 ± 0.1	93.4 ± 0.1	95.3 ± 0.2	87.9 ± 0.0	65.0 ± 15.0	96.2 ± 1.2	84.1 ± 0.8	88.4 ± 0.0	95.5 ± 0.0	91.8 ± 4.6	93.0 ± 0.3	90.6 ± 1.1	96.3 ± 0.3
SpamBase	86.0 ± 0.0	84.5 ± 0.0	82.4 ± 0.0	81.7 ± 0.0	79.1 ± 0.0	82.6 ± 0.4	83.3 ± 0.1	80.1 ± 0.3	80.9 ± 0.5	66.0 ± 0.0	82.3 ± 0.2	84.8 ± 0.8	71.2 ± 3.1	73.6 ± 0.0	85.4 ± 0.7	86.4 ± 0.0	80.8 ± 0.1	84.5 ± 0.4	82.0 ± 0.3
speech	71.4 ± 0.0	48.5 ± 0.0	57.7 ± 0.0	46.3 ± 0.0	49.2 ± 0.0	47.5 ± 0.0	47.6 ± 1.5	50.0 ± 0.0	58.5 ± 1.0	46.1 ± 0.0	54.9 ± 4.6	55.4 ± 0.3	54.3 ± 3.5	52.7 ± 0.0	52.9 ± 2.0	55.1 ± 0.3	49.2 ± 0.1	58.7 ± 2.7	62.8 ± 0.4
Stamps	95.1 ± 0.0	92.3 ± 0.0	91.9 ± 0.0	92.9 ± 0.0	95.2 ± 0.0	93.6 ± 1.5	90.6 ± 0.4	93.6 ± 1.4	79.7 ± 2.7	86.7 ± 0.0	89.0 ± 0.6	81.5 ± 1.1	92.5 ± 1.5	89.8 ± 0.0	89.6 ± 2.2	90.8 ± 1.2	91.9 ± 0.1	96.9 ± 1.6	97.4 ± 1.0
verbal	43.5 ± 0.0	45.7 ± 0.0	58.0 ± 0.0	59.3 ± 0.0	49.6 ± 0.0	55.8 ± 0.7	50.9 ± 1.3	53.6 ± 4.8	61.5 ± 4.7	41.8 ± 0.0	60.8 ± 0.7	46.2 ± 3.4	48.3 ± 9.2	53.0 ± 0.0	66.6 ± 3.5	69.3 ± 5.0	36.7 ± 0.2	82.8 ± 4.3	80.0 ± 0.7
Waveform	76.9 ± 0.0	76.4 ± 0.0	76.7 ± 0.0	72.7 ± 0.0	73.6 ± 0.0	71.6 ± 0.8	73.1 ± 2.9	72.5 ± 1.6	75.3 ± 1.0	60.0 ± 0.0	64.3 ± 1.5	52.2 ± 0.6	82.3 ± 1.9	77.8 ± 0.0	64.9 ± 0.1	75.7 ± 1.8	77.4 ± 0.4	86.7 ± 1.3	80.2 ± 2.8
wine	92.2 ± 0.0	93.3 ± 0.0	92.2 ± 0.0	92.2 ± 0.0	93.8 ± 0.0	92.5 ± 2.7	90.2 ± 4.2	92.6 ± 1.6	97.4 ± 1.8	73.0 ± 0.0	89.2 ± 3.7	95.0 ± 4.0	84.2 ± 7.7	100.0 ± 0.0	94.7 ± 1.7	94.1 ± 0.8	92.8 ± 0.1	99.8 ± 0.3	99.4 ± 0.2
WPBC	52.5 ± 0.0	51.7 ± 0.0	53.5 ± 0.0	51.2 ± 0.0	54.6 ± 0.0	52.6 ± 1.2	49.8 ± 3.5	58.0 ± 1.0	57.5 ± 0.9	47.0 ± 0.0	57.8 ± 0.8	49.8 ± 1.8	51.0 ± 4.8	57.3 ± 0.0	53.8 ± 1.3	51.2 ± 0.6	50.1 ± 0.0	77.5 ± 5.0	64.7 ± 2.5
yeast	43.2 ± 0.0	47.6 ± 0.0	49.5 ± 0.0	48.3 ± 0.0	47.4 ± 0.0	50.0 ± 0.2	50.9 ± 0.3	51.0 ± 1.4	60.2 ± 2.3	45.3 ± 0.0	57.6 ± 3.4	55.3 ± 1.3	55.3 ± 4.2	50.1 ± 0.0	52.0 ± 2.4	59.0 ± 1.0	40.1 ± 0.2	61.2 ± 2.5	63.4 ± 0.9
Avg. AUROC	85.3	86.1	86.3	81.1	82.1	85.9	81.3	82.3	81.0	74.2	81.2	82.0	77.8	85.9	86.2	86.7	81.7	88.5	86.7
Avg. Rank	8.18	7.44	7.82	11.53	10.93	7.86	10.68	11.50	11.02	16.18	11.34	11.61	13.76	7.71	9.12	8.37	11.83	5.79	7.34
p -value	0.037	0.041	0.094	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.010	0.015	0.037	0.000	-	0.008
AUPRC																			
anthyroid	66.2 ± 0.0	72.0 ± 0.0	77.0 ± 0.0	66.9 ± 0.0	65.8 ± 0.0	69.3 ± 7.3	69.5 ± 3.0	77.0 ± 2.8	43.4 ± 7.1	40.8 ± 0.0	48.1 ± 2.6	60.1 ± 0.5	54.7 ± 6.6	85.5 ± 0.0	84.3 ± 0.3	73.8 ± 0.8	65.1 ± 1.3	75.6 ± 3.0	85.7 ± 2.1
cover	34.2 ± 0.0	72.0 ± 0.0	84.8 ± 0.0	48.2 ± 0.0	19.5 ± 0.0	59.1 ± 0.9	2.2 ± 0.0	9.3 ± 1.8	46.6 ± 0.4	18.4 ± 0.0	9.0 ± 3.7	4.6 ± 1.2	2.0 ± 0.0	21.1 ± 0.0	71.8 ± 0.7	93.4 ± 0.1	16.2 ± 0.1	96.5 ± 0.1	84.6 ± 6.8
mammography	41.6 ± 0.0	41.9 ± 0.0	35.0 ± 0.0	41.2 ± 0.0	47.1 ± 0.0	59.7 ± 1.9	38.5 ± 0.8	44.8 ± 2.2	11.1 ± 0.4	54.0 ± 0.0	30.0 ± 0.8	20.2 ± 1.3	36.6 ± 7.2	43.9 ± 0.0	48.0 ± 0.3	39.0 ± 2.3	38.6 ± 0.3	49.6 ± 6.8	59.4 ± 3.5
PageBlocks	84.8 ± 0.0	86.6 ± 0.0	89.9 ± 0.0	86.2 ± 0.0	76.8 ± 0.0	88.2 ± 1.2	87.5 ± 0.4	74.5 ± 1.8	78.5 ± 0.7	66.4 ± 0.0	81.4 ± 1.2	74.7 ± 2.3	78.8 ± 4.4	54.1 ± 0.0	86.4 ± 0.6	86.2 ± 2.7	79.6 ± 0.3	76.6 ± 4.2	90.1 ± 1.2
Pima	77.0 ± 0.0	77.9 ± 0.0	76.0 ± 0.0	76.0 ± 0.0	78.2 ± 0.0	78.1 ± 0.0	75.2 ± 1.0	77.6 ± 0.8	63.0 ± 1.5	65.7 ± 0.0	69.4 ± 2.0	66.6 ± 7.2	74.3 ± 0.5	73.8 ± 0.0	73.4 ± 1.4	74.1 ± 2.1	75.6 ± 0.3	79.5 ± 0.9	80.5 ± 1.0
smtp	58.8 ± 0.0	42.0 ± 0.0	40.4 ± 0.0	61.4 ± 0.0	1.2 ± 0.0	40.8 ± 0.6	28.4 ± 3.8	32.0 ± 2.0	48.1 ± 1.9	52.6 ± 0.0	12.3 ± 12.2	43.9 ± 0.0	50.1 ± 10.9	65.0 ± 0.0	43.6 ± 1.5	52.8 ± 12.3	43.6 ± 6.1	57.6 ± 0.9	57.9 ± 0.5
SpamBase	88.3 ± 0.0	86.6 ± 0.0	83.8 ± 0.0	85.1 ± 0.0	84.1 ± 0.0	85.5 ± 0.3	86.5 ± 0.1	80.5 ± 0.4	83.8 ± 0.8	68.9 ± 0.7	87.0 ± 0.2	74.3 ± 2.6	78.8 ± 0.0	87.2 ± 0.3	88.9 ± 0.2	84.2 ± 0.1	85.3 ± 0.6	80.9 ± 0.6	80.9 ± 0.6
speech	7.5 ± 0.0	3.7 ± 0.0	6.0 ± 0.0	3.6 ± 0.0	4.0 ± 0.0	3.8 ± 0.1	3.1 ± 0.1	3.2 ± 0.0	4.6 ± 0.8	3.8 ± 0.0	4.2 ± 0.9	3.9 ± 0.3	4.9 ± 1.3	5.9 ± 0.0	4.0 ± 0.5	5.0 ± 0.1	3.9 ± 0.1	4.9 ± 0.7	5.1 ± 0.2
Stamps	63.7 ± 0.0	54.2 ± 0.0	55.4 ± 0.0	56.3 ± 0.0	69.4 ± 0.0	59.5 ± 4.2	49.6 ± 1.1	61.8 ± 6.1	33.2 ± 3.4	45.2 ± 0.0	61.3 ± 0.7	36.9 ± 2.0	57.2 ± 3.9	51.5 ± 0.0	54.1 ± 3.7	55.0 ± 2.5	53.6 ± 0.1	78.4 ± 13.0	88.3 ± 4.8
verbal	19.7 ± 0.0	20.3 ± 0.0	29.6 ± 0.0	26.4 ± 0.0	22.2 ± 0.0	24.7 ± 0.1	23.1 ± 0.5	25.3 ± 2.2	33.8 ± 3.2	19.5 ± 0.0	30.3 ± 0.8	20.3 ± 1.0	23.7 ± 5.3	23.2 ± 0.0	33.0 ± 4.2	35.5 ± 5.4	17.4 ± 0.0	51.4 ± 7.4	54.9 ± 2.3
Waveform	27.6 ± 0.0	25.4 ± 0.0	33.8 ± 0.0	11.2 ± 0.0	10.8 ± 0.0	15.8 ± 1.0	10.3 ± 0.4	11.3 ± 0.6	40.0 ± 0.3	7.6 ± 0.0	40.3 ± 1.4	6.8 ± 0.1	32.4 ± 9.4	50.4 ± 0.0	11.5 ± 0.0	10.6 ± 1.7	15.4 ± 0.0	32.3 ± 0.9	14.8 ± 1.4
wine	58.2 ± 0.0	61.5 ± 0.0	52.7 ± 0.0	58.7 ± 0.0	69.1 ± 0.0	65.3 ± 10.9	59.0 ± 13.4	56.6 ± 7.1	85.4 ± 7.7	30.5 ± 0.0	60.3 ± 8.8	73.1 ± 21.0	47.1 ± 14.4	100.0 ± 0.0	72.3 ± 9.7	66.2 ± 4.8	59.8 ± 0.5	99.1 ± 1.5	97.2 ± 1.2
WPBC	38.4 ± 0.0	38.2 ± 0.0	39.0 ± 0.0	37.8 ± 0.0	39.9 ± 0.0	39.8 ± 1.8	39.0 ± 2.0	43.8 ± 1.0	45.5 ± 3.2	35.4 ± 0.0	47.4 ± 2.7	38.9 ± 1.1	40.2 ± 3.0	42.5 ± 0.0	40.0 ± 2.1	37.8 ± 0.6	36.9 ± 0.0	68.7 ± 6.3	52.7 ± 6.4
yeast	48.2 ± 0.0	49.6 ± 0.0	52.4 ± 0.0	51.1 ± 0.0	52.3 ± 0.0	52.1 ± 0.1	51.3 ± 0.0	56.6 ± 1.1	50.4 ± 0.0	55.6 ± 1.9	55.6 ± 1.9	54.4 ± 0.6	54.8 ± 3.1	51.7 ± 0.0	52.2 ± 0.8	56.9 ± 1.4	46.1 ± 0.4	58.5 ± 2.9	61.1 ± 1.3
Avg. AUPRC	63.0	63.6	64.5	56.9	54.3	63.9	57.8	52.8	54.2	45.0	56.8	56.5	55.6	65.1	61.8	66.0	55.6	68.1	66.3
Avg. Rank	8.29	8.27	8.37	11.19	10.73	7.74	10.88	12.85	10.90	15.34	10.09	11.43	12.35	7.37	9.79	7.71	12.51	6.82	7.36
p -value	0.021	0.015	0.078	0.000	0.000	0.041	0.000	0.000	0.003	0.000	0.001	0.000	0.000	0.162	0.005	0.249	0.000	-	0.149

Table 2. Average AUROC (%) on anomaly detection with injected noise experiments over 10 tabular datasets of ADBench.

Metric	KDE	KNN	LOF	OCSVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE	NRDE-M
Avg. AUROC	70.3	69.2	68.9	67.3	65.4	66.9	69.4	62.4	60.8	50.1	69.3	69.5	68.6	65.5	61.7	67.8	65.1	81.8	77.5

Table 3. Average AUROC (%) on outlier detection over 10 tabular datasets. * indicates that we report the performance in their paper.

Metric	KDE	KNN	LOF	OCSVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	ODIM*	NRDE	NRDE-M
Avg. AUROC	40.7	68.2	58.7	65.4	69.2	67.1	63.8	66.6	66.2	62.7	57.6	64.4	61.5	76.1	67.7	69.2	69.1	65.2	79.4	78.2

earlier, tabular data typically consists of features that inherently provide excellent semantic differences. As a result, even Euclidean distance can capture meaningful distinctions between samples. Second, as demonstrated in (Jiang, 2017; Gu et al., 2019), these two methods provide more explicit predictions for datasets with lower dimensions and more samples, which aligns with the experimental results and the curse of dimensionality.

4.3. Results of AD with Anomaly Contamination

In practical AD scenarios, the training set typically contains a small but unavoidable proportion of anomalous samples. To evaluate the robustness and performance of all methods in this scenario, We introduce varying proportions of anomalies into the training set and conduct experiments on these contaminated datasets. The contamination ratio ranges from 1% to 10% of the training set size. We report the average performance of all methods in Figure 4, while the detailed experimental results for each dataset are in Appendix I.3. From the figure, we observe that as the anomaly ratio increases, the performance of all methods decreases. In this

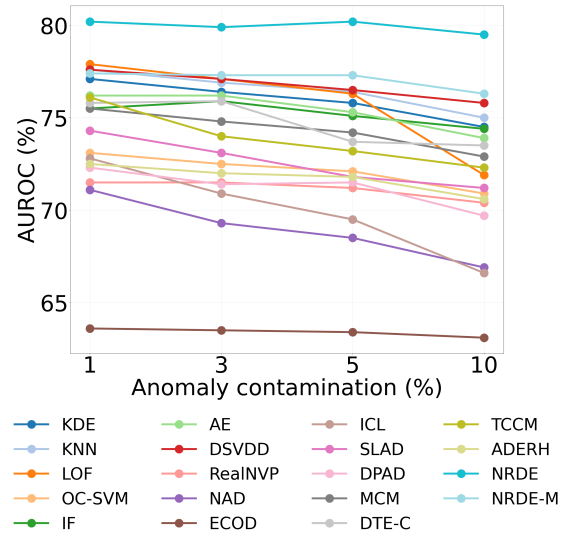


Figure 4. Average AUROC values across 10 datasets of AD experiments with anomaly contamination, contamination ratio ranging from 1% to 10%.

scenario, NRDE consistently achieves superior performance

over other methods, demonstrating its robustness to anomalies in the training set. It should be noted that the AUROC performance of NRDE is affected, with a less significant performance drop than that observed in other methods.

4.4. Results of Anomaly Detection on Noisy Data

In real-world AD, observations can be corrupted by noise. To assess the robustness of different methods under such conditions, we inject additive Gaussian noise into both the training set and the anomalous test samples, i.e., $\mathcal{N}(0, 0.1\mathbf{I}_d)$, while perturbing normal test samples with a stronger noise level $\mathcal{N}(0, 0.2\mathbf{I}_d)$. This setting reduces the separability between normal and anomalous test data, making the detection task more challenging than an equal-noise injection setting. Note that we first normalize the data using z-score normalization and then inject the perturbations. Table 2 reports the average results where NRDE consistently maintains superior performance compared to competing approaches, highlighting its robustness to noise perturbations. We also consider a symmetric variant where anomalous test samples receive stronger noise. In this setting, most methods exhibit improved performance compared to the standard anomaly detection setup, suggesting that the task is less challenging. NRDE still outperforms other methods under this setting. The detailed results are shown in Appendix I.4.

4.5. Results of Outlier Detection

To evaluate the effectiveness of anomaly detection methods in the outlier detection (transductive) setting, we conduct experiments on several datasets where all samples are used for both training and testing. Under this setting, we compare NRDE with the recently proposed ODIM (Kim et al., 2023), which is specifically designed for outlier detection. As shown in Table 3, NRDE outperforms competing methods, indicating strong robustness to the presence of anomalies during training. Detailed results for each dataset are provided in I.5.

4.6. More Results

Further experiments validating our assumptions and motivation are presented in Appendix H. Additional visualization results are provided in Appendix I.1. Ablation studies and hyperparameter (architecture) analysis are reported in Appendix I.6. The training time comparison is included in Appendix I.7, and the training dynamics are shown in Appendix I.8.

5. Conclusion

We proposed a novel and effective method NRDE for anomaly detection in tabular data. Our key observation is that data is typically generated by independent sources,

which can be categorized into pure data sources and noise sources. By distinguishing these sources using the Jacobian matrix, we can approximate the density of the pure data which is unaffected by noise. This allows NRDE to be robust to noise and effectively identify both anomalous data and noisy normal data. We provided theoretical analysis on the estimation error, the reliability of our proposed method, and the time complexity of density-based approaches. Numerical experiments demonstrated that NRDE outperforms 17 baseline methods across 47 real-world datasets. Furthermore, NRDE exhibits robustness to anomalies in the training set and noise inside the data.

Acknowledgements

This work was partially supported by the General Program of the Natural Science Foundation of Guangdong Province under Grant No.2024A1515011771, the National Natural Science Foundation of China under Grant No.62376236, and the Shenzhen Stability Science Program 2023.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Limitations

NRDE leverages the variance across sources to distinguish noise from signal, with noise being isolated through the proposed procedure. This design is primarily grounded in Assumption 6. While this assumption holds for many real-world datasets, there may exist scenarios where it is violated, in which case the effectiveness of NRDE may degrade. We provide a discussion and empirical verification of this limitation in Appendix H. In addition, NRDE may incur substantial computational overhead on extremely high-dimensional data due to the computation of Jacobian matrices. A practical remedy is to first apply dimensionality reduction techniques, as discussed in Appendix I.9.

References

- Ahmed, M., Mahmood, A. N., and Islam, M. R. A survey of anomaly detection techniques in financial domain. *Future Generation Computer Systems*, 55:278–288, 2016.
- Behrmann, J., Vicol, P., Wang, K.-C., Grosse, R., and Jacobsen, J.-H. Understanding and mitigating exploding inverses in invertible neural networks. In *International Conference on Artificial Intelligence and Statistics*, pp. 1792–1800. PMLR, 2021.

- Bergman, L. and Hoshen, Y. Classification-based anomaly detection for general data. *arXiv preprint arXiv:2005.02359*, 2020.
- Breier, J. and Branišová, J. A dynamic rule creation based anomaly detection method for identifying security breaches in log records. *Wireless Personal Communications*, 94:497–511, 2017.
- Breunig, M. M., Kriegel, H.-P., Ng, R. T., and Sander, J. Lof: identifying density-based local outliers. In *Proceedings of the 2000 ACM SIGMOD international conference on Management of data*, pp. 93–104, 2000.
- Cai, J. and Fan, J. Perturbation learning based anomaly detection. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 14317–14330. Curran Associates, Inc., 2022.
- Chandola, V., Banerjee, A., and Kumar, V. Anomaly detection: A survey. *ACM computing surveys (CSUR)*, 41(3): 1–58, 2009.
- Chen, S., Qian, Z., Siu, W., Hu, X., Li, J., Li, S., Qin, Y., Yang, T., Xiao, Z., Ye, W., Zhang, Y., Dong, Y., and Zhao, Y. Pyod 2: A python library for outlier detection with llm-powered model selection. *arXiv preprint arXiv:2412.12154*, 2024.
- Dai, W. and Fan, J. AutoUAD: Hyper-parameter optimization for unsupervised anomaly detection. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=ErQPdaD5wJ>.
- Dai, W., Hwang, K., and Fan, J. Unsupervised anomaly detection for tabular data using deep noise evaluation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(11):11553–11562, Apr. 2025.
- Dinh, L., Sohl-Dickstein, J., and Bengio, S. Density estimation using real nvp. *arXiv preprint arXiv:1605.08803*, 2016.
- Du, M., Li, F., Zheng, G., and Srikumar, V. Deeplog: Anomaly detection and diagnosis from system logs through deep learning. In *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security*, pp. 1285–1298, 2017.
- Durani, W., Leiber, C., Durani, K., Plant, C., and Böhm, C. Anomaly detection by an ensemble of random pairs of hyperspheres. *Advances in Neural Information Processing Systems*, 38:166168–166207, 2026.
- Fu, D. and Fan, J. UniOD: A universal model for outlier detection across diverse domains. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=Eu25AOvORb>.
- Fu, D., Zhang, Z., and Fan, J. Dense projection for anomaly detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 8398–8408, 2024.
- Gu, X., Akoglu, L., and Rinaldo, A. Statistical analysis of nearest neighbor methods for anomaly detection. *Advances in Neural Information Processing Systems*, 32, 2019.
- Gudovskiy, D., Ishizaka, S., and Kozuka, K. Cflow-ad: Real-time unsupervised anomaly detection with localization via conditional normalizing flows. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 98–107, 2022.
- Han, S., Hu, X., Huang, H., Jiang, M., and Zhao, Y. Ad-bench: Anomaly detection benchmark. *Advances in Neural Information Processing Systems*, 35:32142–32159, 2022.
- Hinton, G. E. and Salakhutdinov, R. R. Reducing the dimensionality of data with neural networks. *science*, 313 (5786):504–507, 2006.
- Hyvärinen, A. and Morioka, H. Unsupervised feature extraction by time-contrastive learning and nonlinear ica. *Advances in neural information processing systems*, 29, 2016.
- Hyvärinen, A. and Oja, E. Independent component analysis: algorithms and applications. *Neural networks*, 13(4-5): 411–430, 2000.
- Hyvärinen, A. and Pajunen, P. Nonlinear independent component analysis: Existence and uniqueness results. *Neural networks*, 12(3):429–439, 1999.
- Hyvärinen, A., Hurri, J., and Hoyer, P. O. Independent component analysis. In *Natural Image Statistics: A Probabilistic Approach to Early Computational Vision*, pp. 151–175. Springer, 2001.
- Hyvärinen, A., Sasaki, H., and Turner, R. Nonlinear ica using auxiliary variables and generalized contrastive learning. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pp. 859–868. PMLR, 2019.
- Jiang, H. Uniform convergence rates for kernel density estimation. In *International Conference on Machine Learning*, pp. 1694–1703. PMLR, 2017.
- Jolliffe, I. T. and Cadima, J. Principal component analysis: a review and recent developments. *Philosophical transactions of the royal society A: Mathematical, Physical and Engineering Sciences*, 374(2065):20150202, 2016.

- Jung, T.-P., Makeig, S., Humphries, C., Lee, T.-W., Mckeown, M. J., Iragui, V., and Sejnowski, T. J. Removing electroencephalographic artifacts by blind source separation. *Psychophysiology*, 37(2):163–178, 2000.
- Khayatkhoei, M., Singh, M. K., and Elgammal, A. Disconnected manifold learning for generative adversarial networks. *Advances in Neural Information Processing Systems*, 31, 2018.
- Kim, D., Hwang, J., Lee, J., Kim, K., and Kim, Y. Odin: Outlier detection via likelihood of under-fitted generative models. *arXiv preprint arXiv:2301.04257*, 2023.
- Kobyzev, I., Prince, S. J., and Brubaker, M. A. Normalizing flows: An introduction and review of current methods. *IEEE transactions on pattern analysis and machine intelligence*, 43(11):3964–3979, 2020.
- Li, Z., Zhao, Y., Hu, X., Botta, N., Ionescu, C., and Chen, G. H. Ecod: Unsupervised outlier detection using empirical cumulative distribution functions. *IEEE Transactions on Knowledge and Data Engineering*, 35(12):12181–12193, 2022.
- Li, Z., Huang, Q., Zhu, Y., Yang, L., Amiri, M. M., van Stein, N., and van Leeuwen, M. Scalable, explainable and provably robust anomaly detection with one-step flow matching. *arXiv preprint arXiv:2510.18328*, 2025.
- Liu, B., Tan, P.-N., and Zhou, J. Unsupervised anomaly detection by robust density estimation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 4101–4108, 2022.
- Liu, F. T., Ting, K. M., and Zhou, Z.-H. Isolation forest. In *2008 eighth IEEE international conference on data mining*, pp. 413–422. IEEE, 2008.
- Liu, J., Ma, Z., Wang, Z., Zou, C., Ren, J., Wang, Z., Song, L., Hu, B., Liu, Y., and Leung, V. A survey on diffusion models for anomaly detection. *arXiv preprint arXiv:2501.11430*, 2025.
- Livernoche, V., Jain, V., Hezaveh, Y., and Ravanbakhsh, S. On diffusion modeling for anomaly detection. *arXiv preprint arXiv:2305.18593*, 2023.
- Pang, G., Shen, C., Cao, L., and Hengel, A. V. D. Deep learning for anomaly detection: A review. *ACM computing surveys (CSUR)*, 54(2):1–38, 2021.
- Parzen, E. On estimation of a probability density function and mode. *The annals of mathematical statistics*, 33(3):1065–1076, 1962.
- Pimentel, M. A., Clifton, D. A., Clifton, L., and Tarassenko, L. A review of novelty detection. *Signal processing*, 99:215–249, 2014.
- Pruim, R. H., Mennes, M., van Rooij, D., Llera, A., Buitelaar, J. K., and Beckmann, C. F. Ica-aroma: A robust ica-based strategy for removing motion artifacts from fmri data. *Neuroimage*, 112:267–277, 2015.
- Qiu, C., Pfrommer, T., Kloft, M., Mandt, S., and Rudolph, M. Neural transformation learning for deep anomaly detection beyond images. In *International conference on machine learning*, pp. 8703–8714. PMLR, 2021.
- Ramaswamy, S., Rastogi, R., and Shim, K. Efficient algorithms for mining outliers from large data sets. In *Proceedings of the 2000 ACM SIGMOD international conference on Management of data*, pp. 427–438, 2000.
- Rozner, A., Battash, B., Li, H., Wolf, L., and Lindenbaum, O. Anomaly detection with variance stabilized density estimation. *arXiv preprint arXiv:2306.00582*, 2023.
- Ruff, L., Vandermeulen, R., Goernitz, N., Deecke, L., Siddiqui, S. A., Binder, A., Müller, E., and Kloft, M. Deep one-class classification. In *International conference on machine learning*, pp. 4393–4402. PMLR, 2018a.
- Ruff, L., Vandermeulen, R., Goernitz, N., Deecke, L., Siddiqui, S. A., Binder, A., Müller, E., and Kloft, M. Deep one-class classification. In *International conference on machine learning*, pp. 4393–4402. PMLR, 2018b.
- Ruff, L., Kauffmann, J. R., Vandermeulen, R. A., Montavon, G., Samek, W., Kloft, M., Dietterich, T. G., and Müller, K.-R. A unifying review of deep and shallow anomaly detection. *Proceedings of the IEEE*, 109(5):756–795, 2021.
- Salimi-Khorshidi, G., Douaud, G., Beckmann, C. F., Glasser, M. F., Griffanti, L., and Smith, S. M. Automatic denoising of functional mri data: combining independent component analysis and hierarchical fusion of classifiers. *Neuroimage*, 90:449–468, 2014.
- Schlegl, T., Seeböck, P., Waldstein, S. M., Langs, G., and Schmidt-Erfurth, U. f-anogan: Fast unsupervised anomaly detection with generative adversarial networks. *Medical image analysis*, 54:30–44, 2019.
- Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., and Williamson, R. C. Estimating the support of a high-dimensional distribution. *Neural computation*, 13(7):1443–1471, 2001.
- Shenkar, T. and Wolf, L. Anomaly detection for tabular data with internal contrastive learning. In *International conference on learning representations*, 2022.
- Xia, X., Pan, X., Li, N., He, X., Ma, L., Zhang, X., and Ding, N. Gan-based anomaly detection: A review. *Neurocomputing*, 493:497–535, 2022.

- Xiao, F., Tang, X., and Fan, J. Fairness-aware anomaly detection via fair projection. In Belgrave, D., Zhang, C., Lin, H., Pascanu, R., Koniusz, P., Ghassemi, M., and Chen, N. (eds.), *Advances in Neural Information Processing Systems*, volume 38, pp. 25148–25180. Curran Associates, Inc., 2025a.
- Xiao, F., Wang, Y., Joe Qin, S., and Fan, J. Semi-supervised anomaly detection using restricted distribution transformation. *IEEE Transactions on Neural Networks and Learning Systems*, 36(10):17966–17977, 2025b. doi: 10.1109/TNNLS.2025.3583320.
- Xu, H., Wang, Y., Wei, J., Jian, S., Li, Y., and Liu, N. Fascinating supervisory signals and where to find them: Deep anomaly detection with scale learning. In *International Conference on Machine Learning*, pp. 38655–38673. PMLR, 2023.
- Yin, J., Qiao, Y., Zhou, Z., Wang, X., and Yang, J. Mcm: Masked cell modeling for anomaly detection in tabular data. In *The Twelfth International Conference on Learning Representations*, 2024.
- Zhang, Y., Sun, Y., Cai, J., and Fan, J. Deep orthogonal hypersphere compression for anomaly detection. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=cJs4oE4m9Q>.
- Zheng, Y., Ng, I., and Zhang, K. On the identifiability of nonlinear ica: Sparsity and beyond. *Advances in neural information processing systems*, 35:16411–16422, 2022.
- Zong, B., Song, Q., Min, M. R., Cheng, W., Lumezanu, C., Cho, D., and Chen, H. Deep autoencoding gaussian mixture model for unsupervised anomaly detection. In *International conference on learning representations*, 2018.

Table 4. Organization of the Appendix.

Appendix Section	Content
Appendix A	Proofs of Theoretical Results
<i>A.1</i>	Proof of Theorem 3.1
<i>A.2</i>	Proof of Theorem 3.3
<i>A.3</i>	Connection between $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ and $p_{\bar{\mathcal{X}}}(\mathbf{x}_{\text{pure}})$
Appendix B	Time Complexity Analysis
–	Time Complexity Analysis of Density-Based Methods
Appendix C	Properties of the Regularizer
–	Theoretical Properties of the Regularizer $\mathcal{R}(\cdot)$
Appendix D	Implementation Details
–	Network Architectures, Training Protocol, and Optimization Settings
Appendix E	Hyperparameter Configurations
–	Hyperparameter Search Ranges of baseline methods
Appendix F	Algorithmic Details
–	Detailed Algorithm for NRDE Training and Testing, and Synthetic Data Generation
Appendix G	Dataset Statistics
–	Detailed Statistics of All 47 Datasets
Appendix H	Experiments on Synthetic and Real Datasets to Verify Assumptions and Motivations
<i>H.1</i>	Performance Results When the Variance Difference Assumption Is Violated
<i>H.2</i>	Performance Comparison Between NRDE and Optimal Baselines
<i>H.3</i>	Performance of NRDE Under Opposite Assumptions
<i>H.4</i>	A Practical Diagnostic to Verify the Variance Difference on Real Data
<i>H.5</i>	Performance of Baselines Trained and Evaluated on Pure Data
Appendix I	Additional Experimental Results
<i>I.1</i>	Jacobian Row Norm Visualization
<i>I.2</i>	Standard Anomaly Detection Results
<i>I.3</i>	Anomaly Detection with Anomaly Contamination Results
<i>I.4</i>	Anomaly Detection on Noisy Data Results
<i>I.5</i>	Outlier Detection Experiment Results
<i>I.6</i>	Ablation Studies and Hyperparameter (Architecture) Sensitivity Analysis
<i>I.7</i>	Training Time Comparison between NRDE and RealNVP
<i>I.8</i>	Dynamics of Training Loss and AUROC during the Training Process

A. Proofs of Theorems

A.1. Proof for Theorem 3.1

Lemma A.1. (*Behrmann et al., 2021*) Let $f_{\mathcal{W}_i}$ be a coupling flow, the Lipschitz constant of the forward $f_{\mathcal{W}_i}$ can be locally bounded for $\mathbf{x} \in [a, b]^d$ as:

$$\text{Lip}(f_{\mathcal{W}_i}) \leq \max(1, c_g) + M, \quad (22)$$

Where $\exp(h_{i1}(\mathbf{x})) \leq c_g$ and $M = \max(|a|, |b|) \cdot c_g \cdot \text{Lip}(h_{i1}) + \text{Lip}(h_{i2})$. Similarly, the Lipschitz constant of the reverse

$f_{\mathcal{W}_i}^{-1}$ can be locally bounded for $\mathbf{y}_i \in [a^*, b^*]^d$ as:

$$\text{Lip}(f_{\mathcal{W}_i}^{-1}) \leq \max(1, c_{\frac{1}{g}}) + M^*, \quad (23)$$

Where $M^* = \max(|a^*|, |b^*|) \cdot c_{(\frac{1}{g})} \cdot \text{Lip}(h_{i1}) \cdot c_t + c_{\frac{1}{g}} \cdot \text{Lip}(h_{i2})$

Proof. According to Lemma A.1, here we can assume that $\mathbf{x}_i, \mathbf{y}_i$ are both bounded since data is preprocessed and normalized, then we have the bi-Lipschitz constant of $f_{\mathcal{W}_i}$ are bounded as:

$$\begin{aligned} \tau_i^- \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\| &\leq \|f_{\mathcal{W}_i}(\mathbf{x}^{(i)}) - f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})\| \leq \tau_i^+ \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\|, \\ \frac{1}{\tau_i^+} \|f_{\mathcal{W}_i}(\mathbf{x}^{(i)}) - f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})\| &\leq \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\| \leq \frac{1}{\tau_i^-} \|f_{\mathcal{W}_i}(\mathbf{x}^{(i)}) - f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})\| \end{aligned} \quad (24)$$

The determinant of $\nabla_{\mathbf{x}^{(i)}} f_{\mathcal{W}_i}(\mathbf{x}^{(i)})$ can be calculated as:

$$\log |\det \nabla_{\mathbf{x}^{(i)}} f_{\mathcal{W}_i}(\mathbf{x}^{(i)})| = h_{i1}(\mathbf{x}_{\alpha}^{(i)}) \cdot \mathbf{1}, \quad (25)$$

Suppose $h_{i1}(\mathbf{x}) = \mathbf{W}_{i,L}(\phi(\cdots \phi(\mathbf{W}_{i,2} \phi(\mathbf{W}_{i,1} \mathbf{x})) \cdots))$ and $h_{i2}(\mathbf{x}) = \hat{\mathbf{W}}_{i,L}(\phi(\cdots \phi(\hat{\mathbf{W}}_{i,2} \phi(\hat{\mathbf{W}}_{i,1} \mathbf{x})) \cdots))$ are two neural networks comprising L layers and ϕ represents the activation function. Consider different $\mathbf{x}^{(i)}, \hat{\mathbf{x}}^{(i)}$, denote ρ the Lipschitz constant of ϕ , we have:

$$\begin{aligned} &|\log |\det \nabla_{\mathbf{x}^{(i)}} f_{\mathcal{W}_i}(\mathbf{x}^{(i)})| - \log |\det \nabla_{\hat{\mathbf{x}}^{(i)}} f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})|| \\ &= \|h_{i1}(\mathbf{x}_{\alpha}^{(i)}) - h_{i1}(\hat{\mathbf{x}}_{\alpha}^{(i)})\|_1 \\ &\leq \sqrt{d} \|h_{i1}(\mathbf{x}_{\alpha}^{(i)}) - h_{i1}(\hat{\mathbf{x}}_{\alpha}^{(i)})\| \\ &\leq \sqrt{d} \rho^{L-1} \prod_{l=1}^L \|\mathbf{W}_{i,l}\|_2 \|\mathbf{x}_{\alpha}^{(i)} - \hat{\mathbf{x}}_{\alpha}^{(i)}\| \\ &\leq \sqrt{d} \rho^{L-1} \prod_{l=1}^L \|\mathbf{W}_{i,l}\|_2 \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\| \\ &= \tau_{i\alpha} \sqrt{d} \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\| \\ &\leq \tau_{i\alpha} \sqrt{d} \prod_{j=1}^{i-1} \tau_j^+ \|\mathbf{x} - \hat{\mathbf{x}}\| \end{aligned} \quad (26)$$

Where $\tau_{i\alpha} = \rho^{L-1} \prod_{l=1}^L \|\mathbf{W}_{i,l}\|_2$ is the Lipschitz constant of h_{i1} . Then, we can conclude that $\log |\det(\nabla_x F_{\mathcal{W}}(\mathbf{x}))|$ has a Lipschitz constant:

$$\begin{aligned} &|\log |\det(\nabla_{\mathbf{x}} F_{\mathcal{W}}(\mathbf{x}))| - \log |\det(\nabla_{\hat{\mathbf{x}}} F_{\mathcal{W}}(\hat{\mathbf{x}}))|| \\ &= \left| \sum_{i=1}^T \left(\log |\det \nabla_{\mathbf{x}^{(i)}} f_{\mathcal{W}_i}(\mathbf{x}^{(i)})| - \log |\det \nabla_{\hat{\mathbf{x}}^{(i)}} f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})| \right) \right| \\ &\leq \sum_{i=1}^T |\log |\det \nabla_{\mathbf{x}^{(i)}} f_{\mathcal{W}_i}(\mathbf{x}^{(i)})| - \log |\det \nabla_{\hat{\mathbf{x}}^{(i)}} f_{\mathcal{W}_i}(\hat{\mathbf{x}}^{(i)})|| \\ &\leq \sum_{i=1}^T \tau_{i\alpha} \sqrt{d} \|\mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)}\| \\ &\leq \sum_{i=1}^T (\tau_{i\alpha} \sqrt{d} \prod_{j=1}^{i-1} \tau_j^+) \|\mathbf{x} - \hat{\mathbf{x}}\| \end{aligned} \quad (27)$$

Suppose the true data sources are $[\mathbf{s}_D; \mathbf{0}]$ or $[\mathbf{z}_D; \mathbf{0}]$, and the estimated data sources using $\hat{m}$ are $[\mathbf{s}_{\hat{D}}; \mathbf{0}]$ or $[\mathbf{z}_{\hat{D}}; \mathbf{0}]$. Then according to (24), we have the estimated error between $\mathbf{x}_{\text{pure}}$ and $\hat{\mathbf{x}}_{\text{pure}}$ is bounded by:

$$\|\mathbf{x}_{\text{pure}} - \hat{\mathbf{x}}_{\text{pure}}\| \leq \prod_{i=1}^T \frac{1}{\tau_i} \|[\mathbf{z}_D; \mathbf{0}] - [\mathbf{z}_{\hat{D}}; \mathbf{0}]\| \quad (28)$$

Thus, the estimated error between $\log p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}})$ and $\log p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ is bounded by:

$$\begin{aligned} & |\log p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}) - \log p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})| \\ &= |\log |\det(\nabla_{\mathbf{x}_{\text{pure}}} F_{\mathcal{W}}(\mathbf{x}_{\text{pure}}))| - \frac{\|[\mathbf{z}_D; \mathbf{0}]\|^2}{2} - \log |\det(\nabla_{\hat{\mathbf{x}}_{\text{pure}}} F_{\mathcal{W}}(\hat{\mathbf{x}}_{\text{pure}}))| + \frac{\|[\mathbf{z}_{\hat{D}}; \mathbf{0}]\|^2}{2}| \\ &\leq |\log |\det(\nabla_{\mathbf{x}_{\text{pure}}} F_{\mathcal{W}}(\mathbf{x}_{\text{pure}}))| - \log |\det(\nabla_{\hat{\mathbf{x}}_{\text{pure}}} F_{\mathcal{W}}(\hat{\mathbf{x}}_{\text{pure}}))|| + \frac{|\|[\mathbf{z}_{\hat{D}}; \mathbf{0}]\|^2 - \|[\mathbf{z}_D; \mathbf{0}]\|^2|}{2} \\ &\leq \sum_{i=1}^T (\tau_{i\alpha} \sqrt{d} \prod_{j=1}^{i-1} \tau_j^+) \|\hat{\mathbf{x}}_{\text{pure}} - \mathbf{x}_{\text{pure}}\| + \frac{|\|[\mathbf{z}_{\hat{D}}; \mathbf{0}]\|^2 - \|[\mathbf{z}_D; \mathbf{0}]\|^2|}{2} \\ &\leq \sum_{i=1}^T (\tau_{i\alpha} \sqrt{d} \prod_{j=1}^{i-1} \tau_j^+) \prod_{k=1}^T \frac{1}{\tau_k} \|[\mathbf{z}_D; \mathbf{0}] - [\mathbf{z}_{\hat{D}}; \mathbf{0}]\| + \frac{|\|[\mathbf{z}_{\hat{D}}; \mathbf{0}]\|^2 - \|[\mathbf{z}_D; \mathbf{0}]\|^2|}{2} \end{aligned} \quad (29)$$

This finishes the proof. $\square$

A.2. Proof for Theorem 3.3

Proof. By Assumption 3.2, we have that:

$$|p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) - p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^a)| \geq \varphi \|\mathbf{x}^a - \mathbf{x}^c\| \quad (30)$$

If $\|\mathbf{x}^a - \mathbf{x}^c\| > \frac{\varsigma_1 - \varsigma_2}{\varphi}$, then we have:

$$\begin{aligned} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^a) - p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) &> \varsigma_1 - \varsigma_2 \\ \varsigma_1 - p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) &> \varsigma_1 - \varsigma_2 \\ p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) &< \varsigma_2 \\ p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^c) &< \min_{\mathbf{x} \in \mathcal{D}} p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}) \end{aligned} \quad (31)$$

Now we have the density of $\hat{\mathbf{x}}_{\text{pure}}^c$ is even smaller than the smallest density of pure data from $\mathcal{D}$, thus it can be detected as an anomaly. This finishes the proof. $\square$

A.3. Derivation of $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$ and Its Connection with $p_{\mathcal{X}}(\mathbf{x}_{\text{pure}})$

The support of $\mathbf{x}_{\text{pure}}$ is an m -dimensional manifold M embedded in $\mathbb{R}$. Let $g(\mathbf{z}_D) := F^{-1}(\mathbf{z}_D, \mathbf{0})$. The induced Riemannian metric on the manifold is given by:

$$\mathbf{M}(\mathbf{z}_D) = J_g(\mathbf{z}_D)^{\top} J_g(\mathbf{z}_D) \quad (32)$$

where J_g denote the Jacobian of g , i.e., $\nabla_{\mathbf{z}_D} g(\mathbf{z}_D)$. The volume element on the manifold, relative to the parameter space $\mathbf{s}_D$ is

$$dV = \sqrt{\det [J_g(\mathbf{z}_D)^{\top} J_g(\mathbf{z}_D)]} d\mathbf{z}_D. \quad (33)$$

The probability in the latent space is:

$$\mathbb{P}(\mathbf{z}_D \in B) = \int_B p_{\mathcal{Z}}^D(\mathbf{z}_D) d\mathbf{z}_D. \quad (34)$$

This probability must equal the probability on the manifold M . For a measurable set $A \subset M$:

$$\mathbb{P}(\mathbf{x}_{\text{pure}} \in A) = \int_{g^{-1}(A)} p_{\mathcal{Z}}^D(\mathbf{z}_D) d\mathbf{z}_D \quad (35)$$

We change the variable of integration from $\mathbf{z}_D$ to $\mathbf{x}_{\text{pure}} \in M$ and use the manifold volume element to obtain

$$d\mathbf{z}_D = \frac{d\mathcal{H}^m(\mathbf{x}_{\text{pure}})}{\sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]}}, \quad (36)$$

where $d\mathcal{H}^m$ is the m -dimensional Hausdorff measure on M . Substituting this into the integral, we have

$$\mathbb{P}(\mathbf{x}_{\text{pure}} \in A) = \int_A \frac{p_{\mathcal{Z}}^D(\mathbf{z}_D)}{\sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]}} d\mathcal{H}^m(\mathbf{x}) \quad (37)$$

Therefore, the probability density function on the manifold M with respect to the Hausdorff measure is as follows

$$p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) = \frac{p_{\mathcal{Z}}^D(\mathbf{z}_D)}{\sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]}} = \frac{\mathcal{N}(\mathbf{z}_D; \mathbf{0}, \mathbf{I}_m)}{\sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]}} \quad (38)$$

where the second equality used the fact that $\mathbf{s}_D$ and $\mathbf{s}_N$ are independent.

On the other hand, we have

$$\begin{aligned} p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) &= p_{\mathcal{Z}}(\mathbf{z}_D, \mathbf{0}) |\det J_F(\mathbf{x}_{\text{pure}})| \\ &= c \mathcal{N}(\mathbf{z}_D; \mathbf{0}, \mathbf{I}_m) |\det J_F(\mathbf{x}_{\text{pure}})| \end{aligned} \quad (39)$$

where $c = \frac{1}{2\pi^{(d-m)/2}}$. It follows that

$$\begin{aligned} p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) &= p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) \times c |\det J_F(\mathbf{x}_{\text{pure}})| \sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]} \\ &= p_{\mathcal{X}}(\mathbf{x}_{\text{pure}}) \times c |\det J_{F^{-1}}(\mathbf{z}_D, \mathbf{0})|^{-1} \sqrt{\det [J_g(\mathbf{z}_D)^\top J_g(\mathbf{z}_D)]} \end{aligned} \quad (40)$$

B. Time Complexity of Density-based Methods

Suppose that $F_{\mathcal{W}}$ is a sequence of T flows defined in (2), and h_{i1}, h_{i2} are two MLPs of L layers parameterized by $\{\mathbf{W}_{i,j}\}_{j=1}^L, \{\hat{\mathbf{W}}_{i,j}\}_{j=1}^L$, where $\mathbf{W}_{i,j}, \hat{\mathbf{W}}_{i,j} \in \mathbb{R}^{d_{i,j} \times d_{i,j-1}}, j \in [L]$. Consider a batch of B data points, the time complexity of our method per iteration is $\mathcal{O}(B \sum_{i=1}^T (d_{i,L} \sum_{j=0}^{L-2} d_{i,j} d_{i,j+1}))$, and the space complexity is $\mathcal{O}(B \sum_{i=1}^T \sum_{j=0}^L d_{i,j} d_{i,j+1})$ which primarily arises from the computation of the Jacobian matrix. Here, we also compare the testing time complexity of a few representative density-based methods. We assume that DAGMM (Zong et al., 2018) contains K Gaussians and the encoder and decoder have $\hat{L}$ layers, with i th layer of encoder being $\mathbf{W}_{E,i} \in \mathbb{R}^{d_i \times d_{i-1}}$ and i th layer of decoder being $\mathbf{W}_{D,i} \in \mathbb{R}^{d_{L+1-i} \times d_{L-i}}$. For DPAD (Fu et al., 2024), we assume that the size of its neural network is the same as that of the encoder of DAGMM. Suppose we have one testing data, the time complexity of density-based methods is shown in Table 5.

We notice that traditional density-based methods, such as KNN and KDE, require comparing test data against the entire training set to generate anomaly scores. Consequently, these methods become computationally inefficient as dataset sizes grow, since the time complexity grows linearly with the number of training samples. DPAD encounters a similar issue due to its reliance on KNN, although it mitigates this by employing a neural network for dimensionality reduction. In contrast, methods like DAGMM, RealNVP, and our proposed NRDE primarily utilize neural network outputs for anomaly scoring, which do not depend on the training set.

Table 5. Time complexity comparison of density-based methods in testing stage.

	Testing Complexity
KNN	$\mathcal{O}(nd)$
KDE	$\mathcal{O}(nd)$
LOF	$\mathcal{O}(nd)$
DAGMM	$\mathcal{O}\left(\sum_{l=1}^{\tilde{L}} d_{l-1}d_l + d_{\tilde{L}}^3\right)$
DPAD	$\mathcal{O}\left(\sum_{l=1}^{\tilde{L}} d_{l-1}d_l + d_{\tilde{L}}n\right)$
RealNVP	$\mathcal{O}\left(\sum_{i=1}^T \sum_{j=0}^{L-1} d_{i,j}d_{i,j+1}\right)$
NRDE	$\mathcal{O}\left(\sum_{i=1}^T \sum_{j=0}^{L-1} d_{i,j}d_{i,j+1}\right)$

C. Properties of the regularizer $\mathcal{R}()$

Briefly speaking, our objective is to construct a Jacobian matrix in which the row norms exhibit a clear separation—some being significantly larger than others—so that we can distinguish between pure data sources and noise sources. Consider the derivative $\frac{\partial \mathcal{R}(Q)}{\partial Q_{i,j}} = \frac{\text{sign}(Q_{i,j})}{2\sqrt{\|Q_{i,:}\|}}$, where $\|Q_{i,:}\|$ is the ℓ_1 norm. In this formulation, rows with larger norms receive a smaller penalty from the regularizer, whereas rows with smaller norms receive a larger penalty. This naturally encourages row-wise sparsity and separation. Moreover, unlike the conventional $\ell_{2,1}$ norm, where smaller entries in the same row receive a smaller penalty, our regularizer $\mathcal{R}$ imposes the same penalty on all entries within a given row—avoiding the vanishing-penalty problem for small entries—thereby enhancing both separation and sparsity. Thus, in theory, the formulation is suitable for our task.

D. Implementation Details

Table 6. Network architecture of MLP used in normalizing flow (RealNVP).

Tabular
Dimension_input=2d
Dimension_firstlayer=b
Linear(2d, b), LeakyReLU()
Linear(b, 2d)

All experiments are conducted using the PyTorch framework on a system equipped with an NVIDIA RTX 4090 GPU and an Intel Core i9-12900K CPU. Each experiment is performed 5 times to obtain the mean value and standard deviation. To ensure a consistent network architecture for fair comparison, we employ two 2-layer multilayer perceptrons (MLPs) for RealNVP and NRDE, corresponding to a parameter setting of $T = 2$ in (1) where LeakyReLU is used as the activation function. Note that the outputs of h_{i1} and h_{i2} are actually the split output from the same MLP. The detailed network architecture is shown in Table 6, where $b = 2048$. Additionally, we use Adam as our optimizer and set the batch size to 512 for all experiments, while the number of training epochs is set to 100. Since the scale of the Jacobian norm in different datasets can be largely different, as shown in Figure 6, we use grid search to determine the best hyperparameter configuration on each dataset for NRDE: $lr \in \{0.001, 0.005, 0.01\}$ and $\lambda \in \{0.01, 0.1, 1\}$. To provide a fair comparison between our proposed NRDE and other baseline methods, the same hyperparameter-tuning strategy is also used for all other baseline methods.

E. Hyperparameter configurations of baseline methods

In our experiments, we perform a grid search on each dataset to identify the best hyperparameter configuration for each baseline method, and use these configurations in the following evaluations. The detailed hyperparameter configurations for different methods are summarized in Table 7. Note that ECOD has no hyperparameters.

Table 7. Hyperparameter combinations of traditional and deep AD methods used for grid search.

Method	Hyperparameter 1	Hyperparameter 2	Total Methods
KDE	gamma: [1, 5, 10]	kernel: ['linear', 'rbf', 'sigmoid']	9
LOF	n_neighbors: [5, 10, 20, 30, 50]	-	5
kNN	n_neighbors: [5, 10, 20, 30, 50]	-	5
OCSVM	nu (train error tol): [0.1, 0.3, 0.5, 0.7, 0.9]	kernel: ['linear', 'rbf', 'sigmoid']	15
IF	n_estimators: [50, 100, 150, 200]	max_features: [1, 2, 3]	12
AE	lr: [0.01, 0.005, 0.001]	epoch_num: [10, 50, 100]	9
DSVDD	lr: [0.001, 0.01, 0.005]	epoch_num: [100, 50, 10]	9
RealNVP	lr: [0.001, 0.01, 0.005]	-	3
NeutralAD	lr: [0.001, 0.01, 0.005]	rep_dim: [128, 64, 256]	9
ICL	lr: [0.001, 0.01, 0.005]	rep_dim: [128, 64, 256]	9
ECOD	-	-	1
SLAD	lr: [0.001, 0.01, 0.005]	n_ensemble: [20, 10, 30]	9
DPAD	gamma: [0.1, 0.5, 1]	lambda: [0.01, 0.1, 1]	9
DTE-C	lr: [0.0001, 0.0005, 0.001]	T: [400, 200, 600]	9
MCM	lr: [0.01, 0.005, 0.001]	lambda: [0.1, 1, 10]	9
TCCM	epoch_num: [100, 50, 200]	lr: [0.001, 0.005, 0.01]	9
ADERH	n_estimators: [256, 128, 512]	n: [10, 20, 30]	9

F. Algorithm Details

The detailed algorithm for our proposed NRDE is illustrated in Algorithm 1.

Algorithm 1 Training and Testing Procedure of NRDE

Training stage of NRDE and NRDE-M:

Input: $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, $\lambda > 0$, training epoch B

output: $F_{\mathcal{W}}, \hat{m}$

Initialize the parameters of flow network $\mathcal{W}$

for $b = 1, \dots, B$ **do**

for each batch $\hat{\mathcal{D}}$ **do**

 Obtain the flow output $\{F_{\mathcal{W}}(\mathbf{x})\}_{\mathbf{x} \in \hat{\mathcal{D}}}$

 Update parameters $\mathcal{W}$ using (16)

end for

end for

Obtain $\hat{m}$ using (19)

Testing stage of NRDE and NRDE-M:

Input: $\mathbf{x}^{\text{new}}, F_{\mathcal{W}}, \hat{m}$

output: $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$ or $p_{\bar{\mathcal{X}}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$

Obtain $\mathbf{z}^{\text{new}} = F_{\mathcal{W}}(\mathbf{x}^{\text{new}})$

Obtain pure data $\hat{\mathbf{x}}_{\text{pure}}^{\text{new}}$ using (21)

Obtain anomaly score: $p_{\mathcal{X}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$ in (9) or $p_{\bar{\mathcal{X}}}(\hat{\mathbf{x}}_{\text{pure}}^{\text{new}})$ in (11)

The synthetic data is generated using Algorithm 2. We primarily use Gaussian or uniform distributions to generate data, where the variances of the data sources are significantly larger than those of the noise sources. Specifically, $\mathcal{S}_D = \text{Unif}([-10, 50]^m)$, $\mathcal{S}_N = \text{Unif}([-40, -20]^{d-m})$, $\hat{\mathcal{S}}_D = \text{Unif}([10, 30]^m)$, $\hat{\mathcal{S}}_N = \text{Unif}([-10, 10]^{d-m})$. Both the training and testing normal data are generated using the normal data generative process. For noisy normal data, the data sources are distributed according to $\mathcal{S}_D$, while the noise sources are distributed according to variables distributed in $\mathcal{S}_N$ and are perturbed by $\hat{\mathcal{S}}_N$, introducing anomalies in the noise sources. In the case of anomalous data, the generative process closely resembles that of noisy normal data, where the noise sources are distributed in $\mathcal{S}_N$. However, the data sources are perturbed by variables distributed in $\mathcal{S}_D$ and perturbed by $\hat{\mathcal{S}}_D$, leading to anomalies in the data sources. It is noteworthy that normal

Algorithm 2 Data Generative Process

Normal data generation:

Input: data source distribution $\mathcal{S}_D$, noise source distribution $\mathcal{S}_N$, number of data sources m , data dimension d , mixing matrix $W \in \mathbb{R}^{d \times d}$

output: normal data $\mathbf{x}_i$

Obtain data sources $\mathbf{s}_D \in \mathbb{R}^m$ by $\mathbf{s}_D \sim \mathcal{S}_D$, obtain noise sources $\mathbf{s}_N \in \mathbb{R}^{d-m}$ by $\mathbf{s}_N \sim \mathcal{S}_N$

Generate data using $\mathbf{x} = W[\mathbf{s}_D; \mathbf{s}_N]^T$

Noisy normal data generation:

Input: number of data $\hat{n}$, data source distribution $\mathcal{S}_D$, noise source distribution $\mathcal{S}_N$, noise perturbation distribution $\hat{\mathcal{S}}_N$, number of data sources m , data dimension d , mixing matrix $W \in \mathbb{R}^{d \times d}$

output: noisy normal data $\tilde{\mathbf{x}}$

Obtain data sources $\mathbf{s}_D \in \mathbb{R}^m$ by $\mathbf{s}_D \sim \mathcal{S}_D$, obtain noise sources $\mathbf{s}_N \in \mathbb{R}^{d-m}$ by $\mathbf{s}_N \sim \mathcal{S}_N$, obtain noise perturbation $\hat{\mathbf{s}}_N \in \mathbb{R}^{d-m}$ by $\hat{\mathbf{s}}_N \sim \hat{\mathcal{S}}_N$

Generate data using $\tilde{\mathbf{x}} = W[\mathbf{s}_D; \mathbf{s}_N + \hat{\mathbf{s}}_N]^T$

Anomalous data generation:

Input: number of data $\hat{n}$, data source distribution $\mathcal{S}_D$, data perturbation distribution $\hat{\mathcal{S}}_D$, noise source distribution $\mathcal{S}_N$, number of data sources m , data dimension d , mixing matrix $W \in \mathbb{R}^{d \times d}$

output: anomalous data $\hat{\mathbf{x}}$

Obtain data sources $\mathbf{s}_D \in \mathbb{R}^m$ by $\mathbf{s}_D \sim \mathcal{S}_D$, obtain data perturbation $\hat{\mathbf{s}}_D \in \mathbb{R}^m$ by $\hat{\mathbf{s}}_D \sim \hat{\mathcal{S}}_D$, obtain noise sources $\mathbf{s}_N \in \mathbb{R}^{d-m}$ by $\mathbf{s}_N \sim \mathcal{S}_N$

Generate data using $\hat{\mathbf{x}} = W[\mathbf{s}_D + \hat{\mathbf{s}}_D; \mathbf{s}_N]^T$

data is different from pure normal data in that normal data also contains noise components, whose magnitude is much smaller than that of noisy normal data.

Data shown in Figure 1 is generated using $d = 100, m = 10$. Note that here W is an orthogonal matrix. Since pure normal data are difficult to obtain in real datasets, the normal samples shown in Figure 1 correspond to noisy normal data in (8) whose noise sources are normal. In contrast, the noisy normal samples in Figure 1 correspond to noisy normal data in (8) whose noise sources are anomalous. All models are trained using 10000 normal data, and tested using 1000 normal data, 1000 noisy normal data, and 1000 anomalous data.

G. Statistics of Datasets

In our experiments, we evaluate the performance of 14 methods on 47 widely used real-world datasets spanning multiple domains, including healthcare, audio, language processing, and finance, in a popular benchmark for anomaly detection (Han et al., 2022). The statistics of these datasets are shown in Table 8. These datasets encompass a wide range of sample sizes and feature dimensions, from small to large, providing comprehensive metrics and evaluations for the methods.

H. Experiments on Synthetic and Real Datasets to Verify Assumptions and Motivations

It is noteworthy that verifying the variance-related assumption on real datasets is inherently challenging because the underlying generative process G is not accessible during training. A practical way to indirectly verify our assumptions is to compare NRDE’s performance with that of other baselines on the 47 datasets, particularly with RealNVP (normalizing flow), which estimates the density of data with noise. The fact that NRDE outperforms RealNVP on most datasets indicates that our motivation and underlying assumptions hold for these real datasets. In this section, we include several experiments on both synthetic and real datasets to further verify our assumptions and motivations.

H.1. Performance Results When the Variance Difference Assumption Is Violated

Here, we analyze the performance of NRDE on synthetic datasets where the variance difference is not satisfied. Suppose that the variance of pure data sources is σ_d^2 and that the variance of noise sources is σ_n^2 . We report the performance results on synthetic datasets with different $\frac{\sigma_d^2}{\sigma_n^2}$ in Table 9. As $\frac{\sigma_d^2}{\sigma_n^2}$ decreases, the performance decline of NRDE supports our assumptions and motivations.

Table 8. Statistics of 47 real-world datasets in ADBench.

Data	# Samples	# Features	# Anomaly	% Anomaly	Category
ALOI	49534	27	1508	3.04	Image
annthyroid	7200	6	534	7.42	Healthcare
backdoor	95329	196	2329	2.44	Network
breastw	683	9	239	34.99	Healthcare
campaign	41188	62	4640	11.27	Finance
cardio	1831	21	176	9.61	Healthcare
Cardiotocography	2114	21	466	22.04	Healthcare
celeba	202599	39	4547	2.24	Image
census	299285	500	18568	6.20	Sociology
cover	286048	10	2747	0.96	Botany
donors	619326	10	36710	5.93	Sociology
fault	1941	27	673	34.67	Physical
fraud	284807	29	492	0.17	Finance
glass	214	7	9	4.21	Forensic
Hepatitis	80	19	13	16.25	Healthcare
http	567498	3	2211	0.39	Web
InternetAds	1966	1555	368	18.72	Image
Ionosphere	351	32	126	35.90	Oryctognosy
landsat	6435	36	1333	20.71	Astronautics
letter	1600	32	100	6.25	Image
Lymphography	148	18	6	4.05	Healthcare
magic.gamma	19020	10	6688	35.16	Physical
mammography	11183	6	260	2.32	Healthcare
mnist	7603	100	700	9.21	Image
musk	3062	166	97	3.17	Chemistry
optdigits	5216	64	150	2.88	Image
PageBlocks	5393	10	510	9.46	Document
pendigits	6870	16	156	2.27	Image
Pima	768	8	268	34.90	Healthcare
satellite	6435	36	2036	31.64	Astronautics
satimage-2	5803	36	71	1.22	Astronautics
shuttle	49097	9	3511	7.15	Astronautics
skin	245057	3	50859	20.75	Image
smtp	95156	3	30	0.03	Web
SpamBase	4207	57	1679	39.91	Document
speech	3686	400	61	1.65	Linguistics
Stamps	340	9	31	9.12	Document
thyroid	3772	6	93	2.47	Healthcare
vertebral	240	6	30	12.50	Biology
vowels	1456	12	50	3.43	Linguistics
Waveform	3443	21	100	2.90	Physics
WBC	223	9	10	4.48	Healthcare
WDBC	367	30	10	2.72	Healthcare
Wilt	4819	5	257	5.33	Botany
wine	129	13	10	7.75	Chemistry
WPBC	198	33	47	23.74	Healthcare
yeast	1484	8	507	34.16	Biology

Table 9. AUROC performance of NRDE on synthetic datasets with different $\frac{\sigma_d^2}{\sigma_n^2}$ ratios.

σ_d^2/σ_n^2	9	6	4	2	1	0.5
NRDE	87.3	80.9	77.8	75.4	71.0	65.8

H.2. Performance Comparison Between NRDE and Optimal Baselines

On synthetic datasets where the number of data sources m is known, we compare the performance of NRDE with KDE-O, DSVDD-O, and KNN-O, which are evaluated on datasets generated by data sources only and with NRDE-O, where only the m sources with the smallest Jacobian row norms are used to obtain the pure data $\mathbf{x}_{\text{pure}}$. The results are shown in Table 10. Since NRDE is an approximation of NRDE-O, the fact that its performance is close to, but not as good as, NRDE-O and other ideal baselines supports our claim and motivation.

Table 10. AUROC (%) performance of NRDE and other baselines on the synthetic dataset.

Method	NRDE	NRDE-O	KDE-O	KNN-O	DSVDD-O
AUROC	80.9	86.0	87.2	90.2	87.5

H.3. Performance of NRDE Under Opposite Assumptions

If we make an opposite assumption that the variances of data sources should be smaller, then we will select the sources with the largest Jacobian norms as our data sources. This method is denoted as NRDE-CON. The performance of NRDE-CON and NRDE on several datasets is shown in Table 11, where the results support the assumption in our paper.

Table 11. AUROC (%) performance of NRDE-CON and NRDE on real datasets.

	WPBC	Cardiotocography	vertebral	Waveform	Speech
NRDE-CON	59.5	76.5	70.1	72.1	51.7
NRDE	64.4	87.1	82.7	86.7	58.6

H.4. A Practical Diagnostic to Verify the Variance Difference on Real Data

In this subsection, we provide a simple way to empirically verify the variance difference underlying our assumption as follows. We first train an *unregularized* normalizing flow $F_{\mathcal{W}}$ on the real dataset $\mathcal{D}$ and compute the average absolute Jacobian

$$J = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{D}} |\nabla F_{\mathcal{W}}(\mathbf{x})|.$$

We then take the row norms $\{\|J_i\|\}_{i=1}^d$ and sort them in ascending order, denoted by $\{\|J_i\|\}_{i=1}^d$.

One simple way to measure the gap is to compute the gap $\Delta_i = \frac{J_{(i+1)} - J_i}{J_{(i+1)}}$, since the number of data sources m cannot be obtained, we use the expectation of this gap $\mathbb{E}(\Delta) = \sum_{i=1}^{d-1} \Delta_i / (d-1)$ to measure the variance difference where a large value of $\mathbb{E}(\Delta)$ indicates that the variance difference is pronounced and that NRDE is particularly appropriate in such cases.

We conduct experiments to measure $\mathbb{E}(\Delta)$ on datasets where NRDE shows performance improvements and on datasets where it exhibits performance drops compared to other density-based methods. The results in Table 12 show that datasets with performance improvements tend to have larger values of $\mathbb{E}(\Delta)$, illustrating that $\mathbb{E}(\Delta)$ is an effective diagnostic for quantifying variance differences.

H.5. Performance of Baselines Trained and Evaluated on Pure Data

To further validate our motivation of isolating noise and the variance assumption in (6), we use the proposed NRDE to generate pure data for both the training and test sets via (21). We then train several simple baseline methods using pure training data and evaluate them on the pure test data, and compare their performance with that obtained using the original data.

Table 12. Average gap values on datasets showing significant performance improvement or drop compared to other density-based methods.

Datasets (Improvement)	$\mathbb{E}(\Delta)$
annthyroid	0.23
smtpt	0.58
vertebral	0.16
Pima	0.35
Cardiotocography	0.14
Datasets (Drop)	$\mathbb{E}(\Delta)$
Ionosphere	0.04
landsat	0.03
letter	0.03
optdigits	0.07
pendigits	0.06

As shown in Table 13, we compare the experimental results of baseline methods obtained using their default hyperparameter configurations on pure data with those achieved using best-performing hyperparameter configurations on the original data. The results show that substantially better performance is consistently achieved on the pure data, even without extensive hyperparameter tuning. Notably, this improvement persists even on datasets where NRDE itself is less effective (e.g., *landsat* and *magic.gamma*). These observations suggest that in the pure data space, separability between normal and anomalous samples is substantially enhanced, thereby providing empirical support for our variance-related assumption and the effectiveness of the proposed noise isolation strategy.

Table 13. AUROC and AUPRC performance of different methods when trained and evaluated on pure data generated by NRDE. The numbers in parentheses denote the performance improvement or degradation compared to training and testing on the original data.

AUROC	KDE	KNN	ECOD	NRDE
annthyroid	95.0 (+3.6)	96.2 (+2.1)	91.4 (+12.7)	97.3
Cardiotocography	86.3 (+4.9)	83.9 (+11.4)	83.1 (+4.6)	87.1
landsat	73.7 (+1.0)	75.7 (+0.4)	51.2 (+14.6)	70.4
magic.gamma	79.1 (+3.4)	86.6 (+3.7)	70.3 (+6.9)	82.5
mammography	90.3 (+0.5)	88.5 (+0.7)	92.6 (+1.9)	91.7
Stamps	96.1 (+1.0)	94.6 (+2.3)	92.3 (+5.6)	96.9
vertebral	68.5 (+25.0)	68.4 (+22.7)	76.4 (+34.6)	82.8
Waveform	83.4 (+6.5)	82.9 (+6.5)	63.4 (+3.4)	86.7
Wilt	79.7 (+42.6)	72.7 (+7.5)	70.8 (+30.5)	83.6
WPBC	63.3 (+10.8)	67.1 (+15.4)	54.9 (+7.9)	64.4
AUPRC	KDE	KNN	ECOD	NRDE
annthyroid	75.6 (+9.4)	79.3 (+7.3)	60.1 (+19.3)	85.7
Cardiotocography	78.1 (+6.6)	80.7 (+16.8)	75.7 (+10.0)	77.5
landsat	55.0 (+0.2)	58.6 (+0.6)	33.9 (+6.1)	58.2
magic.gamma	83.5 (+3.1)	88.3 (+2.9)	74.3 (+6.7)	81.9
mammography	50.0 (+8.4)	45.5 (+3.6)	57.9 (+3.9)	59.4
Stamps	84.0 (+20.3)	87.5 (+33.3)	57.1 (+11.9)	88.3
vertebral	32.3 (+12.6)	43.5 (+23.2)	59.9 (+40.4)	54.9
Waveform	33.4 (+5.8)	32.0 (+6.6)	8.4 (+0.8)	14.8
Wilt	42.2 (+34.8)	35.1 (+22.2)	20.1 (+12.0)	25.8
WPBC	60.2 (+21.8)	55.5 (+17.3)	48.0 (+12.6)	52.7

I. More Experimental Results

I.1. Jacobian Row Norm Visualization

In this subsection, we present visualizations of the Jacobian row norms on both synthetic and real-world datasets. Figures 5 and 6 illustrate these results. Notably, even without regularization, the row norms already exhibit clear separability; this distinction becomes even more pronounced when the regularizer is applied.

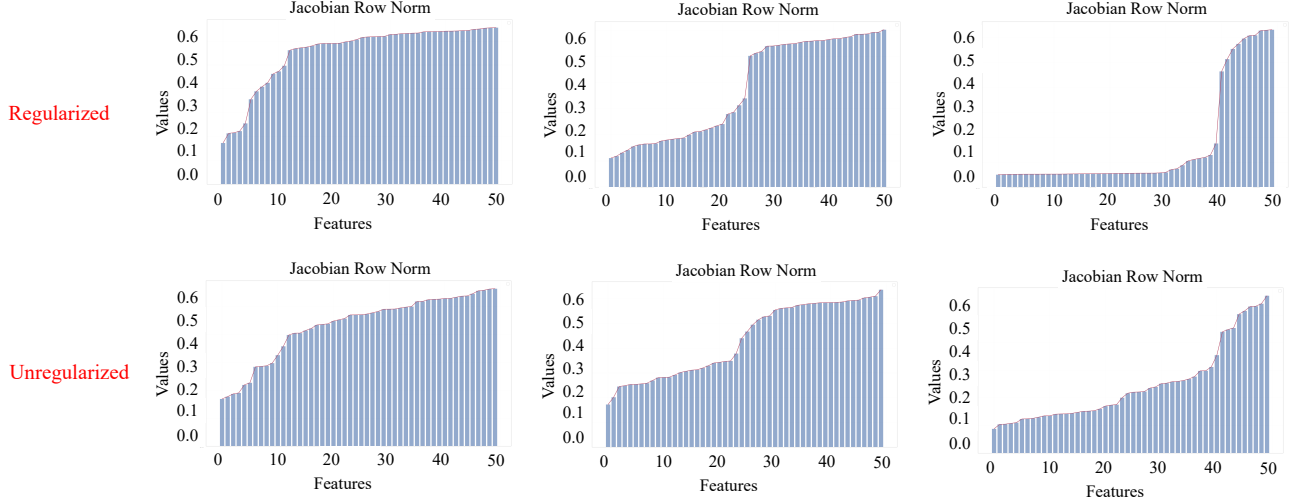


Figure 5. Visualization of the Jacobian matrix row norms on several synthetic datasets with 50 total sources. From left to right, the number of noise dimensions is 40, 25, and 10, respectively. The top row corresponds to the regularized case, whereas the bottom row illustrates the unregularized case.

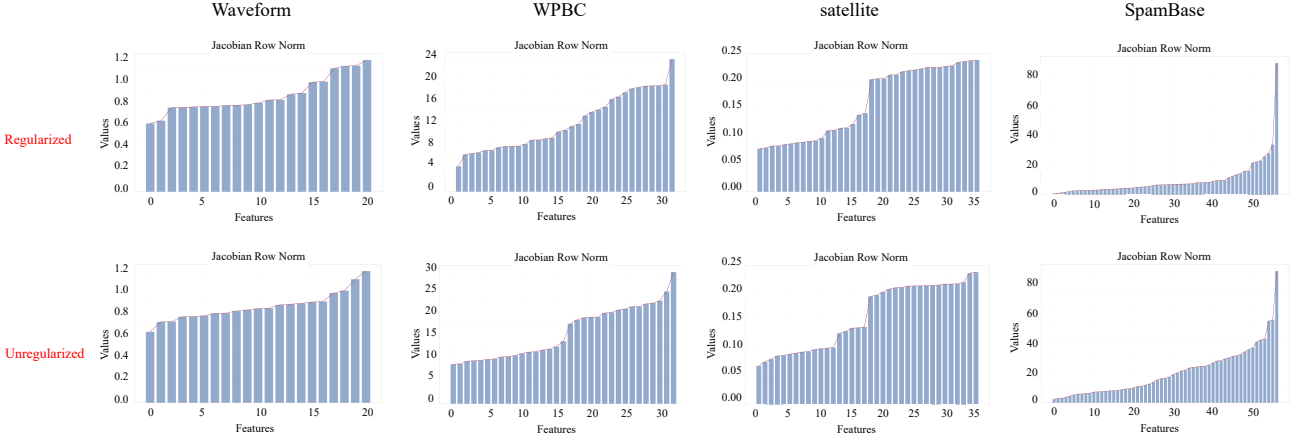


Figure 6. Visualization of the Jacobian matrix row norms on several real datasets. The top row corresponds to the regularized case, whereas the bottom row illustrates the unregularized case.

I.2. Standard Anomaly Detection

In this subsection, we provide the detailed experimental results of AD on 47 real-world datasets. Table 14 shows the detailed AUROC and AUPRC results on 47 datasets. Additionally, box plots of all 19 methods on all the 47 benchmark datasets are provided in Figure 7. Meanwhile, we also provide the experimental results under default hyperparameter setting and under AutoUAD(Dai & Fan, 2025)-tuning setting in Table 15, Table 16 and Table 17, Table 18 respectively.

Noise-Robust Density Estimation for Tabular Data Anomaly Detection

Table 14. Average AUROC (%) and AUPRC (%) with the standard deviation of each method on 47 tabular datasets of ADBench. The best and second-best results are highlighted in **red** and **orange**, respectively. Hyperparameter of each methods on each dataset is tuned using grid search.

AUROC	KDE (1962)	INN (2000)	LOF (2000)	OC-SVM (2001)	IF (2008)	AE (2006)	DSVDD (2018)	RealNVP (2016)	NAD (2021)	ECOD (2022)	ICL (2022)	SLAD (2023)	DPAD (2024)	MCM (2024)	DTE-C (2024)	TCCM (2025)	ADERH (2025)	NRDE	NRDEM
ALOI	56.3 ± 0.0	66.7 ± 0.0	78.6 ± 0.0	55.4 ± 0.0	53.1 ± 0.0	56.7 ± 0.3	55.4 ± 0.1	56.2 ± 0.5	61.3 ± 0.1	53.0 ± 0.0	58.4 ± 1.3	54.7 ± 0.1	56.0 ± 0.5	59.7 ± 0.0	54.5 ± 0.1	56.2 ± 0.5	55.3 ± 0.0	57.2 ± 0.4	53.4 ± 0.5
amthyroid	91.4 ± 0.0	94.1 ± 0.0	95.4 ± 0.0	91.7 ± 0.0	93.1 ± 0.0	93.5 ± 2.3	92.3 ± 1.6	96.1 ± 0.5	87.7 ± 3.2	78.7 ± 0.0	83.5 ± 0.5	87.3 ± 0.4	82.8 ± 4.8	98.2 ± 0.0	97.9 ± 0.1	94.1 ± 0.8	91.7 ± 0.4	97.3 ± 0.5	98.4 ± 0.3
backdoor	90.4 ± 0.0	94.5 ± 0.0	95.8 ± 0.0	66.8 ± 0.0	78.1 ± 0.0	94.8 ± 0.0	95.1 ± 0.7	91.8 ± 0.4	95.2 ± 0.1	84.6 ± 0.0	96.5 ± 0.2	74.6 ± 0.1	91.9 ± 1.5	97.3 ± 0.0	91.7 ± 0.0	96.1 ± 0.4	76.2 ± 0.0	92.1 ± 0.2	91.7 ± 2.7
breastw	99.1 ± 0.0	99.2 ± 0.0	98.4 ± 0.0	99.0 ± 0.0	99.6 ± 0.0	99.2 ± 0.2	99.2 ± 0.1	98.0 ± 0.0	80.1 ± 2.8	99.3 ± 0.0	92.6 ± 1.4	99.3 ± 0.1	98.9 ± 0.3	99.4 ± 0.0	99.0 ± 0.8	99.1 ± 0.2	98.8 ± 0.1	98.9 ± 0.3	98.9 ± 0.2
campuign	77.3 ± 0.0	78.5 ± 0.0	70.0 ± 0.0	77.2 ± 0.0	82.5 ± 0.0	82.6 ± 0.0	77.5 ± 0.4	79.6 ± 0.3	73.7 ± 0.1	78.3 ± 0.0	80.7 ± 0.8	76.0 ± 0.1	55.0 ± 5.0	86.4 ± 0.0	79.7 ± 1.1	79.5 ± 1.3	76.1 ± 0.2	84.9 ± 1.9	79.6 ± 5.4
cardio	96.6 ± 0.0	94.2 ± 0.0	95.3 ± 0.0	97.5 ± 0.0	96.2 ± 0.0	92.4 ± 3.2	96.4 ± 0.1	94.1 ± 0.4	81.5 ± 2.3	93.4 ± 0.0	91.2 ± 0.2	90.3 ± 0.0	87.9 ± 5.1	95.5 ± 0.0	93.7 ± 0.5	96.0 ± 0.2	94.6 ± 0.1	95.3 ± 1.1	96.1 ± 0.7
Cardiolography	81.4 ± 0.0	72.5 ± 0.0	78.2 ± 0.0	83.9 ± 0.0	88.0 ± 0.0	80.7 ± 2.8	83.7 ± 0.5	77.9 ± 1.8	61.3 ± 5.5	78.5 ± 0.0	62.9 ± 1.7	65.0 ± 0.7	72.6 ± 1.3	81.3 ± 0.0	72.0 ± 2.4	77.5 ± 1.5	66.4 ± 0.3	87.1 ± 2.2	83.5 ± 1.4
celeba	67.3 ± 0.0	74.3 ± 0.0	50.0 ± 0.0	79.0 ± 0.0	79.5 ± 0.0	75.3 ± 1.2	76.5 ± 2.0	79.4 ± 0.6	55.8 ± 6.1	75.7 ± 0.0	69.5 ± 0.0	63.9 ± 1.0	50.0 ± 0.0	75.0 ± 0.0	71.9 ± 0.1	73.4 ± 5.4	79.4 ± 0.1	83.6 ± 2.1	80.3 ± 0.5
census	72.7 ± 0.0	71.9 ± 0.0	60.5 ± 0.0	70.2 ± 0.0	51.1 ± 0.0	72.5 ± 0.2	52.4 ± 1.9	72.8 ± 0.3	61.1 ± 2.2	65.9 ± 0.0	66.8 ± 0.0	66.9 ± 1.2	50.0 ± 0.0	68.1 ± 2.0	69.9 ± 0.3	67.9 ± 0.4	69.5 ± 0.4	78.4 ± 1.1	69.2 ± 2.6
cover	95.5 ± 0.0	98.5 ± 0.0	99.3 ± 0.0	98.8 ± 0.0	92.1 ± 0.0	98.4 ± 0.0	49.8 ± 0.5	83.8 ± 0.1	82.2 ± 7.6	92.0 ± 1.1	67.4 ± 3.4	73.1 ± 5.8	50.1 ± 0.0	95.1 ± 0.0	98.1 ± 0.1	98.7 ± 1.0	93.1 ± 0.1	99.9 ± 0.0	98.6 ± 0.1
donors	97.4 ± 0.0	99.9 ± 0.0	99.0 ± 0.0	93.5 ± 0.0	96.0 ± 0.0	92.0 ± 1.2	97.4 ± 0.4	96.0 ± 0.4	99.9 ± 0.0	88.8 ± 0.0	94.0 ± 0.0	87.7 ± 8.4	67.7 ± 10.9	100.0 ± 0.0	98.8 ± 0.1	93.8 ± 0.3	99.6 ± 0.2	98.2 ± 8.2	98.2 ± 8.2
fault	81.2 ± 0.0	79.2 ± 0.0	71.0 ± 0.0	61.9 ± 0.0	65.3 ± 0.0	76.1 ± 0.0	70.2 ± 2.0	50.8 ± 0.1	74.2 ± 0.9	47.4 ± 0.0	79.2 ± 0.6	77.2 ± 0.0	74.2 ± 0.9	77.1 ± 0.0	72.7 ± 1.1	73.8 ± 0.4	72.3 ± 0.0	71.4 ± 2.0	69.0 ± 1.8
fraud	95.8 ± 0.0	96.0 ± 0.0	94.8 ± 0.0	95.6 ± 0.0	95.1 ± 0.0	95.6 ± 0.1	54.3 ± 1.3	54.4 ± 0.1	93.6 ± 0.6	94.9 ± 0.0	93.3 ± 0.4	94.6 ± 0.3	54.8 ± 4.8	94.4 ± 0.0	94.6 ± 0.4	93.9 ± 0.1	95.2 ± 0.1	95.7 ± 0.6	92.6 ± 0.1
glass	86.8 ± 0.0	89.3 ± 0.0	84.0 ± 0.0	79.3 ± 0.0	84.1 ± 0.0	89.5 ± 0.5	83.0 ± 2.3	81.5 ± 1.3	97.8 ± 0.9	69.8 ± 0.0	93.6 ± 1.0	78.7 ± 2.4	84.1 ± 4.4	89.3 ± 0.0	86.7 ± 2.2	88.9 ± 0.3	90.1 ± 0.1	89.2 ± 0.4	91.3 ± 2.1
Hepatitis	84.8 ± 0.0	85.1 ± 0.0	86.0 ± 0.0	85.1 ± 0.0	87.6 ± 0.0	83.9 ± 0.7	83.3 ± 1.8	59.4 ± 0.8	72.6 ± 2.5	71.5 ± 0.0	70.2 ± 2.4	80.3 ± 5.4	72.4 ± 3.6	72.4 ± 0.0	84.8 ± 0.9	85.9 ± 1.0	84.4 ± 0.2	86.7 ± 1.5	85.1 ± 2.4
http	100.0 ± 0.0	100.0 ± 0.0	99.5 ± 0.0	100.0 ± 0.0	99.4 ± 0.0	99.9 ± 0.0	99.8 ± 0.0	99.6 ± 0.0	96.7 ± 0.2	97.8 ± 0.0	74.9 ± 24.8	99.9 ± 0.0	75.9 ± 23.9	100.0 ± 0.0	99.6 ± 0.1	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
InternetAds	97.4 ± 0.0	97.4 ± 0.0	96.3 ± 0.0	96.9 ± 0.0	91.9 ± 0.0	97.8 ± 0.3	96.7 ± 0.1	93.8 ± 0.2	96.6 ± 0.1	75.3 ± 0.0	96.7 ± 1.1	96.5 ± 0.1	89.3 ± 0.0	99.7 ± 0.0	96.7 ± 0.6	96.8 ± 0.1	96.0 ± 0.1	95.3 ± 0.3	87.6 ± 8.6
Ionosphere	72.7 ± 0.0	75.3 ± 0.0	77.8 ± 0.0	47.0 ± 0.0	68.1 ± 0.0	63.1 ± 1.7	68.4 ± 1.1	59.7 ± 2.0	72.6 ± 0.3	36.6 ± 0.0	79.3 ± 0.5	72.6 ± 0.2	70.8 ± 0.5	64.7 ± 0.0	61.0 ± 0.0	61.2 ± 1.2	60.2 ± 0.4	70.4 ± 5.5	74.2 ± 2.5
landsat	91.8 ± 0.0	87.2 ± 0.0	92.2 ± 0.0	61.0 ± 0.0	65.4 ± 0.0	88.5 ± 0.2	67.2 ± 3.7	83.1 ± 0.6	93.2 ± 1.4	56.0 ± 0.0	87.7 ± 0.2	90.2 ± 0.1	77.3 ± 1.2	89.5 ± 0.0	89.9 ± 0.6	93.9 ± 0.2	76.7 ± 0.4	81.6 ± 0.4	68.2 ± 1.4
letter	98.6 ± 0.0	98.6 ± 0.0	98.6 ± 0.0	98.4 ± 0.0	98.1 ± 0.0	98.7 ± 0.1	98.7 ± 0.1	94.3 ± 0.5	92.4 ± 2.0	98.5 ± 0.0	98.7 ± 0.1	98.6 ± 0.0	98.8 ± 0.0	98.6 ± 0.0	98.6 ± 0.0	98.6 ± 0.0	98.7 ± 0.0	98.9 ± 0.7	98.9 ± 0.7
Lymphography	75.7 ± 0.0	82.9 ± 0.0	84.0 ± 0.0	73.8 ± 0.0	77.2 ± 0.0	84.6 ± 0.2	80.7 ± 0.1	79.6 ± 0.5	78.0 ± 2.8	63.4 ± 0.0	78.5 ± 0.1	70.6 ± 0.8	76.6 ± 1.4	83.1 ± 0.0	88.1 ± 0.7	83.5 ± 0.5	70.2 ± 0.1	82.5 ± 0.1	80.3 ± 1.1
magicgamma	98.5 ± 0.0	97.8 ± 0.0	85.0 ± 0.0	88.9 ± 0.0	86.9 ± 0.0	91.5 ± 0.2	89.6 ± 0.2	89.5 ± 0.4	76.1 ± 0.3	97.4 ± 0.0	76.7 ± 1.8	76.0 ± 0.4	78.2 ± 8.9	91.6 ± 0.3	96.3 ± 0.5	86.3 ± 0.5	87.5 ± 0.0	91.7 ± 0.1	92.4 ± 0.7
mnist	95.1 ± 0.0	94.3 ± 0.0	94.1 ± 0.0	91.0 ± 0.0	73.5 ± 0.0	94.4 ± 0.5	91.8 ± 0.6	92.4 ± 0.4	88.2 ± 0.1	74.7 ± 0.0	86.2 ± 0.2	80.9 ± 0.2	82.2 ± 1.6	96.7 ± 0.0	90.7 ± 0.4	93.8 ± 0.1	91.1 ± 0.2	93.2 ± 0.4	84.8 ± 2.0
mnist	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	99.4 ± 0.3	100.0 ± 0.0	95.8 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	83.0 ± 28.6
optdigits	97.4 ± 0.0	94.9 ± 0.0	98.7 ± 0.0	62.6 ± 0.0	86.8 ± 0.0	93.7 ± 0.0	78.4 ± 1.3	94.2 ± 0.7	98.8 ± 0.6	60.4 ± 0.0	96.3 ± 0.5	93.7 ± 2.4	59.9 ± 10.0	96.1 ± 0.0	89.0 ± 0.8	92.1 ± 2.2	63.3 ± 0.2	97.7 ± 0.8	90.5 ± 5.0
PageBlocks	95.0 ± 0.0	96.2 ± 0.0	97.5 ± 0.0	96.1 ± 0.0	94.1 ± 0.0	97.0 ± 0.5	96.4 ± 0.1	92.0 ± 0.9	92.9 ± 0.1	91.4 ± 0.0	91.7 ± 0.9	90.0 ± 1.6	90.8 ± 4.1	79.8 ± 0.0	96.7 ± 0.3	95.1 ± 1.4	94.4 ± 0.0	93.5 ± 0.5	97.2 ± 0.5
pendigits	99.8 ± 0.0	99.8 ± 0.0	99.7 ± 0.0	97.3 ± 0.0	97.5 ± 0.0	99.5 ± 0.0	96.9 ± 1.9	98.5 ± 0.6	97.0 ± 1.1	92.7 ± 0.0	97.6 ± 1.2	94.4 ± 1.9	97.6 ± 1.0	98.2 ± 0.0	98.3 ± 0.3	98.1 ± 0.3	97.6 ± 0.2	99.3 ± 0.2	99.5 ± 0.2
Pima	78.5 ± 0.0	79.0 ± 0.0	77.3 ± 0.0	76.5 ± 0.0	76.9 ± 0.0	76.2 ± 0.8	75.7 ± 0.4	79.3 ± 0.6	63.5 ± 1.2	61.5 ± 0.0	70.2 ± 0.3	64.7 ± 1.2	75.0 ± 0.7	70.3 ± 0.0	73.1 ± 1.5	74.9 ± 3.5	77.2 ± 0.2	80.0 ± 0.3	81.1 ± 0.8
satellite	87.3 ± 0.0	87.3 ± 0.0	86.8 ± 0.0	75.4 ± 0.0	83.2 ± 0.0	82.0 ± 0.2	87.8 ± 0.8	83.6 ± 1.3	79.0 ± 0.2	58.3 ± 0.0	90.1 ± 0.2	87.9 ± 0.3	86.5 ± 0.5	81.7 ± 0.0	86.6 ± 0.6	83.0 ± 0.6	81.1 ± 0.2	85.8 ± 1.2	83.6 ± 1.1
satimage-2	99.9 ± 0.0	99.9 ± 0.0	99.6 ± 0.0	99.7 ± 0.0	99.4 ± 0.0	99.9 ± 0.0	99.9 ± 0.0	98.9 ± 0.0	88.9 ± 0.1	96.6 ± 0.1	99.4 ± 0.1	99.8 ± 0.1	99.2 ± 0.3	99.3 ± 0.0	99.0 ± 0.2	99.4 ± 0.2	99.9 ± 0.0	99.8 ± 0.0	99.4 ± 0.2
shuttle	99.8 ± 0.0	99.9 ± 0.0	100.0 ± 0.0	99.9 ± 0.0	99.7 ± 0.0	99.9 ± 0.0	99.8 ± 0.0	99.8 ± 0.0	100.0 ± 0.0	99.3 ± 0.0	99.8 ± 0.1	99.9 ± 0.0	99.1 ± 0.2	99.9 ± 0.0	99.7 ± 0.0	100.0 ± 0.0	99.8 ± 0.0	99.9 ± 0.0	99.9 ± 0.0
skin	89.1 ± 0.0	99.8 ± 0.0	94.4 ± 0.0	90.3 ± 0.0	90.0 ± 0.0	84.2 ± 0.2	70.4 ± 0.4	90.0 ± 0.1	94.0 ± 2.6	48.8 ± 0.0	90.0 ± 0.0	92.1 ± 1.8	87.7 ± 11.8	92.1 ± 0.0	92.3 ± 0.0	86.0 ± 2.8	91.2 ± 0.1	94.9 ± 0.6	96.2 ± 0.4
smtp	88.2 ± 0.0	93.6 ± 0.0	94.9 ± 0.0	87.2 ± 0.0	93.5 ± 0.0	94.6 ± 0.0	91.9 ± 0.1	93.4 ± 0.1	95.3 ± 0.2	87.9 ± 0.0	65.0 ± 15.0	96.2 ± 1.2	84.1 ± 0.8	88.4 ± 0.0	95.5 ± 0.0	91.8 ± 4.6	93.0 ± 0.3	90.6 ± 1.1	90.2 ± 0.1
SpmBase	86.0 ± 0.0	84.5 ± 0.0	82.4 ± 0.0	87.7 ± 0.0	79.1 ± 0.0	82.6 ± 0.4	83.3 ± 0.1	80.1 ± 0.3	80.9 ± 0.5	66.0 ± 0.0	82.3 ± 0.2	84.8 ± 0.8	71.2 ± 3.1	73.6 ± 0.0	85.4 ± 0.7	86.4 ± 0.0	80.8 ± 0.1	84.5 ± 0.4	82.0 ± 0.3
speech	71.4 ± 0.0	48.5 ± 0.0	57.7 ± 0.0	46.3 ± 0.0	49.2 ± 0.0	47.5 ± 0.0	47.6 ± 1.5	50.0 ± 0.0	58.5 ± 1.0	46.1 ± 0.0	54.9 ± 4.6	55.4 ± 0.3	54.3 ± 3.5	52.7 ± 0.0	52.9 ± 2.0	55.1 ± 0.3	49.2 ± 0.1	58.7 ± 2.7	62.8 ± 0.4
Stamps	95.1 ± 0.0	92.3 ± 0.0	91.9 ± 0.0	92.9 ± 0.0	95.2 ± 0.0	93.6 ± 1.5	90.6 ± 0.4	93.6 ± 1.4	79.7 ± 2.7	88.1 ± 0.0	80.4 ± 0.6	81.5 ± 1.1	92.5 ± 1.5	89.8 ± 0.0	96.8 ± 2.2	98.0 ± 1.2	91.9 ± 0.1	96.9 ± 1.6	97.4 ± 1.0
thyroid	98.3 ± 0.0	98.5 ± 0.0	98.7 ± 0.0	98.3 ± 0.0	99.4 ± 0.0	98.6 ± 0.2	98.4 ± 0.0	98.6 ± 0.1	75.6 ± 1.6	97.7 ± 0.0	85.8 ± 0.9	95.5 ± 1.3	98.0 ± 0.6	95.5 ± 0.0	96.2 ± 0.0	96.2 ± 0.0	98.3 ± 0.4	99.1 ± 0.2	99.4 ± 0.3
verbal	43.5 ± 0.0	45.7 ± 0.0	58.0 ± 0.0	59.3 ± 0.0	49.6 ± 0.0	55.8 ± 0.7	50.9 ± 1.3	53.6 ± 4.7	61.5 ± 4.7	41.8 ± 0.0	60.8 ± 0.7	46.2 ± 3.4	48.3 ± 9.2	53.0 ± 0.0	66.6 ± 3.5	69.3 ± 5.0	36.7 ± 0.2	82.8 ± 4.3	80.0 ± 0.7
vowels	96.5 ± 0.0	97.3 ± 0.0	96.9 ± 0.0	85.0 ± 0.0	75.8 ± 0.0	97.8 ± 0.8	87.5 ± 0.0	90.4 ± 1.1	98.0 ± 0.0	95.3 ± 0.2	95.8 ± 0.3	94.8 ± 0.0	95.9 ± 0.0	99.1 ± 0.0	99.1 ± 0.0	97.0 ± 1.3	95.4 ± 0.0	95.2 ± 0.5	91.6 ± 6.1
Waveform	76.9 ± 0.0	76.4 ± 0.0	76.7 ± 0.0	72.7 ± 0.0	73.6 ± 0.0	71.6 ± 0.8	73.1 ± 2.9	72.5 ± 1.6	75.3 ± 1.0	60.0 ± 0.0	64.3 ± 1.5	52.2 ± 0.6	82.3 ± 1.9	77.8 ± 0.0	64.9 ± 0.1	75.7 ± 1.8	77.4 ± 0.0	86.7 ± 1.3	80.2 ± 2.6
WBC	99.5 ± 0.0	99.5 ± 0.0	99.5 ± 0.0	99.1 ± 0.0	100.0 ± 0.0	99.5 ± 0.0	99.7 ± 0.2	98.8 ± 0.2	67.3 ± 4.5	99.0 ± 0.0	85.7 ± 1.4	99.2 ± 0.5	99.4 ± 0.5	99.5 ± 0.0	98.2 ± 0.4	98.1 ± 0.6	99.0 ± 0.0	99.7 ± 0.1	99.6 ± 0.1
WDBC	99.4 ± 0.0	99.1 ± 0.0	99.7 ± 0.0	99.6 ± 0.0	99.9 ± 0.0	99.5 ± 0.1	100.0 ± 0.0	95.0 ± 0.3	41.5 ± 4.5	97.8 ± 0.0	95.1 ± 2								

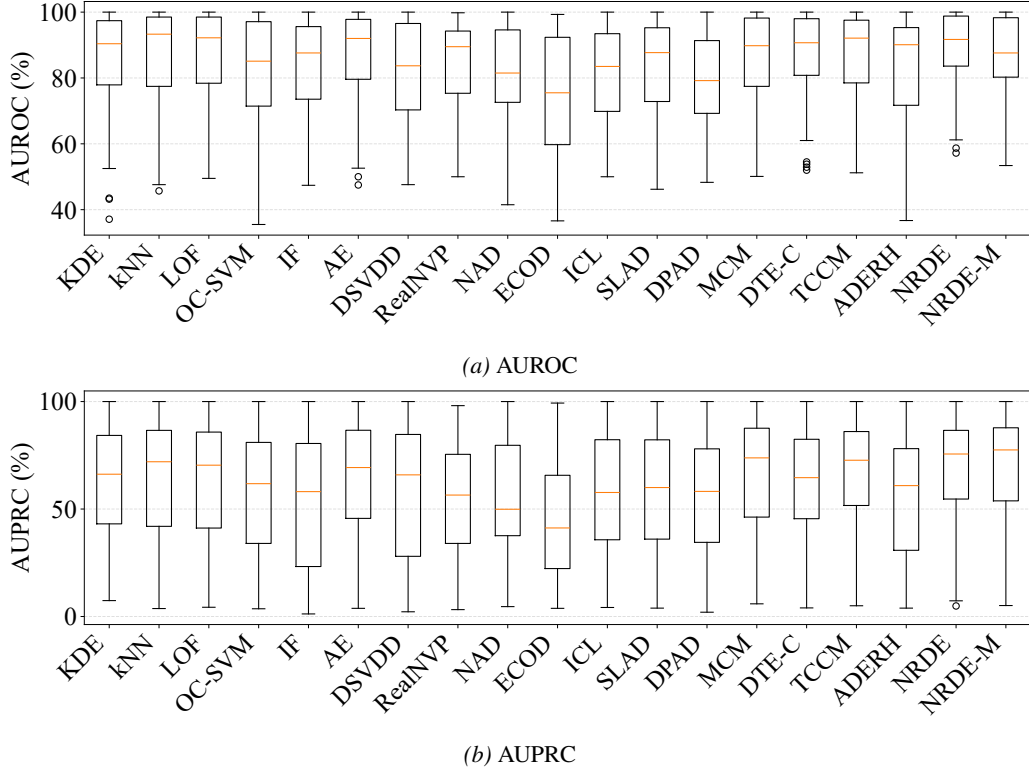


Figure 7. Box plots of 19 methods on 47 benchmark datasets under the standard anomaly detection setting.

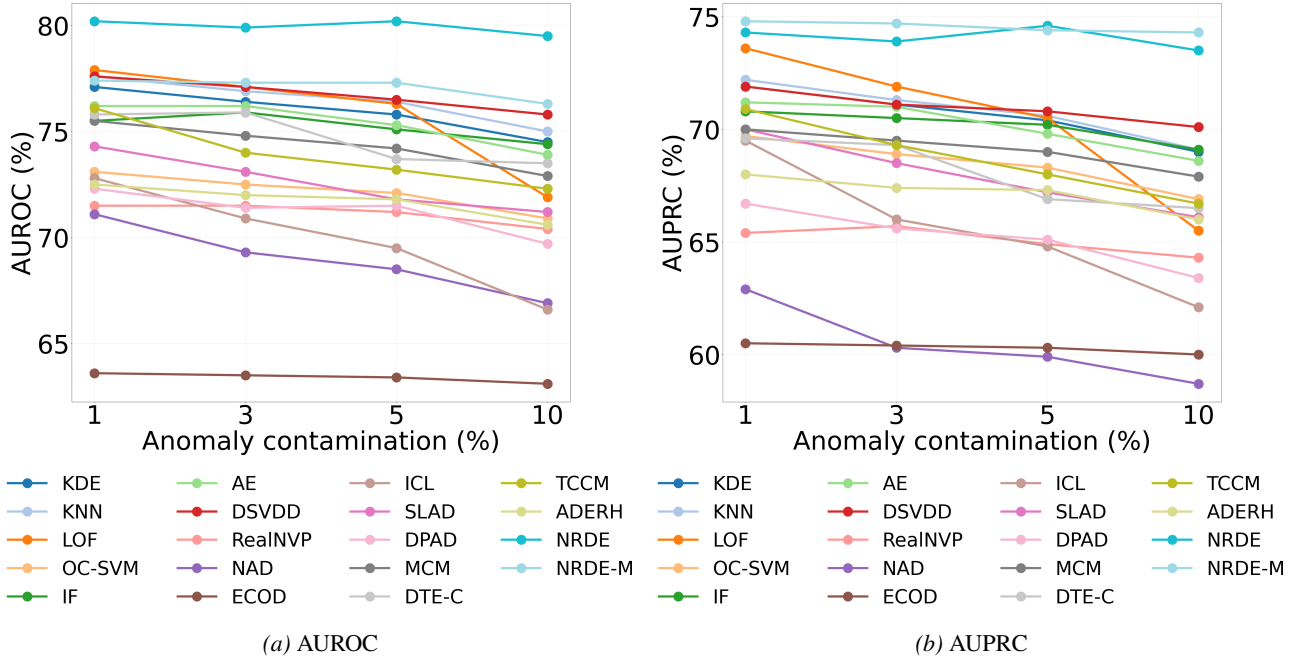


Figure 8. Average performance of all methods under different anomaly contamination ratio.

I.4. Anomaly Detection on Noisy Data

The detailed results of anomaly detection on noisy data are provided in Table 21. We also consider a symmetric variant where anomalous test samples receive stronger noise. In this setting, most methods exhibit improved performance compared to the standard anomaly detection setup, suggesting that adding stronger noise to anomalous samples increases the separability

Table 15. Average AUROC (%) of each method on 47 tabular datasets of ADBench. The best results are marked in **bold**. Each of these methods follow the recommended hyperparameter configuration in their original paper or code without dataset-specific tuning.

Methods	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NeutralAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE
ALOI	56.4	63.4	74.6	55.0	54.4	55.8	51.8	56.2	55.1	53.0	50.2	54.8	51.7	63.2	54.2	54.3	55.1	54.9
anththyroid	91.4	94.1	92.9	90.9	91.8	83.4	79.4	96.1	78.9	78.7	64.0	90.4	91.2	83.9	97.8	82.4	90.6	85.5
backdoor	90.5	93.7	95.7	62.6	75.1	93.5	92.5	91.8	87.1	84.6	95.2	92.5	94.6	96.6	91.8	95.9	75.2	92.1
breastw	98.9	99.1	96.7	99.0	99.5	98.4	99.1	98.0	81.4	99.3	90.2	99.2	98.9	99.0	96.3	98.5	98.7	98.9
campaign	77.3	78.3	69.8	77.1	72.0	80.8	1.1	79.6	62.3	78.3	80.5	76.3	64.2	78.5	76.7	79.5	64.2	81.0
cardio	95.7	93.4	93.0	96.4	94.9	92.4	96.1	94.1	81.0	93.4	83.9	88.7	90.0	90.4	93.6	90.3	94.1	92.8
Cardiotocography	75.0	71.3	72.7	80.7	79.3	73.4	83.4	77.9	58.2	78.5	54.7	58.4	68.0	70.0	72.4	63.3	60.1	68.6
celeba	70.5	68.0	44.9	79.0	70.8	70.9	48.4	79.4	66.4	75.7	69.5	65.2	56.3	65.3	82.7	73.4	79.4	81.0
census	72.0	71.9	60.5	70.2	62.7	71.8	51.9	72.8	72.9	65.9	66.8	68.9	50.7	68.1	69.6	67.9	69.5	78.4
cover	95.5	98.5	98.9	96.2	14.0	98.3	47.6	83.8	85.1	92.0	67.4	79.2	87.9	96.4	96.7	97.2	92.4	87.8
donors	97.4	99.8	98.2	92.0	88.4	93.6	97.7	96.6	95.8	88.8	94.0	96.9	98.7	100.0	98.9	99.8	93.4	100.0
fault	81.2	76.9	62.7	61.4	65.2	73.3	71.9	50.8	73.2	47.4	77.9	79.2	73.5	71.2	71.8	64.7	70.2	68.4
fraud	95.8	96.0	78.3	95.6	95.0	95.7	50.6	54.4	94.9	94.9	83.4	94.6	64.1	95.8	94.4	93.9	95.2	95.7
glass	83.5	89.3	74.0	71.1	81.3	77.9	79.8	81.5	92.1	69.8	90.8	83.5	88.8	86.5	78.4	74.4	88.8	86.5
Hepatitis	79.4	85.0	84.6	84.2	77.8	83.9	80.3	59.4	62.2	71.5	60.1	77.6	83.7	81.2	80.8	79.1	83.5	85.5
http	100.0	99.9	93.0	100.0	99.1	99.8	99.8	99.6	99.0	97.8	50.0	99.9	99.8	99.9	99.5	100.0	100.0	100.0
InternetAds	85.7	73.8	78.7	73.8	43.6	88.2	52.4	81.2	81.6	68.9	78.9	86.7	83.2	82.3	81.1	87.7	70.2	83.9
Ionosphere	97.4	97.0	94.7	96.5	93.6	94.3	78.7	93.8	96.4	75.5	94.1	86.3	97.3	95.4	95.4	95.9	96.0	94.6
landsat	72.7	73.9	75.4	45.7	59.9	57.0	58.1	59.7	70.7	36.6	73.8	69.9	69.7	62.2	58.8	55.5	59.6	55.9
letter	91.8	84.1	86.1	60.9	61.7	80.1	34.2	83.1	92.5	56.0	87.5	90.3	80.2	89.0	89.6	85.0	72.6	77.3
Lymphography	98.6	98.6	98.6	98.4	97.7	98.5	98.5	94.3	82.9	98.5	92.9	98.5	98.3	98.5	97.7	98.4	98.6	95.4
magic.gamma	75.7	82.2	83.2	73.5	77.3	81.8	76.1	79.6	77.5	63.4	71.7	72.4	79.8	82.6	85.8	60.7	69.5	79.9
mammography	88.1	87.6	83.8	88.9	88.3	87.5	81.3	89.5	70.9	90.7	57.3	72.6	84.8	90.7	87.8	79.5	86.9	86.7
mnist	94.8	93.8	92.6	91.0	86.6	93.4	85.1	92.4	77.8	74.7	86.7	91.2	85.8	90.3	84.9	90.0	90.4	92.3
musk	100.0	100.0	1.0	100.0	95.8	1.0	99.9	99.4	100.0	95.8	100.0	100.0	99.9	100.0	100.0	100.0	100.0	100.0
optdigits	97.4	92.7	97.8	60.9	79.6	88.7	32.6	94.2	95.9	60.4	91.7	91.4	75.5	87.8	89.2	57.9	62.6	97.9
PageBlocks	95.0	95.8	96.7	94.4	92.8	94.8	94.7	92.0	93.7	91.4	79.5	87.7	95.4	96.3	96.2	93.6	93.9	93.4
pendigits	99.8	99.8	98.8	96.1	97.0	98.0	88.6	98.5	93.9	92.7	93.8	92.7	94.7	98.6	97.6	94.7	97.0	99.3
Pima	78.1	78.0	73.6	73.1	76.5	73.9	73.0	79.3	56.0	61.5	63.1	54.6	70.2	74.1	70.7	66.8	75.7	78.9
satellite	86.9	87.3	85.1	75.4	80.0	80.3	81.7	83.6	78.4	58.3	83.9	87.0	86.5	82.0	86.1	80.5	81.1	86.4
satimage-2	99.9	99.9	99.6	99.7	99.4	99.8	98.5	98.9	84.9	96.6	96.2	99.7	98.7	98.9	98.9	99.9	99.7	99.7
shuttle	99.8	99.9	99.9	99.7	99.6	99.8	99.2	99.8	99.9	99.3	99.4	99.9	99.9	100.0	99.7	99.9	99.8	99.9
skin	89.1	99.8	92.5	90.3	88.8	83.9	69.6	90.0	88.4	48.8	50.0	91.2	99.3	79.4	92.4	86.0	91.2	92.9
smtp	88.2	93.5	94.2	85.5	90.1	91.4	89.5	93.4	91.0	87.9	53.0	91.9	93.4	83.5	95.2	87.0	92.8	80.0
SpamBase	85.7	83.0	81.7	81.6	83.6	82.0	79.4	80.1	79.2	66.0	81.4	85.3	76.8	81.3	84.5	76.2	79.0	81.1
speech	45.8	48.5	48.9	45.9	46.7	46.8	45.2	50.0	54.3	46.1	49.1	50.7	54.8	49.9	56.1	53.7	48.9	54.4
Stamps	95.1	90.8	87.2	91.2	91.9	89.2	91.9	93.6	74.2	86.7	88.1	73.0	90.9	88.6	67.5	74.5	91.4	94.5
thyroid	98.3	98.5	94.6	98.2	99.0	98.0	97.5	98.6	65.2	97.7	82.2	94.8	96.1	97.9	99.2	83.2	98.2	99.3
vertebral	43.5	42.5	40.0	52.7	42.6	48.0	43.7	53.6	53.9	41.8	54.2	44.4	46.4	47.2	59.2	49.1	33.6	71.0
vowels	96.5	97.3	96.8	83.0	77.7	95.3	41.3	90.4	98.7	59.5	98.2	97.2	93.4	91.5	97.3	85.8	93.6	90.1
Waveform	76.0	76.2	76.6	69.0	72.5	65.8	69.6	72.5	71.5	60.0	59.8	50.2	61.0	69.6	65.6	55.0	70.8	81.5
WBC	98.1	99.4	97.9	99.0	99.7	99.0	99.3	98.8	78.6	99.0	80.0	98.8	98.3	99.1	98.2	98.1	98.5	98.9
WDBC	99.4	99.1	99.4	99.3	99.3	99.4	98.9	95.0	32.5	97.8	83.7	97.8	97.2	97.9	38.9	99.3	99.3	98.7
Wilt	37.1	60.8	70.8	33.9	50.4	56.2	49.9	59.9	80.3	40.3	78.2	66.8	73.1	66.0	86.8	83.5	40.1	88.8
wine	92.2	93.2	92.2	91.2	88.5	85.6	90.2	92.6	78.5	73.0	82.6	92.7	85.3	95.8	92.3	94.1	92.0	99.0
WPCBC	52.5	51.3	50.5	49.1	52.5	49.6	50.0	58.0	59.1	47.0	53.4	50.4	52.1	52.3	48.3	46.3	49.0	71.0
yeast	43.2	46.6	46.7	44.9	42.9	47.9	42.5	51.0	60.1	45.3	55.9	52.7	51.8	45.7	50.5	58.9	39.4	56.4
AVG	84.3	85.1	81.0	79.8	78.2	81.3	74.3	83.6	77.9	73.7	76.1	79.2	81.5	83.4	83.3	80.6	80.5	85.9

between normal and anomalous instances and thereby makes the task easier. The detailed results are shown in Table 22.

I.5. Outlier Detection

To evaluate the effectiveness of our method in outlier detection (transductive learning), we conduct experiments on 10 datasets where all data are used for both training and testing, and compare our method with other AD methods. We provide the detailed experimental results for outlier detection on 10 datasets. We compare our proposed method with traditional and state-of-the-art outlier detection methods. The AUROC and AUPRC results are shown in Table 23.

I.6. Ablation Studies

In this subsection, we investigate how changes in hyperparameters and the architecture of the normalizing flow affect the anomaly detection performance of NRDE. Table 24 and 25 present the performance results of NRDE with different hyperparameter configurations across five datasets. We observe that the regularization term $\mathcal{R}$ contributes positively to overall performance. We also observe that the method is not highly sensitive to changes in λ and learning rate; however, in some datasets, large values of λ may affect the training process and lead to a decrease in performance.

We also include additional experiments to evaluate the sensitivity of our method to architectural choices in Table 26 and Table 27. Overall, NRDE remains robust across different architectures.

Moreover, to validate the effectiveness of our strategy for dynamically selecting $\hat{m}$ across different datasets, we compare

Table 16. Average AUPRC (%) of each method on 47 tabular datasets of ADBench. The best results are marked in **bold**. Each of these methods follow the recommended hyperparameter configuration in their original paper or code without dataset-specific tuning.

Methods	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NeutralAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE
ALOI	10.5	9.8	15.9	7.5	6.6	7.6	7.2	7.1	7.5	6.4	6.1	7.1	7.1	10.9	6.8	9.2	7.2	7.0
anthyroid	66.2	72.0	66.7	65.2	63.8	60.7	54.8	77.0	29.4	40.8	31.3	63.1	64.5	55.0	84.1	52.3	62.4	59.0
backdoor	44.7	45.0	59.9	7.8	9.6	86.8	71.4	77.9	55.8	16.7	89.6	86.0	65.1	81.8	63.2	84.5	12.2	75.2
breastw	98.8	99.1	93.7	98.8	99.5	98.1	99.1	96.7	71.2	99.3	86.3	99.2	98.7	99.0	92.1	98.4	98.2	98.7
campaign	47.7	48.8	39.6	48.9	43.7	49.2	42.5	50.3	28.9	50.0	49.5	48.4	33.0	50.0	48.7	50.5	42.6	57.6
cardio	84.0	76.8	69.3	82.8	78.4	74.7	83.0	71.0	48.9	70.9	60.7	72.7	73.5	73.1	69.5	74.4	80.8	79.6
Cardiotocography	68.1	62.4	59.9	71.0	67.6	65.0	75.1	62.6	40.3	65.7	45.4	54.7	61.5	61.3	61.1	57.4	57.8	55.4
celeba	8.9	9.8	3.7	20.4	12.5	9.5	4.0	13.1	6.6	17.2	8.9	76.1	5.8	7.3	15.7	13.3	18.6	13.7
census	21.6	21.2	14.3	20.5	14.2	21.6	11.9	20.5	23.3	15.5	17.4	19.8	12.1	18.8	18.0	17.9	19.7	28.4
cover	34.2	72.0	83.7	22.0	1.1	52.8	2.0	9.3	29.1	18.4	9.0	9.2	37.4	60.7	67.9	90.2	15.0	22.3
donors	70.9	95.3	76.3	42.4	37.6	49.8	82.3	66.2	64.3	41.2	88.5	65.6	96.6	99.7	77.8	98.0	47.0	94.8
fault	79.8	76.0	64.0	65.2	63.9	72.9	69.3	51.5	70.5	49.4	75.4	78.5	72.5	69.4	72.2	65.9	69.3	70.4
fraud	33.8	31.3	1.1	31.7	21.2	60.5	11.5	3.8	58.8	31.5	30.3	33.3	13.5	80.1	72.8	54.8	27.1	59.7
glass	27.9	29.9	18.8	19.2	22.6	20.8	21.2	24.5	49.7	25.1	38.7	26.1	33.3	26.1	23.4	19.3	29.6	42.1
Hepatitis	59.7	62.1	62.0	61.6	45.9	59.8	54.4	32.0	34.5	40.2	34.3	53.4	59.6	52.4	55.9	53.2	57.1	66.7
http	99.2	99.9	9.6	99.5	45.3	86.8	99.7	55.1	43.3	25.2	0.8	92.9	97.4	87.5	57.7	97.3	99.7	97.6
InternetAds	80.7	65.3	67.3	64.4	27.2	86.1	42.3	56.5	69.6	62.8	62.2	79.6	77.8	73.8	78.4	84.5	60.9	79.8
Ionosphere	97.9	98.3	95.2	97.3	94.0	95.5	80.6	90.9	96.9	77.9	94.4	97.1	97.7	96.6	96.1	96.9	96.8	92.7
landsat	54.8	58.0	70.4	32.1	43.2	40.0	42.5	46.4	49.4	27.8	68.9	48.3	50.6	44.0	39.4	40.8	37.1	43.6
letter	59.9	44.4	49.5	20.7	15.5	36.2	8.6	41.2	69.1	13.8	52.1	57.2	40.6	46.2	64.9	52.3	26.5	37.1
Lymphography	80.0	80.0	80.0	72.0	80.5	76.6	80.0	42.9	28.7	89.7	66.6	77.9	75.6	77.2	65.0	71.7	75.0	48.1
magic.gamma	80.4	85.4	86.2	78.5	80.2	84.9	80.3	83.2	80.5	67.6	77.6	77.9	84.0	86.4	87.9	70.3	76.1	83.5
mammography	43.7	41.9	32.7	41.2	38.8	37.2	34.9	44.8	10.0	54.0	15.9	15.4	36.9	41.2	39.3	42.4	38.4	25.4
mnist	78.7	77.1	72.3	69.4	55.8	74.0	58.1	67.1	49.7	29.9	63.0	74.0	64.8	69.1	58.7	72.4	68.4	66.6
musk	100.0	100.0	99.0	100.0	60.8	99.9	99.2	84.9	100.0	63.2	100.0	100.0	99.9	100.0	100.0	100.0	100.0	100.0
optdigits	49.7	33.6	53.2	6.5	14.1	19.8	4.0	40.5	55.5	7.0	34.0	26.0	16.7	19.9	22.4	7.1	6.7	69.2
PageBlocks	84.8	86.6	87.9	80.0	70.2	82.6	84.4	74.5	78.2	66.4	68.0	71.2	86.9	85.5	84.9	82.4	78.4	78.9
pendigits	96.7	95.9	69.7	47.4	48.9	56.7	27.5	68.0	30.2	38.5	48.3	32.6	58.1	66.6	45.9	65.2	60.5	88.0
Pima	77.0	76.9	73.0	74.3	77.0	73.9	73.4	77.6	56.8	65.7	63.0	58.7	71.0	74.1	70.2	69.6	74.7	79.3
satellite	89.2	89.3	88.5	82.3	84.3	85.8	84.4	86.8	74.5	65.7	87.2	87.8	87.6	85.7	87.7	84.9	84.2	88.2
satimage-2	98.3	97.9	99.6	97.4	93.9	94.2	93.7	57.2	7.3	77.3	82.5	91.8	89.0	61.0	52.4	93.9	98.2	89.6
shuttle	98.1	97.5	99.4	97.5	98.5	97.3	97.0	98.1	99.9	94.3	98.3	97.5	99.2	99.2	94.0	98.3	96.9	98.6
skin	65.0	99.5	73.0	66.3	63.5	64.0	48.6	65.7	67.4	30.3	34.4	80.3	98.3	63.3	70.7	69.9	68.0	73.9
smtp	58.8	42.0	29.3	60.5	0.9	28.9	16.0	32.0	62.8	52.6	6.1	52.1	58.5	43.7	44.1	51.2	49.7	41.7
SpamBase	87.7	86.6	82.9	84.9	87.3	84.8	82.6	80.5	80.8	68.9	86.3	88.9	82.3	84.7	86.2	81.4	83.2	81.9
speech	3.7	3.7	4.5	3.6	3.5	3.6	3.0	3.2	4.0	3.8	3.3	3.8	4.4	4.4	4.9	4.6	3.6	4.3
Stamps	63.7	54.2	44.2	51.0	50.9	47.4	52.8	61.8	26.8	45.2	56.2	33.6	52.8	48.4	26.9	31.2	52.6	79.6
thyroid	73.8	77.4	58.8	73.9	83.7	78.3	78.9	76.4	6.2	62.9	28.8	67.6	60.6	71.9	86.4	58.5	74.8	84.6
vertebral	19.7	20.3	19.6	23.1	19.4	21.6	20.1	25.3	29.8	19.5	26.2	21.4	21.2	20.9	27.1	21.5	16.6	38.4
vowels	77.7	76.3	74.3	44.2	25.4	69.9	6.0	63.6	87.6	14.2	84.5	76.6	64.1	56.4	79.6	42.5	58.2	59.8
Waveform	27.6	27.0	31.7	10.7	10.8	11.1	9.5	11.3	47.4	7.6	29.6	5.7	12.0	20.0	10.3	8.0	15.8	18.4
WBC	85.5	95.7	82.3	91.2	97.8	92.1	94.3	83.2	26.7	99.0	24.3	91.7	86.0	93.7	77.2	86.9	87.2	94.1
WDBC	90.9	81.5	89.9	87.7	86.5	90.0	80.8	36.0	4.0	73.9	34.3	70.3	76.4	66.8	17.1	89.8	88.1	90.2
Wilt	7.4	12.9	17.0	7.0	9.2	10.8	10.1	11.3	51.9	8.1	38.6	17.8	17.3	13.7	29.5	43.8	7.7	48.1
wine	58.2	60.8	52.7	55.0	57.3	42.4	56.5	56.6	33.8	30.5	44.4	64.0	39.8	80.2	59.6	66.2	54.5	94.8
WPBC	38.3	37.5	37.5	36.9	37.8	37.1	37.9	43.8	50.6	35.4	41.3	39	39.2	39.0	36.5	35.2	36.6	65.4
yeast	48.2	49.5	49.8	48.6	48.10.4	40.3	47.0	51.7	57.5	50	55.0	53.1	52.3	49.0	51.2	56.2	46.1	56.1
AVG	62.3	62.2	57.6	54.6	48.9	58.2	51.5	53.6	48.0	45.0	49.9	58.4	58.2	59.9	57.0	59.9	54.6	64.4

NRDE using $\hat{m}$ sources with using different proportions (25%, 50%, 75%, and 100%) of sources with the smallest row norms treated as data sources. As shown in Table 28, the optimal proportions of data sources change for different datasets, yet the performance of NRDE is close to that of using optimal proportions of sources for pure data generation, illustrating the effectiveness of this selection strategy. Meanwhile, using all sources (100%) consistently yields the worst performance across all datasets, which further supports the underlying assumptions and motivations of NRDE.

I.7. Training Time Cost Comparison between NRDE and RealNVP (Normalizing Flow)

Since NRDE and RealNVP share almost the same inference procedure, we report only their training times across datasets of varying dimensionality in Table 29. The main additional time consumption of NRDE compared to RealNVP is the time required for Jacobian matrix computation for each training sample. The results indicate that, in most cases, even for large-scale or high-dimensional datasets, NRDE’s training time remains comparable to RealNVP without a substantial increase.

I.8. Dynamics of Training Loss and AUROC across the Training Procedure

In this subsection, we provide the dynamics of training loss and AUROC across the training procedure for several datasets in Figure 9. As shown in the figure, the decrease in loss is consistent with the improvement in performance.

Table 17. AUROC performance using AutoUAD for hyperparameter tuning on benchmark datasets (best in **bold**).

AUROC	ADERH	DTE	KNN	LOF	TCCM	NRDE
annthyroid	90.7	97.8	94.1	92.9	89.5	95.9
fault	72.5	64.7	79.2	71.0	73.9	69.1
glass	89.8	86.8	83.5	84.0	83.3	87.4
Ionosphere	95.1	95.0	98.0	90.4	96.3	95.4
magic.gamma	69.8	87.7	82.9	82.7	78.0	80.5
mnist	91.3	90.1	94.3	90.6	91.2	92.3
musk	100.0	100.0	100.0	100.0	100.0	100.0
optdigits	59.6	90.0	94.9	98.7	83.1	95.6
PageBlocks	93.9	94.7	96.2	97.4	94.3	93.4
pendigits	97.5	96.3	99.8	99.4	98.4	99.2
satimage-2	99.9	97.1	99.9	99.5	99.4	99.6
shuttle	99.8	99.7	99.9	99.9	99.9	99.8
thyroid	98.3	97.5	98.5	97.7	98.2	99.3
vertebral	37.6	67.0	45.7	49.3	65.6	80.8
vowels	95.2	98.5	97.3	96.6	93.7	92.0
Waveform	68.9	62.5	76.2	76.2	74.2	85.4
WBC	98.5	89.2	99.4	74.9	97.9	99.4
WDBC	99.3	99.2	99.1	99.7	97.7	99.8
Wilt	42.2	91.1	65.2	79.6	83.0	82.2
wine	92.2	94.3	93.3	89.2	88.5	98.5
AVG	88.6	90.0	88.8	88.5	89.3	92.2

Table 18. AUPRC performance using AutoUAD for hyperparameter tuning (best in **bold**).

AUPRC	ADERH	DTE	KNN	LOF	TCCM	NRDE
annthyroid	64.5	83.3	72.0	66.7	55.9	74.5
fault	71.0	64.8	76.0	68.4	73.9	72.7
glass	31.3	30.3	23.4	32.3	24.0	39.9
Ionosphere	96.2	96.4	98.3	90.7	97.1	95.7
magic.gamma	76.5	89.6	85.4	85.6	82.0	84.1
mnist	69.6	62.3	77.1	72.4	70.2	66.6
musk	100.0	100.0	100.0	100.0	100.0	100.0
optdigits	6.2	24.3	33.5	66.6	15.1	55.4
PageBlocks	79.2	83.5	86.6	89.5	84.8	78.8
pendigits	60.4	37.1	95.9	84.5	70.5	79.2
satimage-2	98.3	31.2	98.0	90.3	88.9	95.7
shuttle	96.9	92.6	97.5	98.8	98.9	98.5
thyroid	75.9	62.4	77.4	71.1	72.7	84.5
vertebral	17.6	36.1	20.3	24.3	33.5	50.0
vowels	63.5	89.2	76.3	76.1	76.2	63.1
Waveform	15.5	8.6	27.0	34.8	12.7	27.4
WBC	87.4	39.3	95.7	18.3	86.2	94.3
WDBC	88.6	80.7	81.5	94.3	67.5	98.5
Wilt	8.0	38.1	12.9	26.4	31.9	37.4
wine	52.3	74.8	61.5	42.4	43.1	91.4
AVG	62.9	61.2	67.6	66.7	64.3	74.4

Table 19. Detailed AUROC (%) of each method under anomaly-contamination settings on 10 tabular datasets from ADBench, where the number following each dataset name denotes the anomaly contamination ratio (%). The best and second-best results are highlighted in red and orange, respectively.

Methods	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE	NRDE-M
breastw1	99.2	99.2	98.5	98.9	99.5	99.1	99.3	97.0	75.6	99.4	89.7	99.1	99.2	99.4	97.3	98.8	99.1	98.8	98.9
breastw3	99.2	99.2	98.4	98.9	99.5	99.1	99.1	97.7	73.0	99.4	91.6	99.0	98.7	99.5	96.7	99.0	99.1	99.0	99.1
breastw5	99.2	99.2	98.2	98.9	99.5	99.2	99.3	98.4	72.5	99.4	91.1	99.0	98.7	99.5	96.3	98.7	99.1	99.2	99.2
breastw10	99.2	99.2	97.0	99.0	99.5	98.7	99.0	98.5	75.0	99.3	89.9	97.3	98.5	99.0	95.0	97.5	99.1	99.5	99.4
Cardiotocography1	81.6	72.0	78.0	83.3	84.7	81.8	83.9	63.0	58.2	79.8	58.9	67.8	65.6	79.9	71.8	78.8	67.9	88.7	85.4
Cardiotocography3	81.5	70.3	76.5	81.3	84.9	81.6	84.2	62.8	51.2	79.7	61.3	56.2	63.1	79.6	74.5	74.0	66.1	89.2	88.1
Cardiotocography5	81.1	69.3	75.2	80.2	84.0	81.1	84.4	59.5	48.3	79.5	53.5	54.8	59.3	79.2	71.3	69.2	65.8	88.1	85.4
Cardiotocography10	79.5	65.4	71.5	77.7	80.0	77.6	81.4	62.5	44.6	78.9	48.8	51.4	58.6	77.1	69.9	70.9	62.7	86.9	85.1
Ionosphere1	97.4	98.0	96.2	96.9	92.1	98.0	96.1	90.4	96.7	75.9	96.2	96.3	88.7	96.3	96.8	96.6	96.2	96.0	81.3
Ionosphere3	97.4	98.0	96.1	96.6	91.3	97.9	96.0	91.1	97.2	75.9	95.9	97.8	88.8	96.0	96.9	97.3	96.4	95.3	79.4
Ionosphere5	96.7	97.7	95.7	96.3	92.0	97.5	95.6	90.9	96.1	75.8	95.2	96.0	90.9	96.2	96.8	96.5	95.4	95.1	77.6
Ionosphere10	96.7	97.5	95.3	95.6	90.3	96.9	95.0	91.3	95.8	75.6	94.3	96.4	91.3	95.9	96.1	96.6	95.1	94.7	75.0
landsat1	72.2	74.5	75.6	47.0	65.7	60.6	69.7	55.5	70.6	37.7	76.0	69.9	66.8	67.3	58.6	60.8	62.5	67.1	69.2
landsat3	70.8	71.9	72.4	46.4	64.3	59.9	68.8	54.0	70.0	37.5	73.8	68.4	67.9	62.1	58.2	57.4	61.7	66.4	67.5
landsat5	69.6	70.3	69.8	46.0	64.1	59.8	67.9	53.6	68.6	37.4	71.0	69.3	67.3	59.2	58.6	56.5	62.2	67.3	73.7
landsat10	67.2	67.2	62.6	45.0	65.0	57.1	65.6	53.1	64.8	37.0	72.3	68.9	66.7	57.1	58.7	54.0	59.9	70.6	71.0
magic.gamma1	75.6	83.0	83.9	73.6	75.7	84.2	80.2	77.4	75.8	63.6	71.7	70.4	73.9	82.1	87.3	80.0	70.8	80.3	79.4
magic.gamma3	75.4	82.9	83.2	72.9	77.1	83.1	79.4	75.5	74.5	63.6	69.2	71.9	71.8	81.8	87.2	77.0	70.8	80.4	77.3
magic.gamma5	75.2	82.8	82.3	72.6	74.9	82.8	80.1	75.6	73.7	63.5	63.7	68.8	75.2	80.7	85.1	77.6	71.3	80.5	77.3
magic.gamma10	74.7	82.5	80.5	71.6	74.5	81.7	78.7	74.0	72.5	63.4	63.0	69.4	71.0	80.0	84.6	77.6	71.1	81.1	76.5
Pima1	78.0	78.5	76.5	75.7	76.1	73.5	73.8	76.4	62.2	61.2	67.8	61.1	68.9	68.6	71.9	72.7	76.7	79.8	80.8
Pima3	77.9	78.3	76.4	75.5	75.8	74.0	72.2	77.7	61.6	61.1	63.5	60.7	68.9	69.0	73.1	72.6	76.6	79.8	80.2
Pima5	77.7	78.2	76.2	75.2	74.4	73.2	73.0	77.0	64.1	61.0	66.8	56.6	72.3	68.6	69.8	70.6	76.6	79.2	80.7
Pima10	77.3	78.0	75.6	74.7	74.4	72.5	72.6	75.4	63.4	60.5	63.6	59.8	68.9	68.4	70.5	70.5	76.6	78.3	80.5
satellite1	86.7	87.1	84.9	75.3	82.1	80.5	86.2	74.9	76.7	58.9	77.5	87.9	79.4	77.8	87.0	80.9	81.7	84.1	81.2
satellite3	85.9	86.4	84.4	74.7	82.5	78.8	84.0	76.1	75.3	58.8	68.6	87.7	78.1	76.1	83.5	78.6	81.8	83.7	82.3
satellite5	85.0	85.8	83.4	74.4	82.1	77.1	83.7	77.7	73.0	58.7	64.4	87.7	78.0	75.3	79.4	77.4	81.4	84.1	79.1
satellite10	83.4	84.5	60.9	73.1	81.0	77.6	79.7	74.4	70.6	58.4	55.9	86.6	75.7	73.8	77.2	75.7	80.3	81.3	74.2
SpamBase1	84.4	84.1	81.9	81.1	79.7	81.7	82.7	79.0	79.2	66.8	78.5	83.2	74.8	74.6	84.8	85.6	80.0	81.3	70.8
SpamBase3	81.2	82.9	80.4	79.4	80.4	80.2	81.8	75.4	76.2	66.7	75.2	81.2	71.4	75.6	83.0	83.6	78.4	80.9	72.3
SpamBase5	79.2	81.7	78.9	77.9	79.5	78.9	79.6	75.5	75.3	66.6	72.2	78.7	68.7	75.0	80.4	81.5	77.0	80.2	73.4
SpamBase10	74.7	78.3	74.5	73.6	78.5	74.6	75.2	73.5	70.8	66.2	65.3	73.0	63.6	73.0	77.1	77.9	72.6	78.4	73.9
WPBC1	52.3	51.8	53.6	51.2	53.1	52.7	49.7	54.9	55.9	46.7	56.2	52.1	50.3	56.8	51.5	52.0	48.9	64.7	65.0
WPBC3	51.6	52.1	52.9	50.5	55.5	55.5	52.1	54.4	54.9	46.6	52.9	53.8	50.3	57.2	51.6	49.3	49.1	64.3	64.5
WPBC5	51.4	52.4	53.3	50.7	54.4	52.9	52.4	55.4	55.0	46.8	60.5	53.3	51.0	56.3	50.9	51.3	49.6	67.2	64.0
WPBC10	50.4	51.6	51.5	49.9	53.2	52.0	51.4	54.2	54.9	46.6	55.5	53.4	49.7	53.7	53.4	49.9	48.7	65.1	63.4
yeast1	43.3	47.8	49.7	48.3	46.3	50.0	54.1	46.8	59.7	45.7	55.9	55.0	51.9	50.6	54.4	40.8	40.8	61.3	61.6
yeast3	43.0	47.1	50.6	48.7	48.0	51.4	53.6	50.3	58.9	45.7	56.6	54.2	54.6	51.2	54.4	51.1	40.3	60.4	62.7
yeast5	42.6	46.5	49.7	48.4	45.7	50.5	49.3	48.7	58.2	45.7	56.9	53.6	54.1	52.4	48.3	52.8	40.1	61.1	62.9
yeast10	42.0	45.9	49.3	48.5	47.2	50.0	59.2	46.9	56.9	45.6	57.4	55.4	52.8	51.4	52.9	52.1	40.0	59.4	64.4
AVG1	77.1	77.6	77.9	73.1	75.5	76.2	77.6	71.5	71.1	63.6	72.8	74.3	72.3	75.5	75.8	76.1	72.5	80.2	77.4
AVG3	76.4	76.9	77.1	72.5	75.9	76.2	77.1	71.5	69.3	63.5	70.9	73.1	71.4	74.8	75.9	74.0	72.0	79.9	77.3
AVG5	75.8	76.4	76.3	72.1	75.1	75.3	76.5	71.2	68.5	63.4	69.5	71.8	71.5	74.2	73.7	73.2	71.8	80.2	77.3
AVG10	74.5	75.0	71.9	70.9	74.4	73.9	75.8	70.4	66.9	63.1	66.6	71.2	69.7	72.9	73.5	72.3	70.6	79.5	76.3

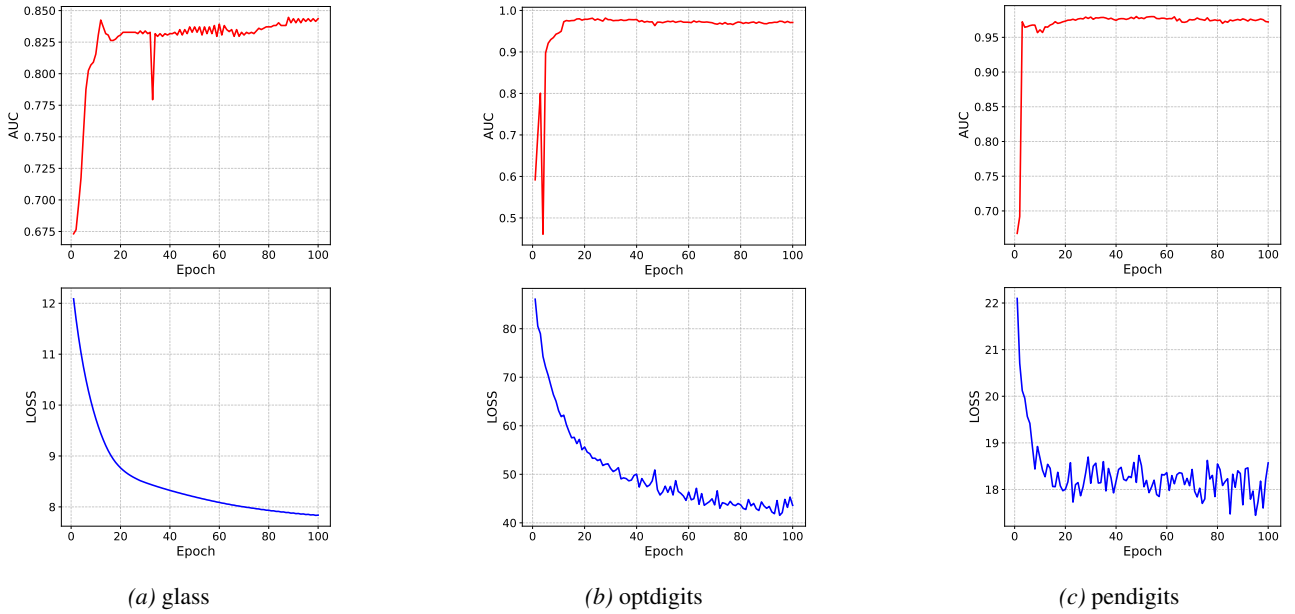


Figure 9. Dynamics of training loss and AUROC across the training procedure for several datasets.

Noise-Robust Density Estimation for Tabular Data Anomaly Detection

Table 20. Detailed AUPRC (%) of each method under anomaly-contamination settings on 10 tabular datasets from ADBench, where the number following each dataset name denotes the anomaly contamination ratio (%). The best and second-best results are highlighted in red and orange, respectively.

Methods	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE	NRDE-M
breastw1	99.1	99.1	98.1	98.6	99.4	99.0	99.3	94.3	64.2	99.4	83.7	99.0	99.1	99.4	94.7	98.5	98.9	98.0	98.5
breastw3	99.1	99.1	97.5	98.7	99.5	99.0	99.1	95.6	60.7	99.4	85.8	99.0	98.3	99.4	92.2	98.8	99.0	98.7	98.7
breastw5	99.1	99.1	95.9	98.7	99.4	99.1	99.2	97.0	59.1	99.3	86.2	98.9	98.4	99.5	91.5	98.3	99.0	98.8	98.9
breastw10	99.1	99.1	91.6	98.9	99.4	98.3	99.1	97.3	61.0	99.3	82.4	96.7	98.1	98.9	90.0	96.3	98.9	99.4	99.2
Cardiotocography1	67.6	59.3	62.6	71.3	72.3	68.3	72.1	42.4	39.5	63.2	52.1	59.5	53.8	62.8	51.4	67.3	57.7	81.7	75.2
Cardiotocography3	67.2	57.1	60.0	67.9	70.0	67.7	71.5	45.5	33.4	62.9	46.0	48.7	49.3	62.2	59.1	61.6	55.0	80.0	79.5
Cardiotocography5	66.6	56.1	58.3	65.7	70.3	66.8	71.3	39.4	32.6	62.7	40.5	43.6	44.3	61.7	53.6	56.8	55.3	75.2	72.8
Cardiotocography10	64.0	52.1	53.3	61.2	65.8	61.3	69.4	40.6	31.5	61.9	34.6	42.2	43.0	60.2	50.5	56.5	51.4	77.2	75.6
Ionosphere1	97.7	98.2	96.5	97.4	92.0	98.4	96.3	85.2	97.1	76.8	96.2	96.9	89.4	96.8	97.0	97.2	96.7	95.8	85.0
Ionosphere3	97.7	98.2	96.4	97.3	90.2	98.1	96.5	86.1	97.2	76.8	95.9	98.1	89.2	96.7	97.5	97.6	96.8	93.8	83.8
Ionosphere5	97.1	97.8	96.0	97.0	92.5	97.8	96.3	85.9	96.3	76.7	95.6	96.6	92.0	96.8	97.1	96.9	96.1	94.9	82.6
Ionosphere10	97.1	97.7	95.5	96.4	90.3	97.2	94.9	86.4	96.0	76.5	95.0	96.8	91.7	96.6	96.2	97.2	95.7	93.6	80.8
landsat1	48.0	51.0	65.2	31.4	41.1	38.1	41.9	40.0	43.3	24.3	64.6	43.4	42.3	44.6	33.6	36.1	35.0	41.1	54.8
landsat3	45.5	47.1	56.6	30.6	39.2	37.5	41.0	37.9	43.1	24.2	58.3	40.7	43.1	39.3	33.5	34.9	34.3	41.8	51.9
landsat5	43.7	43.4	51.3	30.0	39.2	35.8	40.6	36.3	42.2	24.1	57.0	43.3	41.5	36.6	33.2	34.0	35.7	47.0	57.7
landsat10	40.8	40.1	42.3	28.8	37.7	33.8	38.1	35.8	39.7	23.9	57.5	40.0	42.0	35.1	33.6	30.9	33.2	43.8	58.7
magic.gamma1	78.9	84.3	85.5	77.5	77.4	85.7	83.0	80.4	76.2	65.8	75.7	75.1	76.0	84.6	88.4	82.7	75.5	82.8	83.3
magic.gamma3	78.7	84.2	84.1	76.6	78.0	84.6	81.7	78.3	74.5	65.8	72.7	76.1	73.7	84.0	87.7	79.9	75.5	82.2	81.4
magic.gamma5	78.5	84.1	83.4	76.4	76.6	84.1	82.6	78.6	72.1	65.8	66.0	73.2	77.9	83.2	86.0	79.1	75.7	82.7	81.3
magic.gamma10	78.0	83.7	80.9	75.8	76.3	82.8	81.3	77.6	71.0	65.6	65.0	73.7	73.4	82.3	85.1	79.3	75.6	82.6	80.8
Pima1	74.8	75.8	73.5	73.6	75.3	71.9	72.6	72.0	59.0	63.7	66.6	61.1	66.8	70.5	70.9	70.7	73.7	77.7	79.3
Pima3	74.6	75.5	73.2	73.4	74.6	73.0	71.9	73.5	57.2	63.6	61.0	61.2	67.1	69.0	71.3	70.5	73.6	77.9	79.1
Pima5	74.3	75.3	72.8	73.0	74.0	70.3	72.5	71.5	61.2	63.4	66.3	58.2	68.5	69.5	67.7	68.3	73.1	76.3	80.2
Pima10	73.7	74.9	72.1	72.5	73.5	71.0	71.5	71.4	59.4	63.0	62.3	60.3	66.1	69.5	68.9	69.1	73.3	76.8	79.5
satellite1	87.0	87.9	87.2	81.1	83.6	81.8	85.0	79.5	68.6	64.2	79.1	86.9	75.3	77.8	86.1	83.8	83.6	85.0	84.7
satellite3	85.5	86.8	86.2	80.7	84.1	80.1	81.3	80.9	66.1	64.0	69.4	86.7	72.7	76.2	83.2	80.7	83.4	85.2	84.2
satellite5	84.2	86.6	84.7	80.3	83.6	78.8	81.5	81.7	64.6	63.9	66.0	86.6	70.8	75.3	77.3	78.7	83.1	85.0	81.9
satellite10	81.8	85.0	63.2	79.3	82.7	79.7	80.9	78.5	61.4	63.6	61.8	85.0	69.9	73.0	75.2	76.1	82.2	84.4	77.9
SpamBase1	85.7	85.2	82.0	83.3	83.6	83.4	84.5	77.6	79.0	68.2	81.8	85.8	77.8	77.2	86.0	87.1	82.1	79.6	74.8
SpamBase3	83.4	83.9	80.0	81.3	84.6	81.9	82.8	73.9	77.5	68.1	80.0	83.9	74.9	78.5	83.9	85.4	80.3	79.5	75.8
SpamBase5	81.1	82.3	77.8	79.4	83.4	80.0	80.3	73.8	75.9	67.9	75.8	81.7	71.6	78.0	80.4	83.1	78.7	77.7	77.2
SpamBase10	76.8	78.5	72.9	74.7	82.5	75.3	77.7	71.7	71.2	67.5	70.1	75.8	66.3	77.0	77.5	79.2	74.1	77.4	78.4
WPBC1	34.2	34.0	35.1	33.7	34.5	34.3	33.6	36.5	47.2	31.3	41.2	40.4	34.7	37.1	37.9	34.3	32.6	46.3	52.9
WPBC3	33.9	34.1	34.6	33.2	35.5	38.0	35.3	37.0	38.6	31.2	37.3	39.4	35.4	39.9	34.2	33.4	32.5	43.6	51.8
WPBC5	33.9	34.4	34.4	33.5	34.7	35.3	35.9	37.1	41.8	31.3	41.0	38.8	33.7	38.5	34.4	33.9	32.9	51.5	49.9
WPBC10	33.3	33.8	34.0	32.8	33.8	36.7	34.3	36.8	42.6	31.2	37.6	38.0	34.3	36.3	36.4	33.4	32.2	45.9	50.8
yeast1	46.0	47.3	50.3	48.8	48.8	50.6	50.7	46.2	54.7	48.0	54.5	52.0	51.7	49.6	49.6	51.1	44.2	55.5	59.4
yeast3	45.8	47.0	50.4	49.0	49.8	50.1	50.1	48.4	54.7	48.0	53.6	51.5	52.1	49.6	50.2	50.0	44.0	56.2	61.2
yeast5	45.6	46.8	50.0	48.8	48.5	49.9	48.3	47.8	53.6	47.9	53.9	50.7	51.9	51.0	48.1	50.4	43.8	55.8	61.7
yeast10	45.2	46.4	49.5	48.7	49.2	49.6	54.1	47.1	53.0	47.9	55.1	52.3	49.4	49.8	51.4	49.3	43.8	54.4	61.4
AVG1	71.9	72.2	73.6	69.7	70.8	71.2	71.9	65.4	62.9	60.5	69.5	70.0	66.7	70.0	69.6	70.9	68.0	74.3	74.8
AVG3	71.1	71.3	71.9	68.9	70.5	71.0	71.1	65.7	60.3	60.4	66.0	68.5	65.6	69.5	69.3	69.3	67.4	73.9	74.7
AVG5	70.4	70.6	70.5	68.3	70.2	69.8	70.8	64.9	59.9	60.3	64.8	67.2	65.1	69.0	66.9	68.0	67.3	74.6	74.4
AVG10	69.0	69.1	65.5	66.9	69.1	68.6	70.1	64.3	58.7	60.0	62.1	66.1	63.4	67.9	66.5	66.7	66.0	73.5	74.3

Table 21. Detailed AUROC (%) and AUPRC (%) of each method on anomaly detection with injected noise experiments of 10 tabular datasets of ADBench where test normal data are injected with stronger noise. The best and second-best results are highlighted in red and orange, respectively.

AUROC	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	NRDE	NRDE-M
breastw	99.1	99.2	98.8	98.9	99.5	99.0	99.3	96.7	66.4	76.7	94.4	98.4	99.0	99.2	97.5	98.8	98.9	98.7	99.0
Cardiotocography	77.7	68.7	71.6	74.7	70.4	75.1	83.6	50.6	40.8	64.2	60.2	59.0	59.5	67.0	56.4	75.5	62.7	84.3	80.7
Ionosphere	94.3	96.8	95.9	95.4	89.4	95.9	77.1	77.0	91.6	73.9	93.3	93.4	87.8	92.3	96.5	93.6	92.6	94.4	84.3
landsat	47.2	47.1	44.8	35.2	46.6	26.6	49.0	50.0	55.2	30.7	62.8	61.5	54.4	39.8	12.8	42.9	35.9	70.1	69.2
magic.gamma	68.5	68.7	68.2	67.2	67.1	68.0	69.5	67.6	65.0	59.1	64.6	65.2	67.2	68.9	67.5	67.0	62.9	82.2	76.1
Pima	74.5	73.0	72.3	71.4	70.5	70.1	70.8	69.2	54.9	56.7	61.3	55.2	71.5	69.8	67.4	68.5	71.8	79.6	76.7
satellite	74.6	75.1	70.6	67.0	71.5	67.2	74.8	65.2	67.6	55.0	77.6	80.6	75.0	66.7	56.9	70.9	68.4	84.7	79.5
SpamBase	80.8	80.5	79.4	78.8	58.1	79.3	78.6	58.9	68.3	11.5	77.4	79.9	72.3	76.2	78.8	77.4	78.9	83.6	78.0
WPBC	47.3	45.2	47.9	44.5	45.5	44.8	51.0	50.0	45.5	39.7	50.6	51.3	50.5	34.1	41.8	35.9	44.1	75.7	70.5
yeast	38.6	37.5	39.5	40.0	35.9	42.9	40.4	39.2	53.1	33.5	51.0	50.1	49.1	40.7	41.5	47.5	35.2	64.3	61.3
AVG	70.3	69.2	68.9	67.3	65.4	66.9	69.4	62.4	60.8	50.1	69.3	69.5	68.6	65.5	61.7	67.8	65.1	81.8	77.5
AURPC																			
breastw	99.1	99.2	98.7	98.8	99.5	99.0	99.2	94.4	64.1	77.9	93.2	98.4	98.9	99.1	94.9	98.7	98.8	98.6	99.0
Cardiotocography	68.6	62.2	62.3	66.7	58.7	67.1	76.2	36.4	31.6	50.8	51.3	52.6	49.3	61.7	50.6	68.2	58.6	76.1	67.0
Ionosphere	95.9	97.4	96.8	96.6	91.0	96.8	78.2	71.2	92.3	78.8	92.2	95.0	89.4	95.1	97.2	95.7	94.2	95.0	89.0
landsat	34.2	33.4	38.6	28.1	31.9	24.4	35.4	34.3	37.5	25.6	55.7	40.7	39.3	27.8	20.7	33.4	26.1	55.7	51.6
magic.gamma	76.1	76.2	75.9	74.7	73.1	75.5	76.2	74.6	67.6	64.9	71.3	72.6	73.9	76.0	75.5	73.0	72.1	85.3	79.1
Pima	73.6	73.5	71.9	72.0	71.9	69.5	71.2	69.6	55.2	56.6	61.3	56.9	72.2	70.7	67.8	69.1	71.9	79.8	77.8
satellite	81.3	81.3	79.8	77.9	78.7	76.8	79.2	64.2	64.3	64.8	81.6	82.8	75.5	74.2	70.1	77.4	75.8	87.3	83.3
SpamBase	85.2	84.7	83.2	83.6	69.6	83.8	81.5	62.8	71.9	38.2	84.1	84.7	76.0	84.2	83.4	83.8	83.2	85.3	79.9
WPBC	36.2	35.7	36.3	34.9	35.0	35.9	39.1	38.2	37.4	32.5	40.5	40.4	38.9	30.0	34.8	30.9	34.5	67.5	64.2
yeast	44.8	44.0	45.4	45.9	43.3	48.5	45.1	44.6	51.8	41.5	51.8	50.8	50.8	44.9	45.7	48.6	43.6	60.4	57.6
AVG	69.5	68.8	68.9	67.9	65.3	67.7	68.2	59.0	57.4	53.2	68.3	67.5	66.4	66.4	64.1	67.9	65.9	79.1	74.8

Noise-Robust Density Estimation for Tabular Data Anomaly Detection

Table 22. Detailed AUROC (%) of each method on anomaly detection with injected noise experiments of 10 tabular datasets of ADBench where test anomalous data are injected with stronger noise. The best and second-best results are highlighted in red and orange, respectively. * indicates that we report the performance in their paper.

Methods	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	NRDE	NRDE-M
breastw	99.1	99.1	98.6	98.9	99.6	99.1	99.2	94.8	64.8	86.9	92.1	98.9	99.0	99.3	97.8	98.9	99.2
Cardiotocography	73.4	68.8	78.1	78.5	79.1	75.1	75.8	36.1	40.5	74.3	59.9	62.7	51.1	76.9	72.1	83.6	86.0
Ionosphere	98.4	98.7	97.3	98.1	93.2	98.5	77.6	88.0	95.6	85.0	94.6	97.7	89.5	98.3	98.2	95.4	84.0
landsat	72.2	72.3	89.4	51.0	40.0	76.7	44.7	52.6	55.4	29.4	65.4	57.6	54.6	72.9	85.7	77.6	70.0
magic.gamma	82.6	86.3	88.4	83.1	79.7	87.9	82.8	84.4	75.0	70.7	74.2	76.3	76.7	86.0	88.7	85.2	81.7
Pima	79.2	79.5	76.6	76.8	80.2	75.3	75.9	51.7	61.2	70.7	68.3	58.5	75.1	76.6	76.7	80.5	79.1
satellite	94.1	94.3	95.6	87.9	84.3	94.7	82.8	48.1	74.3	68.1	79.4	90.9	87.2	95.0	95.6	91.2	82.8
SpamBase	89.5	88.0	87.4	86.6	96.5	87.6	84.4	59.8	78.7	91.2	87.9	87.6	74.4	92.4	92.4	90.5	84.2
WPBC	42.6	41.5	44.1	41.3	41.1	46.7	44.0	81.0	47.2	37.6	45.5	43.7	39.3	51.1	64.1	75.6	69.1
yeast	52.9	55.3	61.4	57.1	64.1	63.4	50.6	55.9	63.9	61.2	58.9	57.7	56.6	58.5	65.5	61.1	60.3
AVG	78.4	78.4	81.7	75.9	75.8	80.5	71.8	65.2	65.6	67.5	72.6	73.2	70.4	80.7	83.4	84.0	79.6

Table 23. Detailed AUROC and AUPRC performance of outlier detection on 10 datasets. The best and second-best results are highlighted in red and orange, respectively. * indicates that we report the experimental results from their original paper.

AUROC	KDE	KNN	LOF	OC-SVM	IF	AE	DSVDD	RealNVP	NAD	ECOD	ICL	SLAD	DPAD	MCM	DTE-C	TCCM	ADERH	ODIM*	NRDE	NRDE-M
breastw	3.8	98.4	48.5	99.1	99.1	95.9	84.5	98.2	90.3	99.1	66.7	80.6	90.9	99.4	93.3	93.3	98.9	99.1	99.2	98.9
Cardiotocography	49.7	51.9	54.7	74.1	75.2	59.5	75.2	65.7	61.7	78.5	45.4	50.5	54.3	78.9	58.2	65.6	59.2	61.0	78.1	79.3
Ionosphere	19.9	92.8	89.1	85.2	75.9	93.9	57.7	92.6	88.7	72.8	69.8	85.1	68.1	96.4	95.0	94.4	91.7	84.8	95.4	90.4
landsat	63.6	61.8	54.7	42.4	57.0	53.9	65.3	51.3	62.8	36.8	64.1	67.6	57.5	66.0	58.8	54.5	61.5	46.2	72.9	68.2
magic.gamma	33.0	81.1	71.7	68.6	72.3	76.9	72.1	69.9	68.3	63.8	58.7	66.2	62.9	83.2	77.7	71.2	75.0	74.5	80.8	73.8
Pima	33.9	72.7	65.1	68.6	67.8	65.9	62.8	67.9	66.1	59.4	56.1	52.3	61.2	70.7	65.0	68.6	73.6	71.9	75.0	78.8
satellite	39.3	73.1	56.4	66.4	73.9	71.3	68.5	72.1	65.9	58.3	54.5	80.5	65.0	81.6	74.7	76.1	80.2	69.8	81.1	78.7
SpamBase	50.5	56.7	47.9	54.7	73.4	55.7	57.5	50.0	58.6	65.6	47.8	53.1	50.6	76.3	59.2	63.6	58.5	55.0	78.8	76.9
WPBC	51.1	52.9	52.0	49.8	52.4	51.3	49.2	56.8	49.3	48.1	54.8	53.3	52.9	56.6	52.2	55.6	54.1	51.0	73.8	76.2
yeast	61.7	40.6	46.8	45.2	44.5	46.8	44.7	41.6	50.2	44.4	57.9	55.2	51.4	51.9	43.0	48.8	38.7	65.3	58.5	61.2
AVG	40.7	68.2	58.7	65.4	69.2	67.1	63.8	66.6	66.2	62.7	57.6	64.4	61.5	76.1	67.7	69.2	69.1	65.3	79.4	78.2
AUPRC																				
breastw	21.5	95.5	31.6	98.1	98.1	88.1	86.4	94.1	71.8	98.4	63.1	71.3	79.5	99.4	79.0	83.4	97.3	98.8	98.4	97.9
Cardiotocography	22.9	33.6	29.5	41.3	48.5	35.0	47.9	31.6	31.8	50.5	20.7	34.0	25.2	66.2	31.1	37.2	36.9	38.9	51.7	49.4
Ionosphere	23.3	92.9	85.0	82.4	65.6	93.2	45.3	84.9	78.7	64.6	57.9	62.9	58.9	97.1	92.7	93.2	90.5	86.7	92.1	86.4
landsat	37.1	25.8	25.2	17.5	22.8	21.7	28.8	22.7	28.5	16.3	42.7	29.1	24.9	49.0	23.9	22.0	24.2	17.9	37.1	35.5
magic.gamma	27.1	74.1	56.4	62.1	62.8	69.2	63.1	63.4	52.3	53.4	44.7	59.2	50.8	86.7	66.5	59.3	68.2	69.3	71.8	62.9
Pima	27.7	52.9	45.1	50.4	51.3	48.6	47.0	48.2	47.9	46.4	41.4	38.1	43.1	73.9	45.4	51.5	53.7	49.1	59.4	61.6
satellite	34.2	59.2	39.3	65.5	65.8	58.3	54.7	72.2	42.7	52.6	41.3	63.6	48.1	86.5	57.1	64.5	69.3	65.2	74.0	67.2
SpamBase	37.6	41.7	38.9	40.8	62.2	41.7	42.9	39.9	43.3	51.8	37.0	40.2	39.6	80.4	44.2	49.0	42.8	41.0	68.0	63.7
WPBC	23.8	23.9	23.2	22.9	24.3	26.1	23.4	26.3	25.9	21.8	30.9	25.1	25.2	42.0	27.7	26.1	24.2	23.6	52.8	49.6
yeast	48.0	30.1	32.7	31.5	34.3	32.0	30.8	29.3	34.4	33.3	41.2	38.2	36.8	52.4	30.8	32.9	29.3	28.7	38.7	40.1
AVG	30.3	53.0	40.7	51.3	53.6	51.4	47.0	51.3	45.7	48.9	42.1	46.2	43.2	73.3	49.9	51.9	53.6	51.9	64.4	61.4

Table 24. Detailed AUROC and AUPRC performance of NRDE with different values of learning rate lr .

AUROC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$lr = 0.001$	85.4	80.8	80.0	85.8	59.2
$lr = 0.005$	87.1	79.1	78.3	77.3	64.4
$lr = 0.01$	81.6	76.8	76.4	61.3	57.2
AUPRC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$lr = 0.001$	79.0	80.9	79.5	87.9	45.2
$lr = 0.005$	79.6	77.7	79.0	77.8	49.3
$lr = 0.01$	67.5	77.1	77.0	59.5	42.8

I.9. Experimental results of using dimension reduction methods for high-dimensional data before NRDE

The main computational bottleneck of NRDE compared to RealNVP is the Jacobian matrix computation for each training sample, and this time-cost gap increases with data dimensionality. However, NRDE’s inference time remains virtually identical to standard RealNVP. The overhead may become prohibitive on high-dimensional data. However, for such high-dimensional cases, the computational cost can be easily mitigated by employing dimensionality reduction methods like PCA before applying NRDE. Table 30 demonstrates that NRDE remains effective even on low-dimensional (128) data processed via Kernel PCA.

Table 25. Detailed AUROC and AUPRC performance of NRDE with different values of hyperparameter λ .

AUROC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$\lambda = 0$	85.9	84.2	76.7	79.4	61.6
$\lambda = 0.01$	86.9	80.7	80.0	85.8	63.8
$\lambda = 0.1$	87.1	80.2	79.2	83.1	64.4
$\lambda = 1$	82.1	77.1	79.1	84.0	64.0
AUPRC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$\lambda = 0$	78.3	85.2	80.4	78.2	47.0
$\lambda = 0.01$	80.9	80.9	79.5	87.9	51.2
$\lambda = 0.1$	79.5	80.7	79.8	85.8	49.3
$\lambda = 1$	66.3	76.8	79.9	86.4	48.5

Table 26. Detailed performance of NRDE with different numbers of coupling layers (T).

AUROC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$T = 2$	87.1	80.8	80.0	85.8	64.4
$T = 3$	83.3	83.8	80.4	85.3	71.1
$T = 4$	82.2	84.2	80.6	83.3	62.6
AUPRC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$T = 2$	79.5	80.9	79.5	87.9	49.4
$T = 3$	71.9	85.1	81.7	87.2	60.0
$T = 4$	65.3	84.8	80.1	86.0	52.2

Table 27. Detailed performance of NRDE with different width of coupling layers (b).

AUROC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$b = 512$	85.3	81.0	78.8	84.4	71.5
$b = 1024$	84.2	81.6	80.2	84.2	73.6
$b = 2048$	87.1	80.8	80.0	85.8	64.4
AUPRC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
$b = 512$	77.2	81.1	78.5	86.8	64.8
$b = 1024$	75.2	81.5	80.8	86.9	67.1
$b = 2048$	79.5	80.9	79.5	87.9	49.4

Table 28. Detailed performance of the proposed NRDE using different proportions of sources as data sources. The best performance is marked in red.

AUROC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
NRDE (25%)	86.7	81.5	79.8	82.5	63.4
NRDE (50%)	87.3	82.3	81.3	84.3	61.4
NRDE (75%)	86.2	81.8	81.1	83.0	62.0
NRDE (100%)	81.1	81.3	78.1	81.7	60.2
NRDE	87.1	84.5	80.0	85.7	64.4
AUPRC	Cardiotocography	SpamBase	Pima	Satellite	WPBC
NRDE (25%)	80.3	82.6	79.4	84.9	48.0
NRDE (50%)	76.3	82.2	81.4	86.9	45.7
NRDE (75%)	73.9	82.0	79.7	86.1	47.1
NRDE (100%)	61.9	81.4	75.8	84.9	45.0
NRDE	79.5	85.3	79.5	87.9	49.3

Table 29. 100-epoch training time cost (s) comparison between NRDE and RealNVP on different dimensional datasets from ADBench.

Dataset	NRDE	RealNVP
annthyroid	4.12	2.03
glass	1.58	0.59
mammography	5.46	2.72
Pima	1.68	0.61
vertebral	1.45	0.59
Cardiotocography	4.29	0.90
fraud	388.80	75.21
satellite	30.31	18.8
satimage-2	31.40	19.04
shuttle	46.26	30.05
mnist	71.34	20.84
musk	70.41	18.03

Table 30. Experimental results of using dimension reduction methods for high-dimensional data before NRDE

Data (Dimension)	Original	Low-dimensional (128)
Speech (400)	58.7	56.5
InternetAds (1555)	83.9	80.0